Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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Abstract

This paper uses graduate admissions to study dynamic selection under costly evaluation. With a privacy-protecting synthetic data protocol, I relate acceptance to success measures based on job placements. I estimate structural models of an admissions committee's decision problem capturing first-round filtering and matriculation, test whether observed decisions are optimal given the number of application files read, and quantify gains from deviating to the models' chosen sets. The committee's decisions align with many applicants' estimated success probabilities, but some applicant characteristics may be misweighted. Reweighting these characteristics increases the admitted cohort's average success probability without reading more files.

Keywords: Costly Evaluation, Dynamic Optimization, Education Economics, Graduate Admissions, Optimal Selection, Structural Estimation, Synthetic Data

JEL Codes: C61, C81, D81, I23

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1 Introduction

How do decision-makers select among a set of options in which each option's true quality is unobserved and costly to evaluate in terms of time and effort? In many such problems, decision-makers rely on a coarse, low-cost screening of all the options, followed by a precise, high-cost evaluation of the subset of options that survive the screening. Moreover, if the selected options have agency over whether to match with the decision-maker, final selections are often staggered over time through a deferred process to ensure the number of matches meets a target number.

Admissions and hiring are two prominent examples of such selection problems. There, committees first screen all the applications to filter out unqualified applicants, then read a subset of promising applications in full detail, then potentially interview the most promising applicants, and finally make early and late offers, depending on how many early offers are accepted. Other examples of selection with costly evaluation, and in some cases deferred offers, include conducting multi-stage clinical trials for medicines, messaging virtually on dating apps before meeting in real life, and simulating product prototypes on a computer before physically building them.

A fundamental question in such settings is how decision-makers should interpret the available information at each stage of evaluation to select either the best subset of options or all options above a quality threshold. A related question is whether observed decision rules reflect optimal dynamic selection given evaluation costs and other constraints, or whether those rules systematically deviate from optimal behavior. One could imagine a hiring committee undervaluing a skill or discriminating against a demographic. Moreover, if there are deviations, which decision rules should decision-makers implement instead? Answering these questions requires moving beyond identification of the determinants of success and explicitly modeling the decision problem.

Graduate admissions is a meaningful setting to study these questions, as evidenced by economics papers from Krueger and Wu (2000) to Bai, Esche, MacLeod, and Shi (2022). Graduate programs, especially PhD programs, invest at least a few years of time, along with tens of thousands of dollars of funding, into students' development. In addition, high-ranking academic placements boost a program's reputation, and so increase its ranking from sources such as U.S. News. Better students also lead to better coauthoring opportunities for faculty, and since people with graduate degrees often find themselves in influential positions, such as professorships, studying how best to admit them is important for programs and society. In particular, if schools are undervaluing applicants with certain characteristics (e.g. from a particular demographic), future applicants with such characteristics may receive opportunities they would not have otherwise. For example, while Chetty, Deming, and Friedman (2026) focus on undergraduate admissions, they find that applicants from high-income families have higher admissions rates at Ivy-Plus colleges even if their standardized test scores are no higher than those of less fortunate applicants.

Moreover, graduate admissions is a setting with costly evaluation. Admissions committees observe for all applicants a limited set of objective characteristics, such as standardized test scores, but typically lack the time to read every file in detail. As such, applicants are initially filtered based on coarse signals, with more intensive evaluation reserved for the subset of applicants with promising objective characteristics. Final decisions may also involve waitlists or other forms of deferred acceptance, as uncertainty about applicant quality and matriculation status resolves over the period between the application and admissions deadlines. These features make graduate admissions well-suited for modeling dynamic selection under costly evaluation. Doing so for one economics PhD program, I find that while the committee's decisions are much better than random chance and align with many applicants' estimated success probabilities, the committee misweights some applicant characteristics, and correctly weighting them would significantly increase the admitted cohort's average success probability without reading any more files.

This paper addresses two specific research questions. Both require first determining which *ex-ante* factors in application files affect admission and success in graduate school. Following the literature, success may be defined in multiple ways, including program completion, and job placement outcomes. I study these factors as comprehensively as possible by extracting all available data from application files, including recommendation letters, application essays, and CVs. My data consist of all applicants to a research university's economics PhD program from 2015 to 2019, for which I observe both the application files and the outcomes. In Online Appendix A, I supplement these data with the objective characteristics of all applicants to that university's economics, government, history, linguistics, and psychology PhD programs from 2020 to 2023 to compare admissions across programs. Because these data are confidential, I implement a privacy-protecting research protocol that uses synthetic data for model development and validation (see Section 2). This protocol also serves as a template for researchers who wish to work with sensitive data in general. It also has a side benefit, namely that I had to commit to my specifications on the synthetic data and could adjust them only sparingly after seeing results from the actual data.

The first research question asks whether admissions committees optimize or make mistakes when admitting applicants. Specifically, if a factor predicts admission but not success, the committee may be overvaluing it, whereas if a factor predicts success only, the committee may be undervaluing it. For instance, the committee may be putting too much or too little weight on elite undergraduate institutions, standardized test scores, or other applicant characteristics. To determine whether the committee is optimizing, it is necessary to model the committee's optimization problem, estimate the parameters, and formally test the hypothesis of optimization, which to date, no other paper has done. To make such models tractable, per Bai et al. (2022), I reduce the committee's portfolio choice problem (i.e. from N applicants, accept the optimal subset of N_A applicants) to a cutoff rule problem (rank the N applicants by their expected success probabilities and select the N_A above the cutoff). I estimate multiple structural models in increasing order of complexity, modeling such features as first-round filtering and matriculation. That said, PhD program quality also affects success (Angrist, Diederichs, & Ellison 2026), so it is important to account for that lest the model encourage the committee to put more weight on a variable that only predicts success by way of predicting admission into a highly-ranked program.

Perhaps as Simon (1956) suggests, if committees are not optimizing, they may be satisficing (i.e. aiming for a satisfactory result), namely because reading every application file would require too much effort.[1] This idea lends itself to the second research question of whether committees can make better decisions without having to expend additional effort, and if so, how much better of a cohort they can admit. To answer this question, I use the structural models to calculate counterfactual deviation gains from adopting the optimal admissions strategy, while respecting the committee's time constraint of not being able to read every application file in detail. For inference, I perturb the success parameter estimates about their asymptotic distribution (Krinsky & Robb 1986) when comparing the expected success probability for the cohort admitted by the model's decision rule vs. the committee's. I then show that the fraction of perturbations in which the model's cohort does not improve on the committee's is an asymptotically valid p-value, though in a future version of this paper, I will present a version of the optimality test that is robust to sampling error (Andrews, Kitagawa, & McCloskey 2023). Perhaps committees have other priorities than maximizing success probabilities, such as admitting a diverse class across demographics, but I show how the models can be extended to factor in those priorities.

The structural estimates show that for many applicants, the success probabilities implied by the committee's decisions align with the success probabilities estimated from outcomes. However, the committee may have misweighted some applicant characteristics, and the deviation gains test rejects the null hypothesis of no gains. Holding fixed the number of files the committee read in each round and the number of applicants it admitted in each year, so that the counterfactual committee only re-ranks the applicants, the model's acceptance set's average success probability is greater than the committee's by about 9 percentage points. When the models account for matriculation, so that the relevant set is the matriculation set, the average success probability increases by about 10 percentage points. Because the committee's own subjective score for each application enters the models as a proxy for information in the files that my variables miss, these gains are unlikely to reflect private information. Consistent with this hypothesis, the committee's actual matriculants succeeded at a rate no higher than their estimated success probabilities, so if the committee has information that the model lacks, it does not appear to be using it. In Monte Carlo simulated datasets, the same test finds larger gains when the simulated committee misweights applicant characteristics but not when it behaves nearly optimally (see Appendix D). That said, any changes in the actual data-generating process across cohorts, however minor, will reduce the magnitude of the deviation gains when applied in practice. Therefore, 9–10 percentage points for the 2015–2019 application years is unlikely to have been a gain that this committee could have achieved using success probability parameters estimated from before those years.

To determine which variables the committee may have misweighted, I reweight one variable at a time in the committee's estimated decision rule, holding the remaining weights fixed, and calculate the gain from the best reweighting. No individual variable accounts for more

[1] Satisficing is distinct from rational inattention (see Section 1.1), which assumes that decision-makers are optimizing and so embeds the choice of how much effort to allocate into an optimization framework.

than three percentage points of the gain, so the committee did not make any large mistakes. The committee underweighted age, work experience, and English proficiency as measured by TOETL/IELTS scores. In additon, in application essays, positive language was underweighted, whereas economics-specific language was overweighted. The committee also penalized applicants who listed more coding languages on their CVs even though coding did not negatively impact success. Meanwhile, the committee overweighted elite U.S. undergraduate institutions, though only for the acceptance set, as accepted applicants from elite U.S. universities matriculated at lower rates. However, most of these individual misweightings are not statistically significant, and they may not hold going forward. For example, if applicants start using AI to write their essays, then it is unlikely that any essay-specific results from 2015–2019 will hold today.

Moreover, the committee's constraint on how many application files it reads does not affect the success of its acceptance or matriculation set. I first graph the average success probability of the model's acceptance and matriculation sets as a function of the number of files read. Reading every application file rather than the roughly 40 percent the committee actually read raises the model's admitted cohort's average success probability by only about one percentage point. The committee's own decision rule, meanwhile, yields nearly the same average success probability regardless of how many files it reads. Given that the university in this study is not necessarily representative of all graduate programs, some of this paper's results might not generalize to other settings. However, the methods are generalizable to admissions, hiring, and any other setting where decision-makers must decide which options to select based on *ex-ante* information.

1.1 Related Literature

While there have been earlier papers on graduate education, Ehrenberg and Mavros (1995) were the first to include predictors of success found in application files. They analyze Cornell PhD students, including economics, and find that verbal but not quantitative GRE scores are predictive of program completion. Subsequent research implies the reason is likely that their sample is restricted to only the applicants who attended Cornell, nearly all of whose quantitative GRE scores were concentrated in the top percentiles. Two years later, Attiyeh and Attiyeh (1997) use a sample across 48 universities and multiple programs from 1990 to 1991 to study determinants of admission. They find that all else equal, committees prioritized underrepresented minorities.

Krueger and Wu (2000) and Grove and Wu (2007) then study admission and success at a single top economics department in 1989, including all applicants in their success regressions and tracking placement outcomes for non-matriculants. They find that admissions committee scores, GRE scores, and the prominence of recommendation letter writers predict good placements, but there is a lot of uncertainty. Finally, Athey, Katz, Krueger, Levitt, and Poterba (2007) analyze the determinants of success for Chicago, Harvard, MIT, Princeton, and Stanford economics PhD students from 1990 to 1999, adding graduate course grades into their models. While GRE scores and other *ex-ante* factors predict good placements, they do not when controlling for grades.

Stock, Finegan, and Siegfried (2006) begin a series of papers by compiling a 586 student dataset of Fall 2002 economics PhD matriculants. They start by looking at attrition rates in the first two years and find that low GRE scores predict attrition, but RAships and shared department office space help programs retain students. Siegfried and Stock (2007) compile another, larger dataset of everyone who earned a U.S. economics PhD from 1966 to 2003 and observe that top liberal arts colleges feed economics PhD programs at disproportionally high rates.

Stock, Finegan, and Siegfried (2009a) and Stock, Finegan, and Siegfried (2009b) follow up on the progress of the students in Stock et al. (2006) and find similar results but emphasize that there is a lot of uncertainty in predicting completion after five years. Stock, Siegfried, and Finegan (2011) and Stock and Siegfried (2014) track the students after eight years and summarize their multi-paper study, concluding that while some factors like high GRE scores, a liberal arts undergraduate institution, and financial support predict completion, uncertainty remains very high. Finally, Stock and Siegfried (2015) add more recent years to Siegfried and Stock (2007)'s larger dataset, noting that foreign undergraduate institutions have become more common over time.

Meanwhile, Grove, Dutkowsky, and Grodner (2007) study Syracuse economics PhD students, who have different characteristics than students attending top-ranked programs. They also incorporate factor analysis into their regressions, finding that different attributes predict success at different stages. For example, GRE scores predict passing comprehensive exams, whereas research motivation (a factor from their factor analysis) predicts program completion. Schlauch and Startz (2018) study economics PhD candidates from top 50 programs posting their CVs for the 2016–2017 job market and find that while RA experience predicts attending a highly-ranked PhD program, a masters degree does not. This result tracks with the increasing prevalence of predocs in economics. Meanwhile, Jones, Schuhmann, Soques, and Witman (2020) do not run regressions but rather survey individuals involved in admissions at 132 economics PhD programs. They find that higher-ranked programs care about different factors than lower-ranked programs. For example, they care more about undergraduate rankings. Also, while recommendation letters matter for admission, programming skills usually do not, though they may matter for success.

Bai et al. (2022) represent the frontier of economics graduate admissions research. They study economics PhD applicants at a single top program from 2013 to 2019 and like Krueger and Wu (2000) collect data on admissions and placement outcomes for all applicants. They start by proving that a committee's portfolio choice problem simplifies to a cutoff rule problem, which is more tractable. I build on this result by directly modeling the cutoff rule problem and estimating the parameters, which is necessary to make claims as to whether committees are optimizing. In addition, they perform textual analysis of recommendation letters and find that subjective factors, such as letters, matter more for both admission and success more than objective factors, such as GREs. As such, the models in this paper incorporate subjective as well as objective factors.

This paper also relates to a broader methodological literature on optimal dynamic selection with costly evaluation, as well as a literature on synthetic data and privacy protection in empirical economics. Within the former literature, earlier papers develop classic models of sequential search where agents face uncertainty about option quality and must decide when to stop searching given search costs. McCall (1970) and Mortensen (1970) model job search as a dynamic stopping problem in which workers optimally accept offers that exceed a reservation wage. Weitzman (1979) extends this logic to settings in which decision-makers can acquire information about multiple alternatives at a cost, which he denotes as "Pandora's Problem." Together, these papers establish that when information is costly, optimal decision rules are often threshold-based.

A more recent group of papers studies costly evaluation outside of search problems. In models of rational inattention, agents are fully optimizing but face constraints on their ability to process information. Sims (2003) introduces rational inattention as a framework in which information acquisition is costly, leading agents to optimally trade off decision accuracy with information costs. Matějka and McKay (2015) show that rational inattention can generate probabilistic discrete choice rules in line with multinomial logit models, while Caplin and Dean (2015) provide a revealed-preference characterization of optimal information acquisition that separates rational inattention from actual mistaken decisions. These models differ from Simon (1956)'s concept of satisficing, which proposes heuristic, "good enough" decision rules. In contrast, rational inattention models retain the mathematical framework of rationality but endogenize information acquisition. Notably, unlike this paper, none of these papers estimate their models on data.

This paper differs from both rational inattention and satisficing. I take the information environment faced by decision-makers as given, since admissions committees, at least in this framework, do not choose how much information is available at each stage of evaluation. Instead, they interpret the information they observe at each stage and decide how to act on it. The contribution of this paper is to model this dynamic selection problem explicitly, estimate parameters governing the resulting decision rules, and determine whether observed behavior is optimal conditional on respecting the committee's constraints. By doing so, this paper makes optimality testable and allows for the computation of counterfactual decision rules and deviation gains.

There are also papers that use structural models to determine whether decision-makers behave optimally in contexts other than dynamic selection. For example, Cho and Rust (2010) find that a large car rental company could have increased profits by holding onto their rental cars for longer and charging lower prices to consumers who rent the older cars. In addition, Anderson, Rosen, Rust, and Wong (2025) find that some of the world's best professional tennis players could have increased their service game win probabilities by choosing different serve direction strategies than the ones they were using. This paper provides a framework to determine whether decision-makers, such as admissions committees, optimally solve dynamic selection problems.

The second methodological literature addresses synthetic data and privacy protection. Rubin (1993) introduces synthetic data as a tool for protecting confidentiality by replacing sensitive microdata with simulated observations drawn from an estimated data-generating process. Before synthetic data, privacy protection often involved masking, or adding random noise to the data. Masking, however, induces a trade-off between privacy and information loss (Little 1993), and the rows in a masked dataset still correspond to real subjects. More recent work, such as Abowd and Schmutte (2015), emphasizes the challenges posed by restricted access to data. Meanwhile, Koenecke and Varian (2020) propose that researchers publicly release synthetic data, such as those generated by the Synthetic Data Vault (Patki, Wedge, & Veeramachaneni 2016), when they cannot release the actual data. Their proposed protocol differs from this paper's protocol, as this protocol is designed for a setting where the *researcher* cannot access the actual data.

The rest of the paper proceeds as follows. Section 2 describes the data and synthetic data protocol. Section 3 presents regression-based evidence of the determinants of acceptance, matriculation, and success, and shows why comparing the determinants of acceptance with those of success is insufficient to test optimality. Section 4 develops and estimates structural models of admissions that incorporate first-stage filtering and matriculation probabilities. Section 5 implements the deviation gains test, which rejects no gains. Section 6 concludes.

2 Data

This paper uses two datasets: one consisting of all applicants to a research university's economics PhD program from 2015 to 2019, and another of all applicants to that university's economics, government, history, linguistics, and psychology programs from 2020 to 2023. Unlike the graduate school dataset, the economics department dataset contains subjective variables (i.e. those derived from textual documents such as CVs and recommendation letters) and program outcomes data (e.g. post-program job placements). Therefore, I estimate this paper's structural models on the economics department dataset only. However, the graduate school dataset, which still contains objective variables and each applicant's admission status, is useful for making regression-based comparisons across programs. All results from the graduate school dataset are in Online Appendix A. In this section, I describe the variables in the economics department dataset and the synthetic data protocol, which serves as a template for conducting research on confidential data.

2.1 Variables

There are three types of variables: dependent variables, objective independent variables, and subjective independent variables. Table 6 of Appendix A contains a full list of the variables. The dependent variables consist of measures of both admission and success. For admission, such measures include binary variables of whether the applicant makes it past the first-round filtering stage, whether they are accepted into the program, whether they are waitlisted, whether they attend the university's open house, whether they matriculate at the university, and whether they attend (and were therefore accepted at) a different university's PhD program. I also have the U.S. News American and global rankings of the program they attended. If they attended a U.S. program, I use their American ranking, whereas if they attended a non-U.S. program, I use their global ranking times 3/8. The reason is that the maximum American ranking is 147, the maximum global ranking is 400, and I assign 150 to anyone who did not attend a PhD program. This procedure ensures that all U.S. and non-U.S. programs are ranked on the same scale.

For success in economics, I have a binary variable of whether they complete the program, along with variables for the type and ranking of their eventual job placement. Placement types include academic, government, consulting, and technology. For academic placements, I use the department's IDEAS/RePEc American and global rankings with a plus five penalty for business schools (Krueger & Wu 2000). For postdocs, I use a 2/3 to 1/3 weighted average of the university's ranking and the maximum ranking, which is 138 for American and 340 for global. For non-tenure-track positions, I use a 1/3 to 2/3 weighted average. For unranked tenure-track positions, I assign the maximum ranking plus one; for unranked postdocs and non-tenure-track positions, I assign 150 for American and 350 for global. Ultimately, for U.S. placements, I use the American ranking, whereas for non-U.S. placements, I use the global ranking times 3/7.

Meanwhile, for non-academic placements, I construct academia-equivalent rankings shown in Appendix A. Following Krueger and Wu (2000), the most prestigious non-academic placements (World Bank, IMF, Federal Reserve Board of Governors, European Central Bank, Amazon Science, Google Research, and Microsoft Research) are given a ranking of 40, with lower rankings down to 135 for less prestigious placements.[2] Per Eberhardt, Facchini, and Rueda (2026), economics PhD graduates, even those with Nobel laureate advisors, are increasingly taking non-academic jobs, especially in technology. As in Krueger and Wu (2000) and Bai et al. (2022), to avoid selection bias, I use publicly available LinkedIn profiles, graduate program websites, personal websites, PandaInUniv (2026), and García Guzmán (2026)'s *The Econ PhD placements dataset* to obtain outcomes data for all applicants, not just matriculants. The minority of applicants whose outcomes data are not available receive the lowest possible ranking of 150.

Regarding independent variables from the application files, I distinguish objective variables from subjective variables. Objective variables are those variables that the application software compiles into a spreadsheet for the committee's convenience, whereas subjective variables are those that the applicant uploads as pdf files that cannot be easily converted into a spreadsheet.[3] Because the admissions committee does not have time to read everyone's file, it uses the objective variables to filter out a predetermined percentage of the applicants before reading any files. For example, it filters out applicants with low quantitative GRE scores. For this reason, the structural models in this paper treat objective variables separately from subjective ones.

Objective variables include demographics; such as application year, age, gender, and a region of the world categorical variable, in which the categories are U.S., International Asian, and International Other. There are also performance variables, such as verbal, quantitative, and analytical

[2]I use an average of the rankings provided by ChatGPT 5.5, Claude Opus 4.7, and Gemini 3 in temporary chat mode. The prompt and a link to screenshots of the outputs are available in Appendix A.

[3]Bai et al. (2022) define objective variables as "verifiable performance measures" of applicant quality and subjective variables as committee member assessments. For most variables, our definitions coincide. While they do not categorize demographics, such as gender and nationality, I categorize demographics as objective variables.

GRE percentiles; undergraduate GPA; TOEFL/IELTS scores for non-native English speakers; two binary variables for having a graduate degree and work experience respectively; and a categorical variable for the type of undergraduate institution (elite U.S. university, elite U.S. liberal arts college, other U.S. college/university, elite foreign university, and other foreign university), where elite is always defined as top 50. Moreover, recommender variables, namely the number of professor (Krueger & Wu 2000) and prolific recommenders the applicant has, measure how well-connected the applicant is. A prolific recommender is one who writes a letter for at least one other applicant in the sample, which Bai et al. (2022) find predictive of acceptance and success.

Meanwhile, subjective variables in economics come from recommendation letters, application essays, CVs, and supplemental forms.[4] I use textual processing algorithms on the letters to measure the overall word count and rates of positive, negative, standout, ability, research, grindstone, teaching, communal, and agentic words (Bai et al. 2022). From application essays, the algorithms extract the word count; rates of positive, negative, economics, and research words; and whether the applicant mentioned a professor's name from this university. The last three are my analogues of Grove et al. (2007)'s *Specific Topic*, *Mention Paper*, and *Specific Member*.[5] From CVs, there are the word count, a binary variable for whether the applicant has a publication or working paper, and the number of programming languages the applicant listed. Jones et al. (2020) find that economics committee members, especially at high-ranking programs, do not put much weight on coding skills. But given the increasing prevalence of empirical research that uses techniques like webscraping and machine learning, along with the increasing use of AI tools in economics, such skills have become more commonplace. There is also a supplemental form, in which the applicant lists the number of math courses they took, as well as the names of advanced math courses such as real analysis. I addition, I have a categorical variable of the applicant's indicated field of interest (e.g. econometrics, macroeconomics, etc.). Finally, for applicants whose files the committee read, I have the subjective score from 0 (worst) to 10 (best) that their assigned reader gave them relative to the scores that reader gave other applicants. This score is a proxy for information that the committee can observe but I cannot. The committee score is available for only the applicants whose files were read, and the other subjective variables from the textual documents are available for only about one-fifth of applicants. Per the 2015–2019 committee chair, the availability of these documents is mostly random, though the fraction of Round 1 survivors is higher for the applicants with available documents than for the full application pool.

[4]The department saved 430 of the applicants' textual documents, 102 of which were not transitioned to Round 2. For those without textual documents, I perform mean imputation and set *Missing Application Pdf* = 1.

[5]Grove et al. (2007)'s *Specific Topic* is a binary variable that equals 1 if the essay contains at least one economics topic. *Mention Paper* is a binary variable that equals 1 if the essay mentions any paper (including papers like undergraduate term papers that are not publications or working papers), and is designed to indicate a "demonstrated interest in doing economics research." Finally, *Specific Member* is the same as my binary specific professor variable.

2.2 Synthetic Data Protocol

In this section, I describe the synthetic data protocol used to protect the applicants' privacy, which I employ in conjunction with a university administrative office that is responsible for stewarding the data. While a handful of recent papers, such as Stanley and Totty (2024) and Carr, Wiemers, and Moffitt (2025) use synthetic data, the synthetic data in those papers had already been made available to researchers by the U.S. Census Bureau. In contrast, this paper's protocol covers all the steps from creation of the synthetic data to obtaining the final results on the actual data.

The steps of the protocol are as follows. First, the administrative office receives the data from the graduate admissions office and individual departments, such as economics. They remove all direct identifiers, such as names and email addresses, and run algorithms I provide to extract variables from textual documents, such as recommendation letters. Second, they use Patki et al. (2016)'s publicly-available Synthetic Data Vault (SDV) to generate the synthetic data. The SDV estimates a Gaussian copula to capture correlations between variables and then simulates a new, synthetic dataset from the copula using a random number generator. Crucially, the rows in the synthetic dataset do not correspond to real subjects, which prevents identity disclosure (Chapelle & Falissard 2023). The SDV works best when the final versions of the variables are already created (e.g. turning a list of country names into a categorical variable), so without seeing any actual data, I work with the office to create the variables before implementing the SDV.

However, there remains the possibility of attribute disclosure. For example, if an applicant from an unusual country has an unusually low GRE percentile, information about that applicant may leak into the synthetic dataset. As such, if any cell in the synthetic data has fewer than five other cells from that variable with the same value, the variable's observations are binned and in each bin, replaced with the bin's sample mean. Only at this point do I receive the synthetic data. I use the data to finalize my estimation code, which I then give to the administrative department. The department takes that code, runs it on the actual data, and returns the parameter estimates.

3 Regression Analysis

In this section, I present regression results for the economics department dataset with dependent variables for acceptance, matriculation, and success. For success, I have two dependent variables. The first variable is a binary variable for a top 100 or top 100 equivalent placement. *Top 100 Placement* $= 1$ for applicants who attained a top 100 U.S. placement, an international placement that maps to a top 100 U.S. placement, or a non-academic placement with a top 100 academia-equivalent ranking. Per Table 7 in Appendix B, *Top 100 Placement* $= 1$ for 19.71% of applicants vs. 34.18% of matriculants, indicating that the committee's matriculation set is much better than random chance would predict. The second variable is *Placement Rank*, which ranges from 1 to 150. For applicants whose placements I could not find, I set *Top 100 Placement* to 0, *Placement Rank* to 150, and *Missing Placement Rank* to 1. Likewise, for applicants whose PhD program I could not find, I set *Program Rank* to 150 and *Missing Program Rank* to 1.

Meanwhile, for the subjective variables in recommendation letters, I use principal component analysis to reduce the dimensionality of the letters part of the state space from 9 to 2. The loadings are in Table 1, and they show that PC1 can be interpreted as the degree to which the letters contain words from the dictionaries in general, with an emphasis on positive and ability words. Per Table 8 in Appendix B, the correlation between PC1 and the rate of positive words is 0.8114, and that between PC1 and the rate of ability words is 0.7486. Meanwhile, PC2 can be interpreted as the degree to which the letters are factual and unflattering. Its loadings are positive for negative, research (largest positive loading), grindstone, and teaching words and negative for positive, standout (largest negative loading), ability, communal, and agentic words.

The regression results are in Tables 2a and 2b.[6] I find that different independent variables best predict each dependent variable. For *Accept*, *Committee Score* is the best predictor; a one point increase in *Committee Score* makes an applicant 7.88 percentage points more likely to be accepted, and including it increases the Pseudo-R^2 from 0.2957 to 0.8108. However, other significant predictors of acceptance include demographics, *GRE Quantitative Percentile*, *Work Experience*, and *CV Coding*. Interestingly, *CV Coding*'s average marginal effect (AME) is negative, meaning that for each additional coding language an applicant indicates on their CV, they are 0.44 percentage points less likely to be accepted. Despite the reduced sample size for *CV Coding*, it is still significant at the 10% level. Perhaps having more coding languages on a CV is associated with being from more of an industry background, which the committee might find less appealing. Meanwhile, women are 1.63 percentage points more likely to be accepted, and applicants from East or South Asia are 2.14 percentage points less likely to be accepted. While *Female* and *International Asian* do not significantly predict success, I will show that the committee's decision rule with respect to these two demographics does not negatively affect the average success probability of the acceptance or matriculation set. Therefore, if this decision rule is part of a diversity objective, it is nonetheless still consistent with a success-maximizing objective.

Meanwhile, predictors of acceptance often negatively predict matriculation for applicants who are accepted. For example, *International Asian*'s AME is significantly negative for *Accept* but significantly positive for *Matriculate*. In contrast, *GRE Quantitative Percentile*'s AME is significantly positive for *Accept* but significantly negative for *Matriculate*. Presumably, applicants with high GRE quantitative scores have more outside acceptances and so are less likely to matriculate at this university. However, *Work Experience* is significantly positive for both *Accept* and *Matriculate*, whereas *CV Coding* is significantly negative for both. That said, the best predictor of matriculation is whether the accepted applicant attended the department's open house. Those who attended were 26.15 percentage points more likely to matriculate than those who did not. Attending the open house is a positive signal that the applicant is interested in this university.

[6]In Tables 9a and 9b in Appendix B, I present the regression results on synthetic data. For some variables, the synthetic estimates are similar to the actual estimates, but the synthetic estimates are generally less significant.

Finally, the best predictor of success is *Program Rank*. Applicant characteristics equal, attending a PhD program ranked 10 places better (i.e. 10 places closer to rank 1) increases the probability of a top 100 placement by 1.1 percentage points and improves the expected placement rank (i.e. toward rank 1) by 0.8 places. However, other significant predictors include *Winsorized Age*, *TOEFL/IETLS*, *Work Experience*, and *Essay Positive Terms (%)*.[7] Curiously, a high *Undergraduate GPA* significantly predicts failure when controlling for *Program Rank*. If GPA does not provide much signal, which is plausible because GPAs are not standardized across institutions, but admissions committees in general select positively on them, then applicants with equal characteristics who were selected into the same program tier may exhibit a negative correlation between GPA and success. However, this paper's structural models are meant for pre-selection decision-making, so I follow Bai et al. (2022) and remove GPA after this section.

In Table 10 of Appendix B, I present lasso logistic regression results with 5-fold cross-validation for variable selection. For acceptance and matriculation, lasso selects most of the variables, but it selects fewer variables for success. In the table's sixth column, with *Top 100 Placement* as the dependent variable and controlling for *Committee Score* and *Program Rank*/*Missing Program Rank*, lasso selects the following ten variables: *Winsorized Age*, *Undergraduate GPA*, *TOEFL/IELTS*, *Work Experience*, *Letters Terms (%) PC1*, *Essay Economics Terms (%)*, *Essay Positive Terms (%)*, *Missing Committee Score*, *Program Rank*, and *Missing Program Rank*.

Readers familiar with Table 18 of Bai et al. (2022) may notice that their success regressions without controlling for PhD program rank fit substantially better than this paper's success regressions without *Program Rank* and *Missing Program Rank*. It is only after including *Program Rank* and *Missing Program Rank* that the Pseudo-R^2s in Table 2a become comparable to those in Bai et al. (2022), though it is worth noting that adding PhD program rank controls does not futher increase their Pseudo-R^2s by much. This pattern occurs because they define success as attaining a top 10 or top 20 assistant professorship. It is very hard to attain such a professorship without attending a top-ranked PhD program. Per their Table 17, 83% of top 10 assistant professor placements in their data came from top 10 economics PhD programs, an additional 8% came from non-economics PhD programs which may have also been top 10 in their disciplines, and only one applicant attained a top 10 placement from a PhD program ranked lower than this paper's university. It is also the case that variables in application files, such as GRE scores, predict acceptance at top PhD programs. Therefore, their application file variables predict placement indirectly by predicting program rank, and adding program indicators contributes little because the file has already revealed most of that information. In contrast, while attending a PhD program is necessary for a top 100 or top 100-equivalent placement, attending a top-ranked program is not. Graduates of programs across many tiers attain it, in part because non-academic research institu-

[7]The age distribution of applicants drops off sharply past the early 30s to the point that a quadratic *Age* term is unidentified. To avoid extrapolating *Age*'s positive AME to the small fraction of much older applicants, I winsorize *Age* at its 95th percentile, which is 33 years old. Also, I use percentiles to convert TOEFL's 0–120 scores to IELTS's 1–9 scores, ensuring that both tests are on the same scale when combining them into one variable.

tions (e.g. IMF, Federal Reserve, etc.) account for many of this paper's top 100 placements. As such, application file variables do not indirectly predict success as they do in Bai et al. (2022), making it necessary to control for program rank to increase the Pseudo-R^2 in Table 2a.

Table 1: Principal Component Loadings

	PC1	PC2
Positive	0.5044	-0.1769
Negative	0.1479	0.3759
Standout	0.2115	-0.5082
Ability	0.4667	-0.0177
Research	0.1214	0.6418
Grindstone	0.3879	0.3075
Teaching	0.3708	0.0743
Communal	0.1284	-0.1983
Agentic	0.3768	-0.1324
Proportion of Variance	0.2763	0.1456

I briefly address inference vs. prediction, as well as sample selection. Both are important when running regressions with success measures as dependent variables. Regarding inference vs. prediction, omitted independent variables make it challenging to obtain consistent estimates in a success regression. For example, I do not observe the effort that students put in after matriculating. Even effort, let alone any other variables about the applicants that I do not observe, can bias success parameter estimates upward if effort is correlated with the variables corresponding to those parameters and correlated with success irrespective of those variables.

However, there are reasons for reassurance. First, I incorporate *Committe Score* as an independent variable, which proxies variables that the committee observes and I do not. I also follow Bai et al. (2022) and include as a proxy of effort the number of "grindstone terms," such as depend* and diligen*, in recommendation letters. In addition, in this paper, prediction matters as much as inference. Mullainathan and Spiess (2017) distinguish $\hat{\beta}$ (e.g. the effect on educational outcomes of hiring an additional teacher) from $\hat{y}$ (e.g. which teacher to hire) problems. For graduate admissions, if *Work Experience* predicts success, then the committee should consider it regardless of whether the committee knows the mechanism through which it affect success.

Moreover, the determinants of success conditional on applying (see Krueger and Wu 2000, and Grove and Wu 2007) are different from those conditional on matriculating (see Athey et al. 2007). Bai et al. (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples, a theoretical finding consistent with the significance of the quantitative GRE being weaker in studies on matriculants only. This problem is akin to height not predicting success in a sample of professional basketball players. Like Bai et al. (2022), I collect placement outcomes for all applicants, not just those who attended this university. But there are other measures of success, such as course grades and passing comprehensive examinations, that are only available for matriculants, and that have been used in papers like Athey et al. (2007).

Table 2a: Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.3826	-0.5049	-0.2577	0.0230	0.0301	0.0885	1.8038	1.7607	-0.6618
	(0.2365)	(0.4703)	(0.6853)	(0.1708)	(0.1711)	(0.1887)	(4.2583)	(4.2504)	(3.6266)
2017	-0.0954	-1.3247***	0.1357	0.0829	0.0711	0.1460	2.7746	3.0235	0.7473
	(0.2564)	(0.4881)	(0.6732)	(0.1707)	(0.1721)	(0.1934)	(4.2295)	(4.2188)	(3.7157)
2018	-0.2206	-3.7609***	0.9476	0.0624	-0.0392	0.0142	2.7265	5.2685	4.9770
	(0.2648)	(0.6518)	(0.6200)	(0.1761)	(0.1807)	(0.2011)	(4.4414)	(4.5011)	(3.8786)
2019	-0.1059	-3.1670***	0.5909	0.0594	0.0144	0.0366	5.7085	6.6750	5.2241
	(0.2653)	(0.7114)	(0.6855)	(0.1821)	(0.1830)	(0.2052)	(4.6400)	(4.6266)	(4.1424)
Winsorized Age	-0.0218	-0.0080	-0.1567**	0.0240	0.0242	0.0420*	0.3596	0.3632	-0.1277
	(0.0342)	(0.0592)	(0.0778)	(0.0207)	(0.0205)	(0.0236)	(0.5457)	(0.5410)	(0.4839)
Female	0.3759**	0.8784**	-0.3445	-0.0880	-0.1026	-0.0256	1.8085	2.0921	-1.6375
	(0.1657)	(0.3813)	(0.3794)	(0.1126)	(0.1130)	(0.1252)	(2.8776)	(2.8750)	(2.5117)
U.S.	-0.4982*	-0.0681	1.0348*	0.1167	0.1396	0.1520	-0.6372	-0.9759	-0.7059
	(0.2624)	(0.4538)	(0.6098)	(0.1903)	(0.1909)	(0.2236)	(4.7990)	(4.7913)	(4.3464)
International Asian	-1.1569***	-1.1507***	1.1559**	0.1695	0.1949	-0.1005	-7.8212**	-8.2008**	3.3007
	(0.2129)	(0.4146)	(0.4904)	(0.1379)	(0.1395)	(0.1546)	(3.5528)	(3.5708)	(3.1577)
GRE Verbal Percentile	0.0264***	0.0079	-0.0107	0.0049	0.0030	0.0007	-0.1958**	-0.1470*	-0.0330
	(0.0054)	(0.0130)	(0.0118)	(0.0031)	(0.0032)	(0.0036)	(0.0787)	(0.0806)	(0.0735)
GRE Quantitative Percentile	0.1329***	0.0589**	-0.1033**	0.0245***	0.0184**	0.0047	-0.8498***	-0.7196***	-0.2428*
	(0.0202)	(0.0275)	(0.0464)	(0.0077)	(0.0075)	(0.0084)	(0.1567)	(0.1578)	(0.1460)
GRE Analytic Percentile	0.0076**	0.0094	-0.0119	-0.0020	-0.0027	-0.0021	0.0656	0.0821	0.0160
	(0.0039)	(0.0085)	(0.0095)	(0.0025)	(0.0025)	(0.0029)	(0.0669)	(0.0671)	(0.0618)
Undergraduate GPA	1.2726*	1.0509	-0.7209	-0.3976	-0.4722	-0.8687*	4.8040	6.9311	17.2597**
	(0.6975)	(1.2453)	(1.2760)	(0.3912)	(0.3848)	(0.4442)	(10.1461)	(10.0615)	(7.7507)
TOEFL/IELTS	0.4236	-0.1165	-0.4434	0.1778	0.1437	0.3240*	-2.2504	-1.2960	-7.6837*
	(0.2924)	(0.5976)	(0.5525)	(0.1571)	(0.1559)	(0.1757)	(4.3852)	(4.3431)	(3.9285)
Elite U.S. University	0.7328***	-0.1822	-0.4980	-0.0159	-0.0853	-0.1523	-2.3790	-0.3552	-0.3450
	(0.2774)	(0.5918)	(0.5799)	(0.2133)	(0.2133)	(0.2344)	(5.2934)	(5.2869)	(4.4943)
Elite U.S. Liberal Arts College	0.0167	0.5446	0.9907	0.0731	0.0803	-0.1010	-3.9470	-3.8047	-3.4006
	(0.4568)	(0.6896)	(0.8447)	(0.3370)	(0.3361)	(0.3936)	(8.8571)	(8.7916)	(8.2332)
Elite Foreign University	0.1244	-0.5507	-0.0673	0.1210	0.0751	0.0969	-2.7705	-1.5414	-2.2142
	(0.5319)	(0.9725)	(1.1224)	(0.3239)	(0.3259)	(0.3526)	(7.4748)	(7.4679)	(6.1380)
Other Foreign University	0.5323*	0.6328	0.7373	0.1483	0.1091	0.1587	-6.7584	-5.7186	-9.1260**
	(0.3022)	(0.6505)	(0.6347)	(0.1964)	(0.1970)	(0.2216)	(4.9679)	(4.9620)	(4.3486)
Missing Undergraduate Institution	0.3260	0.2482	-0.6851	0.2292	0.2082	0.1770	-7.9862	-7.5382	-8.1758*
	(0.3111)	(0.6016)	(0.7260)	(0.2017)	(0.2022)	(0.2345)	(5.1133)	(5.1129)	(4.5085)
Graduate Degree	0.1726	-0.2058	-0.2954	-0.0406	-0.0921	-0.1835	-3.2803	-2.0242	2.6215
	(0.1788)	(0.3845)	(0.5017)	(0.1231)	(0.1273)	(0.1370)	(3.1662)	(3.2087)	(2.7452)
Work Experience	0.7785***	0.7519**	1.1183**	0.2298*	0.1772	0.2839**	-3.5942	-2.2559	-6.4380**
	(0.1915)	(0.3400)	(0.5291)	(0.1253)	(0.1270)	(0.1422)	(3.2005)	(3.2301)	(2.9093)
#Professor Recommenders	-0.0377	-0.1797	0.3587	-0.0472	-0.0513	-0.0547	0.8765	0.9972	0.7371
	(0.1079)	(0.2434)	(0.3115)	(0.0749)	(0.0754)	(0.0862)	(2.0492)	(2.0495)	(1.7160)
#Prolific Recommenders	0.2208***	0.2081	-0.2993	0.1011*	0.0899*	0.0600	-1.4623	-1.1967	0.1726
	(0.0811)	(0.1721)	(0.2060)	(0.0528)	(0.0533)	(0.0585)	(1.3680)	(1.3670)	(1.1704)
Missing GRE	-1.5399	1.0141		-0.4515	-0.3258	0.0174	11.5202	8.5434	-3.7106
	(0.9821)	(1.1356)		(0.3510)	(0.3544)	(0.3797)	(8.6385)	(8.6602)	(6.1802)
Missing Undergraduate GPA	-0.3432	-0.2798	0.0798	-0.1010	-0.0639	-0.0513	2.4296	1.5256	-0.8710
	(0.2210)	(0.4413)	(0.4943)	(0.1565)	(0.1585)	(0.1756)	(3.9808)	(3.9842)	(3.3572)
Missing TOEFL/IELTS	0.4719**	0.3743	-0.0734	-0.1261	-0.1447	-0.0065	4.0029	4.3290	-2.3792
	(0.2062)	(0.4223)	(0.5891)	(0.1267)	(0.1282)	(0.1443)	(3.3209)	(3.3244)	(2.9657)
Committee Score		4.2419***	-0.2608		0.0943**	0.0059		-2.5088**	0.6884
		(0.4534)	(0.2250)		(0.0456)	(0.0488)		(1.1860)	(1.0132)
#Math Courses	-0.0949	-0.2706	0.0755	0.0626	0.0662	0.0008	-2.3828	-2.4550	0.9116
	(0.0773)	(0.2829)	(0.1680)	(0.0759)	(0.0766)	(0.0885)	(1.8033)	(1.8149)	(1.5082)
Advanced Math	0.6333**	1.3229	-1.3706**	-0.1932	-0.2078	-0.2275	3.0400	3.4570	1.9957
	(0.3036)	(0.9203)	(0.5621)	(0.2605)	(0.2639)	(0.2949)	(6.4446)	(6.4377)	(5.5819)
Letters Length/100	0.1358***	0.1084	-0.0910	0.0067	-0.0018	-0.0150	-0.2593	-0.0078	0.7206
	(0.0518)	(0.1424)	(0.0712)	(0.0335)	(0.0339)	(0.0388)	(0.8254)	(0.8325)	(0.7084)
Letters Terms (%) PC1	0.1507	0.2939	0.4716**	0.0362	0.0251	0.1260	1.6082	1.8374	-0.5612
	(0.0923)	(0.2912)	(0.2051)	(0.0732)	(0.0748)	(0.0915)	(2.0561)	(2.0540)	(1.8186)
Letters Terms (%) PC2	-0.0913	-0.4980	0.2724	-0.1226	-0.1199	-0.1420	0.7118	0.5616	0.1155
	(0.1464)	(0.4143)	(0.2604)	(0.1162)	(0.1177)	(0.1272)	(2.7992)	(2.8081)	(2.3450)
CV Length/100	0.0361	0.0559	-0.1986	0.0301	0.0274	0.0352	-1.0456	-1.0120	-0.8208
	(0.0617)	(0.1186)	(0.1337)	(0.0562)	(0.0563)	(0.0593)	(1.3224)	(1.3130)	(1.0252)
CV Coding	-0.0439	-0.2369*	-0.2911**	0.0716	0.0710	0.0992	-1.9601	-1.8941	-2.0142
	(0.0765)	(0.1381)	(0.1390)	(0.0532)	(0.0532)	(0.0654)	(1.5459)	(1.5358)	(1.3384)
CV Publication/Working Paper	0.6162*	0.8309	-0.1668	0.2940	0.2834	0.1310	-11.5166	-10.8849	-4.6204
	(0.3421)	(0.9326)	(0.5756)	(0.2998)	(0.3036)	(0.3383)	(7.2873)	(7.3131)	(6.7001)
Essay Length/100	0.1747*	0.2225	0.1776	-0.0008	-0.0008	-0.0711	-0.4758	-0.5351	1.8425
	(0.0933)	(0.3920)	(0.1536)	(0.0751)	(0.0764)	(0.0821)	(1.8648)	(1.8721)	(1.6240)
Essay Specific Professor	0.3842	0.4170	0.6845	0.2144	0.1774	0.2872	-6.8688	-5.9104	-7.5240
	(0.2908)	(0.7014)	(0.5119)	(0.2407)	(0.2410)	(0.2789)	(6.0223)	(5.9982)	(5.2718)
Essay Economics Terms (%)	-0.0169	0.0070	0.2576**	-0.1126**	-0.1150**	-0.0822	3.6540***	3.7277***	2.1702*
	(0.0547)	(0.0866)	(0.1159)	(0.0533)	(0.0531)	(0.0618)	(1.2540)	(1.2496)	(1.1179)
Essay Research Terms (%)	-0.0018	-0.0459	-0.1868	0.0697	0.0642	0.0150	-3.7084**	-3.6795**	-1.0575
	(0.0736)	(0.1690)	(0.1451)	(0.0633)	(0.0639)	(0.0674)	(1.6220)	(1.6209)	(1.3903)
Essay Positive Terms (%)	-0.0202	-0.0700	-0.0504	0.0724	0.0772*	0.0977*	-0.9256	-0.9929	-0.7266
	(0.0512)	(0.1044)	(0.1331)	(0.0453)	(0.0458)	(0.0542)	(1.1634)	(1.1496)	(0.9752)
Essay Negative Terms (%)	-0.0151	-0.0886	0.0019	-0.0386	-0.0454	-0.0472	0.8164	0.9436	0.9507
	(0.0770)	(0.1557)	(0.1071)	(0.0663)	(0.0663)	(0.0748)	(1.6317)	(1.6259)	(1.3985)
Missing Committee Score		-0.5168			-0.3458**	-0.2199		9.2234***	2.8589
		(0.6383)			(0.1397)	(0.1519)		(3.4582)	(3.0258)
Missing Application Pdf	-1.0107***	0.1926	-0.7407	0.1461	0.2686	0.2416	-5.1611	-8.4497	-6.8690
	(0.3188)	(0.8088)	(0.6084)	(0.2442)	(0.2479)	(0.2763)	(6.0328)	(6.1404)	(5.4325)
Open House			1.8384***						
			(0.4115)						
Program Rank						-0.0087***			0.1813***
						(0.0021)			(0.0404)
Missing Program Rank						-3.1432***			89.4721***
						(0.4077)			(5.6947)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.2957	0.8108	0.2877	0.0308	0.0353	0.2430	0.0089	0.0097	0.1065

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 2b: Logistic Regression AMEs

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.0274	-0.0094	-0.0367	0.0035	0.0046	0.0112	0.8694	0.8485	-0.3038
	(0.0171)	(0.0085)	(0.0976)	(0.0262)	(0.0261)	(0.0238)	(2.0523)	(2.0482)	(1.6650)
2017	-0.0068	-0.0246***	0.0193	0.0127	0.0108	0.0184	1.3374	1.4571	0.3431
	(0.0184)	(0.0090)	(0.0958)	(0.0262)	(0.0263)	(0.0244)	(2.0386)	(2.0331)	(1.7054)
2018	-0.0158	-0.0699***	0.1348	0.0096	-0.0060	0.0018	1.3142	2.5391	2.2847
	(0.0189)	(0.0086)	(0.0857)	(0.0270)	(0.0276)	(0.0254)	(2.1405)	(2.1686)	(1.7791)
2019	-0.0076	-0.0589***	0.0841	0.0091	0.0022	0.0046	2.7516	3.2169	2.3982
	(0.0190)	(0.0109)	(0.0970)	(0.0279)	(0.0279)	(0.0259)	(2.2355)	(2.2284)	(1.8977)
Winsorized Age	-0.0016	-0.0001	-0.0223**	0.0037	0.0037	0.0053*	0.1734	0.1751	-0.0586
	(0.0024)	(0.0011)	(0.0108)	(0.0032)	(0.0031)	(0.0030)	(0.2629)	(0.2606)	(0.2222)
Female	0.0269**	0.0163**	-0.0490	-0.0135	-0.0157	-0.0032	0.8717	1.0082	-0.7517
	(0.0118)	(0.0068)	(0.0534)	(0.0173)	(0.0172)	(0.0158)	(1.3867)	(1.3851)	(1.1534)
U.S.	-0.0356*	-0.0013	0.1472*	0.0179	0.0213	0.0192	-0.3071	-0.4703	-0.3241
	(0.0188)	(0.0084)	(0.0864)	(0.0292)	(0.0291)	(0.0282)	(2.3132)	(2.3090)	(1.9951)
International Asian	-0.0828***	-0.0214***	0.1644**	0.0260	0.0298	-0.0127	-3.7699**	-3.9523**	1.5152
	(0.0148)	(0.0073)	(0.0691)	(0.0212)	(0.0213)	(0.0195)	(1.7114)	(1.7195)	(1.4490)
GRE Verbal Percentile	0.0019***	0.0001	-0.0015	0.0008	0.0005	0.0001	-0.0944**	-0.0708*	-0.0151
	(0.0004)	(0.0002)	(0.0017)	(0.0005)	(0.0005)	(0.0005)	(0.0379)	(0.0388)	(0.0337)
GRE Quantitative Percentile	0.0095***	0.0011**	-0.0147**	0.0038***	0.0028**	0.0006	-0.4096***	-0.3468***	-0.1114*
	(0.0014)	(0.0005)	(0.0063)	(0.0012)	(0.0011)	(0.0011)	(0.0757)	(0.0762)	(0.0670)
GRE Analytic Percentile	0.0005**	0.0002	-0.0017	-0.0003	-0.0004	-0.0003	0.0316	0.0395	0.0073
	(0.0003)	(0.0002)	(0.0013)	(0.0004)	(0.0004)	(0.0004)	(0.0322)	(0.0323)	(0.0283)
Undergraduate GPA	0.0910*	0.0195	-0.1026	-0.0610	-0.0721	-0.1096**	2.3156	3.3404	7.9232**
	(0.0499)	(0.0235)	(0.1820)	(0.0600)	(0.0587)	(0.0558)	(4.8911)	(4.8499)	(3.5564)
TOEFL/IELTS	0.0303	-0.0022	-0.0631	0.0273	0.0219	0.0409*	-1.0847	-0.6246	-3.5273*
	(0.0209)	(0.0111)	(0.0776)	(0.0241)	(0.0238)	(0.0221)	(2.1145)	(2.0935)	(1.8033)
Elite U.S. University	0.0524***	-0.0034	-0.0708	-0.0024	-0.0130	-0.0192	-1.1467	-0.1712	-0.1584
	(0.0198)	(0.0111)	(0.0816)	(0.0327)	(0.0326)	(0.0296)	(2.5514)	(2.5479)	(2.0632)
Elite U.S. Liberal Arts College	0.0012	0.0101	0.1409	0.0112	0.0123	-0.0127	-1.9025	-1.8336	-1.5611
	(0.0327)	(0.0126)	(0.1197)	(0.0517)	(0.0513)	(0.0496)	(4.2692)	(4.2369)	(3.7802)
Elite Foreign University	0.0089	-0.0102	-0.0096	0.0186	0.0115	0.0122	-1.3354	-0.7428	-1.0164
	(0.0381)	(0.0182)	(0.1597)	(0.0497)	(0.0497)	(0.0445)	(3.6022)	(3.5986)	(2.8173)
Other Foreign University	0.0381*	0.0118	0.1049	0.0228	0.0167	0.0200	-3.2576	-2.7560	-4.1894**
	(0.0216)	(0.0119)	(0.0890)	(0.0301)	(0.0301)	(0.0279)	(2.3945)	(2.3912)	(1.9955)
Missing Undergraduate Institution	0.0233	0.0046	-0.0975	0.0352	0.0318	0.0223	-3.8495	-3.6329	-3.7531*
	(0.0222)	(0.0111)	(0.1014)	(0.0309)	(0.0308)	(0.0296)	(2.4639)	(2.4635)	(2.0698)
Graduate Degree	0.0123	-0.0038	-0.0420	-0.0062	-0.0141	-0.0231	-1.5811	-0.9755	1.2034
	(0.0128)	(0.0071)	(0.0712)	(0.0189)	(0.0194)	(0.0172)	(1.5257)	(1.5461)	(1.2606)
Work Experience	0.0557***	0.0140**	0.1591**	0.0353*	0.0271	0.0358**	-1.7325	-1.0872	-2.9554**
	(0.0135)	(0.0064)	(0.0700)	(0.0192)	(0.0194)	(0.0178)	(1.5432)	(1.5570)	(1.3344)
#Professor Recommenders	-0.0027	-0.0033	0.0510	-0.0072	-0.0078	-0.0069	0.4225	0.4806	0.3384
	(0.0077)	(0.0045)	(0.0437)	(0.0115)	(0.0115)	(0.0109)	(0.9881)	(0.9881)	(0.7878)
#Prolific Recommenders	0.0158***	0.0039	-0.0426	0.0155*	0.0137*	0.0076	-0.7048	-0.5767	0.0792
	(0.0058)	(0.0031)	(0.0285)	(0.0081)	(0.0081)	(0.0074)	(0.6595)	(0.6589)	(0.5373)
Missing GRE	-0.1102	0.0188		-0.0693	-0.0497	0.0022	5.5529	4.1174	-1.7034
	(0.0706)	(0.0212)		(0.0538)	(0.0541)	(0.0479)	(4.1627)	(4.1727)	(2.8358)
Missing Undergraduate GPA	-0.0246	-0.0052	0.0114	-0.0155	-0.0098	-0.0065	1.1711	0.7352	-0.3999
	(0.0158)	(0.0082)	(0.0703)	(0.0240)	(0.0242)	(0.0221)	(1.9187)	(1.9201)	(1.5412)
Missing TOEFL/IELTS	0.0338**	0.0070	-0.0104	-0.0194	-0.0221	-0.0008	1.9295	2.0863	-1.0922
	(0.0147)	(0.0078)	(0.0839)	(0.0194)	(0.0195)	(0.0182)	(1.6007)	(1.6020)	(1.3613)
Committee Score		0.0788***	-0.0371		0.0144**	0.0007		-1.2091**	0.3160
		(0.0045)	(0.0322)		(0.0069)	(0.0062)		(0.5716)	(0.4649)
#Math Courses	-0.0068	-0.0050	0.0107	0.0096	0.0101	0.0001	-1.1485	-1.1832	0.4185
	(0.0055)	(0.0053)	(0.0239)	(0.0116)	(0.0117)	(0.0112)	(0.8692)	(0.8747)	(0.6922)
Advanced Math	0.0453**	0.0246	-0.1950**	-0.0296	-0.0317	-0.0287	1.4653	1.6660	0.9161
	(0.0216)	(0.0168)	(0.0793)	(0.0399)	(0.0402)	(0.0372)	(3.1061)	(3.1022)	(2.5621)
Letters Length/100	0.0097***	0.0020	-0.0129	0.0010	-0.0003	-0.0019	-0.1250	-0.0038	0.3308
	(0.0037)	(0.0026)	(0.0100)	(0.0051)	(0.0052)	(0.0049)	(0.3978)	(0.4012)	(0.3253)
Letters Terms (%) PC1	0.0108	0.0055	0.0671**	0.0056	0.0038	0.0159	0.7752	0.8855	-0.2576
	(0.0066)	(0.0054)	(0.0283)	(0.0112)	(0.0114)	(0.0115)	(0.9905)	(0.9892)	(0.8348)
Letters Terms (%) PC2	-0.0065	-0.0093	0.0387	-0.0188	-0.0183	-0.0179	0.3431	0.2706	0.0530
	(0.0105)	(0.0077)	(0.0371)	(0.0178)	(0.0179)	(0.0160)	(1.3495)	(1.3535)	(1.0765)
CV Length/100	0.0026	0.0010	-0.0282	0.0046	0.0042	0.0044	-0.5040	-0.4877	-0.3768
	(0.0044)	(0.0022)	(0.0190)	(0.0086)	(0.0086)	(0.0075)	(0.6374)	(0.6328)	(0.4706)
CV Coding	-0.0031	-0.0044*	-0.0414**	0.0110	0.0108	0.0125	-0.9448	-0.9129	-0.9246
	(0.0055)	(0.0026)	(0.0193)	(0.0082)	(0.0081)	(0.0082)	(0.7454)	(0.7403)	(0.6139)
CV Publication/Working Paper	0.0441*	0.0154	-0.0237	0.0451	0.0433	0.0165	-5.5511	-5.2459	-2.1210
	(0.0244)	(0.0173)	(0.0820)	(0.0460)	(0.0463)	(0.0427)	(3.5094)	(3.5215)	(3.0760)
Essay Length/100	0.0125*	0.0041	0.0253	-0.0001	-0.0001	-0.0090	-0.2293	-0.2579	0.8458
	(0.0067)	(0.0073)	(0.0219)	(0.0115)	(0.0117)	(0.0103)	(0.8988)	(0.9022)	(0.7451)
Essay Specific Professor	0.0275	0.0077	0.0974	0.0329	0.0271	0.0362	-3.3108	-2.8484	-3.4539
	(0.0208)	(0.0130)	(0.0724)	(0.0369)	(0.0368)	(0.0351)	(2.9044)	(2.8921)	(2.4207)
Essay Economics Terms (%)	-0.0012	0.0001	0.0366**	-0.0173**	-0.0175**	-0.0104	1.7613***	1.7965***	0.9962*
	(0.0039)	(0.0016)	(0.0162)	(0.0082)	(0.0081)	(0.0078)	(0.6048)	(0.6027)	(0.5134)
Essay Research Terms (%)	-0.0001	-0.0009	-0.0266	0.0107	0.0098	0.0019	-1.7875**	-1.7733**	-0.4855
	(0.0053)	(0.0031)	(0.0206)	(0.0097)	(0.0098)	(0.0085)	(0.7813)	(0.7808)	(0.6382)
Essay Positive Terms (%)	-0.0014	-0.0013	-0.0072	0.0111	0.0118*	0.0123*	-0.4461	-0.4785	-0.3335
	(0.0037)	(0.0019)	(0.0189)	(0.0069)	(0.0070)	(0.0068)	(0.5609)	(0.5541)	(0.4475)
Essay Negative Terms (%)	-0.0011	-0.0016	0.0003	-0.0059	-0.0069	-0.0060	0.3935	0.4548	0.4364
	(0.0055)	(0.0029)	(0.0152)	(0.0102)	(0.0101)	(0.0094)	(0.7865)	(0.7837)	(0.6420)
Missing Committee Score		-0.0096			-0.0528**	-0.0277		4.4451***	1.3124
		(0.0120)			(0.0212)	(0.0191)		(1.6677)	(1.3892)
Missing Application Pdf	-0.0723***	0.0036	-0.1054	0.0224	0.0410	0.0305	-2.4877	-4.0722	-3.1533
	(0.0227)	(0.0149)	(0.0858)	(0.0375)	(0.0379)	(0.0348)	(2.9090)	(2.9614)	(2.4962)
Open House			0.2615***						
			(0.0512)						
Program Rank						-0.0011***			0.0832***
						(0.0003)			(0.0185)
Missing Program Rank						-0.3965***			41.0729***
						(0.0512)			(2.6010)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

As such, for these measures, one can implement a Heckman (1979) correction. Notationally, let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$. y is a measure of success, x are variables in applications, and d is matriculation. The first step is to run a probit regression of d on z and compute the inverse Mills ratio $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$. The second step is to run OLS or heckprobit of y on $(x, \hat{\lambda})$ to obtain $\hat{\beta}$. For a Heckman instrument, one needs a variable that affects d but not y. An example is a variable measuring the applicant's enthusiasm about the program, such as whether they included a specific professor's name in their application essay. In Section 4, I discuss the implementation of Heckman corrections in a subset of this paper's structural models.

In addition, in Section B, I implement a Wald test of whether the determinants of acceptance are the same across this university's graduate programs, and I find that they are not the same. Could I similarly implement a Wald test of whether the determinants of acceptance are the same as those of success within a program? For example, does the quantitative GRE have the same AME for acceptance as for success? Intuitively, provided the committee wants to select applicants based on a "success" model, predictors of admission only may be overvalued, and vice versa. Such a test would be not work, however. First, there may be unwanted rejections of the null. For example, the committee may value diversity if it does not hurt success probabilities too much, in which case it may favor demographics not strictly more likely to succeed. There may also be unwanted non-rejections. If there is discrimination in the job market, the committee may rationalize discriminating in the same way, which would make the test not reject the null.

But more importantly, this test would not account for the mechanics of the admissions process. As a simple example, suppose the following conditions hold. First, *Work Experience* has a large positive effect on success, and *Graduate Degree* has a smaller positive effect. Second, 25% of applicants have both characteristics and therefore the highest success probability, 25% have *Work Experience* only (second-highest), 25% have *Graduate Degree* only (third-highest), and 25% have neither (lowest). Third, this university accepts half the applicants, and the other half attend a lower-ranked program, which ensures that everyone's outcomes are observable. Fourth, unlike applicant quality, program quality does not affect outcomes, at least in this example.

Under these suppositions, an optimal committee will accept the applicants with *Work Experience* and ignore *Graduate Degree*. As such, an acceptance regression will have a degenerate *Work Experience* AME of 1 and a *Graduate Degree* AME of 0, while a success regression will have AMEs as follows: $1 >$ *Work Experience* $>$ *Graduate Degree* > 0. Also, for the acceptance AMEs to equal the success AMEs, the committee would suboptimally have to accept applicants without work experience. Therefore, unlike with, for instance, wages equaling marginal revenue product in a competitive labor market which allows for cross-regression comparisons, the acceptance and success AMEs need not be equal for the committee to be optimal. In turn, to draw conclusions about optimality, the relationship between the AMEs must be interpreted through models of the committee's selection problem, which are the basis for Section 4.

4 Structural Analysis

There are two main reasons to estimate structural models of the admissions committee's decision problem. First, in order to make any claims about whether committees are optimizing, it is necessary to define what optimization is by explicitly modeling their problem. Second, structural models can uncover feasible counterfactual deviation decision rules to help committees admit stronger applicants, and the models can quantify any deviation gains from such rules.

The question then is how to set up the models. Bai et al. (2022) prove that if all the variables in an application file can be reduced to a single number (i.e. an expected success probability),[8] then accepting the optimal subset of applicants is the same as using a cutoff rule. A cutoff rule problem is far more tractable than a portfolio choice problem because I do not have to consider all the subsets of the hundreds of the applicants in each year who can be accepted; I only have to rank the applicants and choose the ones above the cutoff. Because committees may be more selective in some years than others, I allow the cutoff probability to vary across years.

After generalizing into a two-round dynamic framework Bai et al. (2022)'s proof that the committee's portfolio choice problem reduces to a cutoff rule problem, I start by modeling the committee's problem as follows. In Round 1, the committee observes only each applicant's objective variables, such as GRE scores. Based on those variables, it can transition the applicant to Round 2, or it can issue an immediate rejection. If the applicant survives Round 1, the committee then takes the time to read the application file and access all variables, both objective and subjective (e.g. recommendation letters). Finally, it accepts or rejects the applicant.

Rather than setting up a two-round model, why do I not rank the applicants based on their success probabilities conditional on all data and compare this one-round model's ranking to the committee's? The reason is that a one-round model is unfair to the committee because the committee does not have time to read everyone's files (though in the future, perhaps committees can use textual processing algorithms or local LLMs to "read" files quickly and generate additional objective variables without too much effort). Moreover, two-round models are useful not just for graduate admissions but for any other setting with first-stage filtering, such as hiring, etc.

I start with two different two-round models: dynamic and static. In Round 2 of the dynamic model, the committee accepts the applicants who survived Round 1 and have success probabilities greater than the Round 2 cutoff probability. In Round 1, it forms a conditional expectation about those who survive to Round 2, meaning that the Round 1 cutoff is the level of that applicant's expected Round 2 value. As such, the Round 1 cutoff can take a value greater than one. In contrast, in the static model, the committee ranks the applicants based on their objective variables, reads the files of those whose expected success probabilities are above the first cutoff probability, re-ranks the surviving applicants based on all their data, and accepts those above the second cutoff

[8] The assumption that committees select based on expected success may have become more realistic in light of the 2023 Supreme Court ruling against race-based affirmative action in *Students for Fair Admissions v. Harvard*.

probability. I refer to this model as static because each round is a separate static problem. The static model is useful for its simplicity and interpretable cutoff probabilities in both rounds.

The dynamic model appears more sophisticated because in Round 1, it favors applicants whose objective variables predict good subjective variables. However, the static model also correctly favors those applicants. While the static Round 1 success probability parameter estimates are inconsistent because the subjective variables are omitted, the fitted values estimate the success probability averaged over the subjective variables given the objective ones. Therefore, the component of the subjective variables that is predictable from the objective variables is absorbed into the estimates. Only the myopic model, which is identical in setup to the dynamic model except that the parameters for the objective variables in the distribution of the subjective variables are restricted to zero, does not favor such applicants in Round 1. The static model also better describes what real-world committees do, as they do not solve Round 2 conditional expectations.

Later in this section, when I present maximum likelihood estimation results of these models, I include the Akaike Information Criterion (AIC) $= 2[K - \log(\hat{\mathcal{L}}^R)]$, which can be used to compare each restricted model with its unrestricted counterpart. K is the number of parameters, and $\hat{\mathcal{L}}^R$ is the restricted likelihood's value; lower AIC values are preferred. AIC values are comparable only between the restricted and unrestricted specifications of the same model. The reason is that the static objective is a composite likelihood over decisions and placement outcomes, with every outcome entering once in each of the Rounds 1 and 2 success blocks, whereas the dynamic likelihoods also include the densities of the subjective variables conditional on the objective variables.

Still later in this section, I present dynamic and static models in which not every accepted applicant matriculates, and which are also derived from two-round portfolio choice models, albeit models in which the objective is to maximize the sum of the success probabilities of the matriculants rather than the acceptances. In Round 2, the committee still accepts the applicants who survived Round 1 and have success probabilities greater than the Round 2 cutoff probability. No weight is placed on matriculation probabilities in Round 2 because while likely matriculants are more likely to produce a positive payoff, they are also more likely to take up cohort spots and so are more "expensive." However, weight is placed on matriculation probabilities in Round 1 because if the committee chooses to read the files of only the applicants who are most likely to succeed, it could find itself in a situation in Round 2 in which the only accepted applicants who matriculate are those with strong objective variables and weak subjective variables. This situation becomes increasingly likely the lower-ranked the committee's university is.

4.1 Portfolio Choice to Cutoff Rule

Bai et al. (2022) prove that selecting the best N_A applicants out of N is equivalent to selecting the N_A whose success probability exceeds a cutoff.[9] Let x be a vector of objective variables

[9]To relax the assumption that the committee only cares about success, its objective can be generalized to maximizing the sum of a convex combination of success probabilities and other factors, such as diversity.

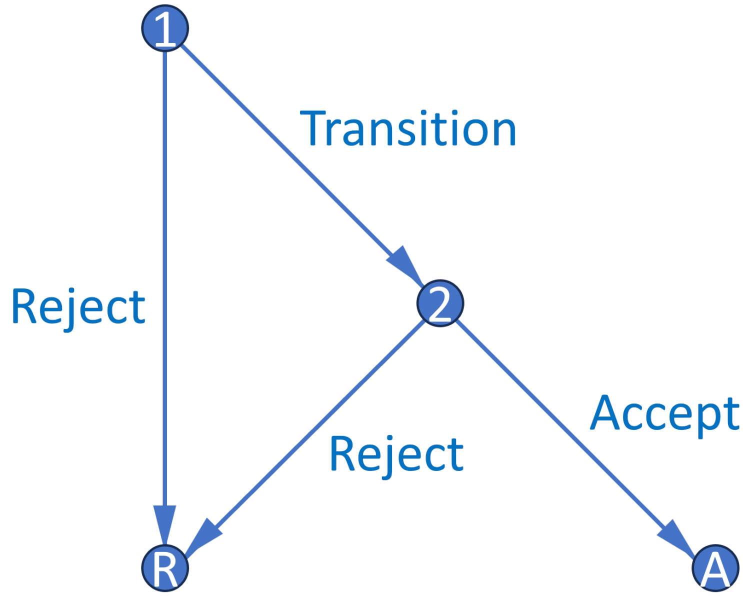


Figure 1: Decision Graph

(e.g. GRE scores), and z be a vector of subjective variables (e.g. variables from recommendation letters). Also, let d_n denote the accept or reject decision for the nth applicant, and y_n denote that applicant's outcome, where $S =$ success and $F =$ failure. The committee solves a constrained optimization problem, in which it maximizes the sum of the success probabilities of the accepted applicants under the constraint that it cannot accept more than N_A applicants. For simplicity, suppose there is one round in one year, and all accepted applicants matriculate:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n,z_n)\mathbb{1}(d_n = A), \text{ such that } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A. \tag{1}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_n = A$, which means that the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned}\mathcal{L}(d_1,\ldots,d_N;\lambda) &= \sum_{n=1}^{N} P(y_n = S|x_n,z_n)\mathbb{1}(d_n = A) + \lambda\left\{N_A - \left[N - \sum_{n=1}^{N} \mathbb{1}(d_n = R)\right]\right\} \\ &= \sum_{n=1}^{N} \left[P(y_n = S|x_n,z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right] + \lambda(N_A - N).\end{aligned} \tag{2}$$

Suppose without loss of generality that the applicants are sorted in decreasing order such that $P(y_1 = S|x_1,z_1) \geq \cdots \geq P(y_N = S|x_N,z_N)$. To ensure that the committee accepts exactly N_A applicants, it must be the case that $\lambda^* \in \left[P(y_{N_A+1} = S|x_{N_A+1},z_{N_A+1}), P(y_{N_A} = S|x_{N_A},z_{N_A})\right]$. That is, in this setup, λ^* is set identified.[10] Then the solution is to accept all applicants whose success probability exceeds λ^* and reject the rest, which means that λ^* is a cutoff probability:

$$\forall n \in \{1,\dots,N\},\, d_n(x_n,z_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n,z_n) > \lambda^* \\ \{A,R\} & \text{if } P(y_n = S|x_n,z_n) = \lambda^* \\ R & \text{if } P(y_n = S|x_n,z_n) < \lambda^*. \end{cases} \tag{3}$$

If $P(y_n = S|x_n,z_n) = \lambda^*$, then the committee is indifferent between acceptance and rejection, which means that in the event of a tie at the margin, the committee will randomly accept enough applicants to reach N_A acceptances. Also, while λ^* is set identified in this setup, I show in the proof of Theorem 2 in the next subsection how to point identify it with admissions data by modeling a committee that compares each applicant's success probability to a single-valued cutoff, and then by exploiting the observed committee's exact selectivity. Finally, given a large number of applicants, λ^* can be interpreted as the marginal accepted applicant's success probability. Therefore, I denote it going forward as $P(y^* = S)$, or the success probability of that applicant.

This version of the portfolio choice problem contains just one round. But the committee's actual problem has two rounds: first-round filtering on x only, along with second-round final selections on x and z. In Appendix E.1, I generalize, for both the dynamic and static setups, the one-round portfolio choice problem to a two-round portfolio choice problem, and show that it also reduces to a two-round cutoff rule problem. The Round 2 cutoff rule problem is:

$$\max_{(d_{2,1},\dots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|x_n,z_n)\mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \le N_A. \tag{4}$$

Its solution with $\lambda_2^* = P(y_2^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1},z_{N_A+1}), P(y_{N_A} = S|x_{N_A},z_{N_A})\right]$ is:

$$\forall n \in \{1,\dots,N_T\},\, d_{2,n}(x_n,z_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|x_n,z_n) > \lambda_2^* \\ \{A,R\} & \text{if } P(y_n = S|x_n,z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|x_n,z_n) < \lambda_2^*. \end{cases} \tag{5}$$

I show in Appendix E.1 that under an independence assumption, the set-identified interval for λ_2^* collapses to a point, denoted $\overline{\lambda_2^*}$. Let $EV_{2,n}(x_n) = \mathbb{E}[\max\{P(y_n = S|x_n,z_n), \overline{\lambda_2^*}\}|x_n] = \int_{\tilde{z}_n} \max\{P(y_n = S|x_n,\tilde{z}_n), \overline{\lambda_2^*}\} dF_{z|x}(\tilde{z}_n|x_n)$. Then the Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\dots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \le N_T. \tag{6}$$

[10]If there were a continuum of applicants as in Bai et al. (2022)'s proof, the set's length would be 0, and so λ^* would be point identified. But all the models in this paper have N applicants, as there are in the data.

Its solution with $\lambda_1^* = L(y_1^* = S) \in \left[EV_{2,N_T+1}(x_{N_T+1}), EV_{2,N_T}(x_{N_T})\right]$ is:

$$\forall n \in \{1,\ldots,N\},\, d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{7}$$

That said, it is possible to add the preference shocks directly to the portfolio choice problem and carry them through the derivation, in which case the result is exactly the cutoff rule problems in this section. For notational simplicity, I do not do so throughout this paper, but I prove in Appendix F that for the one-round portfolio choice problem without matriculation (i.e. the portfolio choice problem I first presented), the preference shocks can be carried through the derivation:

Theorem 1. *For large N, the committee's portfolio choice problem with preference shocks has a cutoff rule solution with logistic conditional choice probabilities.*

Proof. See Appendix F.

4.1.1 Dynamic Model

Figure 3 portrays the dynamic model, whose notation is as follows. $r \in \{1,2\}$ is the round, x is a vector of objective variables, z is a vector of subjective variables, t is the year, $\bar{k}$ is this university's U.S. News program rank (k is the applicant's observed future program rank), and ε_r is a type 1 extreme value (T1EV) shock with scale σ. The shocks let the model fit the data, as a real-world committee will not always accept the applicants with the highest success probabilities.

But what do the shocks represent? Mechanically, ε_r is the part of the committee's decision that the observables $(x,z,t,\bar{k})$ do not explain. It is convenient to call ε_r a preference shock, but the label should not be read as pure idiosyncratic taste. The shock reflects my ignorance of what the committee conditioned on, not randomness in the committee's behavior, and is therefore a residual defined relative to the observables. As such, anything the committee responded to that is absent from the observables is folded into ε_r, which means that what ε_r contains is an empirical matter. The more of the committee's information I observe, the closer ε_r comes to genuine taste, and the more I do not observe, the more ε_r stands in for private information that the committee acted on. This gap matters most when such variables also predict success, since the model then omits determinants of success. I discuss private information and its implications in Section 5.

Starting in Round 2, the committee chooses $d_2 \in \{\text{Accept}, \text{Reject}\}$. If it accepts an applicant, it gets the applicant's success probability $P(y = S|x,z,t,\bar{k})$; the probability of succeeding at this university is a function of applicant characteristics, the year, and this university's quality. Whereas if the committee rejects the applicant, it gets the Round 2 cutoff probability $P(y_2^* = S|t)$, which is the success probability of t's marginal accepted applicant, whose success probability is conditional on $(x,z,t,\bar{k})$. There are various measures of success, from completing the program to obtaining a job placement above a threshold ranking. The Round 2 value function is:

$$v_2(x,z,t,\bar{k},\varepsilon_2) = \max_{d_2\in\{A,R\}} u_2(x,z,t,\bar{k},d_2)+\varepsilon_2(d_2), \text{ such that:} \tag{8}$$

$$u_2(x,z,t,\bar{k},d_2) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_2=A)+P(y_2^*=S|t)\mathbb{1}(d_2=R).$$

Meanwhile in Round 1, the committee chooses $d_1 \in \{\text{Transition}, \text{Reject}\}$. If it transitions an applicant to Round 2, it gets $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]$, which is the expected value of Round 2, where z is integrated out conditional on (x,t). However, if it rejects the applicant, it gets $L(y_1^*=S|t)$, or the expected Round 2 value of t's marginal transitioned applicant. Because this value can exceed 1, it cannot be interpreted as a probability. The Round 1 value function is:

$$v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,\bar{k},d_1)+\varepsilon_1(d_1), \text{ such that:} \tag{9}$$

$$u_1(x,t,\bar{k},d_1) = \mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]\mathbb{1}(d_1=T)+L(y_1^*=S|t)\mathbb{1}(d_1=R), \text{ such that:}$$
$$\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma\log\left(\sum_{\tilde{d}_2\in\{A,R\}} e^{u_2(x,\tilde{z},t,\bar{k},\tilde{d}_2)/\sigma}\right)dF(\tilde{z}|x,t).$$

Each round's conditional choice probabilities (CCPs) then have the closed-form expression:

$$P(d_r|x,(z),t,\bar{k}) = \frac{\exp(u_r(x,(z),t,\bar{k},d_r)/\sigma)}{\sum_{\tilde{d}_r\in D_r}\exp(u_r(x,(z),t,\bar{k},\tilde{d}_r)/\sigma)}. \tag{10}$$

To estimate this model, the first step is to set up the unrestricted likelihood function. I parameterize the unrestricted CCPs (P_{d_1},P_{d_2}), success probability (P_y), and cutoffs ($L_{y_1^*}$,$P_{y_2^*}$) with the following flexible exponential and logistic specifications containing parameters θ:

$$P_{d_1} = P\big(d_1=T\big|x,t;\theta_T\big) = \frac{\exp(f_T(x,t)\theta_T)}{1+\exp(f_T(x,t)\theta_T)}, \quad P_{d_2} = P\big(d_2=A\big|x,z,t;\theta_A\big) = \frac{\exp(f_A(x,z,t)\theta_A)}{1+\exp(f_A(x,z,t)\theta_A)} \tag{11}$$

$$P_y = P\big(y=S\big|x,z,t,\bar{k};\theta_S\big) = \frac{\exp(f_S(x,z,t,\bar{k})\theta_S)}{1+\exp(f_S(x,z,t,\bar{k})\theta_S)} \tag{12}$$

$$L_{y_1^*} = P\big(y_1^*=S\big|t;\theta_{S_1^*}\big) = \exp(f_{S_1^*}(t)\theta_{S_1^*}), \quad P_{y_2^*} = P\big(y_2^*=S\big|t;\theta_{S_2^*}\big) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1+\exp(f_{S_2^*}(t)\theta_{S_2^*})}. \tag{13}$$

Letting F_j be the conditional CDF for each z_j, I parameterize the integral in Equation (9) as:

$$\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}_1}\cdots\int_{\tilde{z}_J}\bigg[\,\cdot\,\bigg]dF_1(\tilde{z}_1|\tilde{z}_2,\ldots\tilde{z}_J,x,t)\cdots dF_J(\tilde{z}_J|x,t). \tag{14}$$

Depending on whether each z_j is continuous, binary, or multinomial, I use either: $f_{z|x,t}(z_j|x,t;\beta_{F_{z|x,t,j}},\sigma^2_{F_{z|x,t,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,t,j}}}}\exp\left[\frac{-1}{2\sigma^2_{F_{z|x,t,j}}}(z_j-g(x,t)\beta_{F_{z|x,t,j}})^2\right]$, $P(z_j=1|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}{1+\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}$, or $P(z_j=\ell|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta^{\ell}_{F_{z|x,t,j}}\right)}{1+\sum_{i=1}^{L_j-1}\exp\left(g(x,t)\theta^{i}_{F_{z|x,t,j}}\right)}$ for $\ell=1,\ldots,L_j-1$, respectively.

Let N_T be the number of N total applicants who survive to Round 2. Also, T = transition, A = accept, R = reject, S = success, and F = failure. Finally, z^c = *Committee Score*, whereas $z^{\neg c}$ = all the other subjective variables. The unrestricted likelihood is then:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F\right) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}$$
$$\prod_{n=N_T+1}^{N} \left(1 - P_{d_{1,n}}(\theta_T)\right) \prod_{n=1}^{N} f\left(z_n^{\neg c} | x_n, t_n; \theta_F\right) P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \quad (15)$$

I estimate the restricted likelihood by substituting the restricted CCPs for the unrestricted ones. To save computation time, I use (θ_F^U, θ_S^U) as an initial guess of (θ_F^R, θ_S^R). There is also a complication with the data, namely that *Committee Score* is unobservable for the $N - N_T$ of N applicants who were rejected in Round 1.[11] The missing data pose a problem when estimating $f(z_n | x_n, t_n; \theta_F)$ in the likelihood because *Committee Score* is part of z. Because I cannot obtain *Committee Score* for everyone, I tried estimating the following Heckman-corrected likelihood:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \Phi(x_n, t_n; \theta_H) f\left(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} \prod_{n=N_T+1}^{N} \left(1 - \Phi(x_n, t_n; \theta_H)\right)$$
$$f\left(z_n^{\neg c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right) \left(1 - P_{d_{1,n}}(\theta_T)\right) \prod_{n=1}^{N} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \quad (16)$$

The probit selection block is $\Phi(x_n, t_n; \theta_H)$, and the inverse Mills ratio is $\frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}$. Unfortunately, there are no available Heckman instruments that affect selection but not the distribution of z (year dummies affect z as well as selection). Therefore, identification must come from the nonlinearity of the inverse Mills ratio. This identification is weak, and the corrected results are nearly identical to the uncorrected ones, so the uncorrected likelihood is my preferred specification.

After estimating the unrestricted likelihood, I estimate the restricted likelihood $\mathcal{L}^R$ by substituting in the model's CCPs, which are functions of the success probability (θ_S) and structural $(\sigma, \theta_{S_1^*}, \theta_{S_2^*})$ parameters. Because I can observe whether the committee read each applicant's file and whether the committee accepted each applicant, I identify the cutoff parameters $(\theta_{S_1^*}, \theta_{S_2^*})$ for each year by how selective the committee is that year. Without loss of generality, every time $d_2 = A$, the term $\frac{\exp(P(y=S|x,z,t,\bar{k})/\sigma)}{\exp(P(y=S|x,z,t,\bar{k})/\sigma)+\exp(P(y_2^*=S|t)/\sigma)}$ enters $\mathcal{L}^R$. This term decreases in $P(y_2^* = S|t)$, so every time an applicant is accepted, it puts downward pressure on $P(y_2^* = S|t)$, considering the objective is to maximize the likelihood. Intuitively, the committee's selectivity identifies the cutoff probability. Also, if the committee engages in sorting, or selecting the applicants with the highest success probabilities $P(y = S|x, z, t, \bar{k})$, that improves the likelihood. Formally:

[11] For those applicants, I perform mean imputation for *Committee Score*. Also, in the restricted Round 2 CCPs, I set their *Missing Committee Score* = 0 so as not to penalize them for not being in the committee's transition set.

Theorem 2. *For simplicity, suppose there is one round. The maximum likelihood estimate $\hat{P}_{y^*} = \hat{P}(y^* = S|t)$ is such that the number of acceptances N_A equals the expected N_A. Also, for any $\hat{P}_{y^*}$ and N_A, $\mathcal{L}^R$ is highest when the committee accepts the N_A highest $P_{y_n} = P(y_n = S|x_n, z_n, t_n, \bar{k}_n)$.*

Proof. See Appendix F.

When estimating this model, it is necessary to use public outcomes data to track success for all applicants, including those who did not matriculate at this university. Bai et al. (2022) prove that success parameter estimates will be biased toward zero if the sample is restricted to matriculants only. The reason is that all matriculants must have been accepted, and accepted applicants tend to have high values of the determinants of success. For example, conditional on matriculating, the relationship between the quantitative GRE score and success is weakened because nearly all matriculants have very high scores (analogous to height not predicting success in the NBA).

Along with tracking success for all applicants, it is also necessary to track which, if any, graduate program each applicant attended in order to avoid assuming that success depends only on applicant characteristics. For instance, a rejected applicant who succeeded at some other university might not have succeeded at this one, and so the committee may not have made a mistake by rejecting them. If a variable predicts success only through its effect on matriculation at a higher-ranked university, then this university should not use that variable in admissions. In estimation, I use each applicant's observed future program rank k in the success probability block and this university's fixed program rank in the restricted conditional choice probabilities.

In Table 3, I present the dynamic model's estimated cutoffs with 95% confidence intervals, as well as the estimated T1EV scale parameter σ and the AIC. While the Round 1 cutoffs are not probabilities, they and the Round 2 cutoffs show that the committee was similarly selective across years. $\hat{\sigma}$ is also small, meaning that the committee's decisions are relatively deterministic with respect to the variables in the application files. A larger set of the unrestricted and restricted parameter estimates is in Tables 11 and 12 of Appendix C. I also present estimation results on simulated data in Tables 20 and 21 of Appendix D, which illustrate features I embed in the simulated data-generating process, namely that the simulated committee's weights are more aligned with the success probability parameters in the optimal simulated dataset than the suboptimal one.

4.1.2 Static Model

I now present the static model. Relative to the dynamic model, this model omits z in the Round 1 success probability specification, meaning that z's parameter estimates are inconsistent (though crucially, the fitted values are still correct). In addition, for any multiyear model, such as those in Online Appendix C, the Round 1 value function must be dynamically linked to the Round 2 value function, meaning that the static framework cannot work across years. However, this model has two advantages. First, it more accurately describes how real-world committees behave. Second, the Round 1 cutoff can be interpreted as a probability and compared to the Round 2 cutoff

Table 3: Restricted Dynamic Cutoffs

	R1 Cutoff Level	R2 Cutoff Probability	Sigma
2015	0.3731	36.58%	
	(0.3067, 0.4539)	(29.81%, 43.92%)	
2016	0.3795	37.21%	
	(0.3025, 0.4761)	(29.20%, 45.99%)	
2017	0.4126	40.27%	
	(0.3563, 0.4778)	(34.52%, 46.30%)	
2018	0.3836	37.68%	
	(0.3271, 0.4497)	(31.79%, 43.97%)	
2019	0.3887	38.14%	
	(0.3268, 0.4624)	(31.89%, 44.81%)	
Sigma			0.0020
			(0.0013)

probability. The static model's Round 2 value function is the same as the dynamic's:

$$v_2(x,z,t,\bar{k},\varepsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x,z,t,\bar{k},d_2) + \varepsilon_2(d_2), \text{ such that:} \tag{17}$$

$$u_2(x,z,t,\bar{k},d_2) = P(y = S|x,z,t,\bar{k})\mathbb{1}(d_2 = A) + P(y_2^* = S|t)\mathbb{1}(d_2 = R).$$

But unlike in the dynamic model, if the committee transitions an applicant in Round 1, it gets $P(y = S|x,t,\bar{k})$, or the applicant's success probability unconditional on z, which approximates the expected Round 2 value in the dynamic model (see Appendix E.1) and may differ from $P(y = S|x,z,t,\bar{k})$. Meanwhile, if the committee rejects the applicant, it gets $P(y_1^* = S|t)$, or the success probability of t's marginal transitioned applicant, whose success probability is conditional on x only. $P(y_1^* = S|t)$ can be less or greater than $P(y_2^* = S|t)$ depending on how much the committee values its time vs. admitting the best possible applicants. The Round 1 value function is:

$$v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x,t,\bar{k},d_1) + \varepsilon_1(d_1), \text{ such that:} \tag{18}$$

$$u_1(x,t,\bar{k},d_1) = P(y = S|x,t,\bar{k})\mathbb{1}(d_1 = T) + P(y_1^* = S|t)\mathbb{1}(d_1 = R).$$

The restricted CCPs are the same as in the dynamic model:

$$P(d_r|x,(z),t,\bar{k}) = \frac{\exp(u_r(x,(z),t,\bar{k},d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(x,(z),t,\bar{k},\widetilde{d}_r)/\sigma)}. \tag{19}$$

Rather than write out the specifications for the unrestricted CCPs, success probabilities, and cutoff probabilities, I reference Equations (11)–(13). That said, there are two differences from the dynamic framework. First, there are two estimated success probabilities (P_{y_1}, P_{y_2}), and second, the Round 1 cutoff is now a probability rather than a level. The unrestricted likelihood is then:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} \prod_{n=N_T+1}^{N} \left(1 - P_{d_{1,n}}(\theta_T)\right)$$
$$\prod_{n=1}^{N} \left[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})\right]^{\mathbb{1}(y_n=S)} \left[(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))\right]^{\mathbb{1}(y_n=F)}. \quad (20)$$

The static model's estimated cutoff probabilities, $\hat{\sigma}$, and AIC are in Table 4, and the unrestricted and restricted estimates are in Table 13 and 14 of Appendix C (with simulated estimates in Tables 22 and 23 of Appendix D). Both rounds' cutoffs are probabilities,[12] and in all five years, the Round 2 cutoff is greater than the Round 1 cutoff. This ordering is expected because both forecasts are scaled to the same outcomes over the same applicant pool, Round 2 accepts about a tenth of that pool, and Round 1 transitions between one third and three fifths of it. The committee read 59% of the files in 2018 vs. 35–43% in the other years, but 2018's Round 1 cutoff is not lower than the other years' cutoffs because the committee's decisions by themselves identify only which applicants fell above and below the cutoff. The success probabilities are necessary to identify the cutoff's level, but they do so relatively weakly, as the confidence intervals show.

Table 4: Restricted Static Cutoffs

	R1 Cutoff Probability	R2 Cutoff Probability	Sigma
2015	29.36%	39.20%	
	(23.20%, 36.38%)	(32.75%, 46.05%)	
2016	29.17%	40.53%	
	(22.21%, 37.27%)	(33.55%, 47.92%)	
2017	37.06%	41.69%	
	(32.10%, 42.29%)	(34.83%, 48.90%)	
2018	38.12%	39.30%	
	(33.98%, 42.44%)	(31.80%, 47.33%)	
2019	34.04%	39.62%	
	(28.67%, 39.85%)	(32.54%, 47.16%)	
Sigma			0.0064
			(0.0035)

The myopic model is a special case of the dynamic model in which the parameters for x in every block of $f_{z|x,t}$ are restricted to equal zero, while the intercepts, parameters for t, dependence parameters among the components of z, and variance parameters remain unrestricted. In this model, if an objective variable is associated with z in addition to being associated with success in its own right, the committee puts no extra weight on that variable in Round 1. The myopic estimates on actual data are in Tables 15 and 16 of Appendix C, and those on simulated data are

[12] The cutoffs are with respect to restricted success probabilities. The average restricted success probability of the acceptance set is greater than its average unrestricted counterpart because the restricted model assumes the accepted applicants are likeliest to succeed. As such, the restricted success probability parameter estimates shift accordingly.

in Tables 24 and 25 of Appendix D. The myopic model is the one model where the restricted AIC is worse than the unrestricted AIC, which is evidence that the committee is not myopic.

4.2 Portfolio Choice with Matriculation

In the portfolio choice problem in Section 4.1, as well as in Bai et al. (2022), there is an assumption that every accepted applicant will matriculate. But in practice, many accepted applicants do not matriculate and instead attend another university. For committees at the very top universities, matriculation probabilities may be less of a concern, as accepted applicants who do not matriculate likely do so for idiosyncratic reasons (e.g. choosing Harvard over MIT). But at the majority of universities, the accepted applicants with the highest success probabilities will often have relatively low matriculation probabilities, as the lower-ranked universities were likely safety options from the outset. The question then is whether the committee should select applicants based on matriculation probabilities as well as success probabilities, or whether it should still accept the applicants with the highest success probabilities irrespective of matriculation.

As such, I generalize the portfolio choice problem to incorporate matriculation probabilities. The committee now accepts the subset of applicants with the highest sum of success times matriculation probabilities subject to a capacity constraint on the expected number of matriculants. For each applicant n, the committee makes an acceptance or rejection decision given:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)\mathbb{1}(d_n = A),$$
$$\text{such that: } \sum_{n=1}^{N} P(m_n = M|x_n, z_n)\mathbb{1}(d_n = A) \leq N_M, \tag{21}$$

where each applicant's matriculation decision m is to either matriculate or decline, $P(m = M|x, z)$ is each applicant matriculation probability, and N_M is the maximum number of matriculants. Due to the linearity of expectations, the objective function is equivalent to the expected sum of the success probabilities of the accepted applicants who matriculate. Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_n = A$, which means that the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned}\mathcal{L}(d_1,\ldots,d_N;\lambda) &= \sum_{n=1}^{N} P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)\mathbb{1}(d_n = A)\\ &\quad + \lambda\left\{N_M - \left[\sum_{n=1}^{N} P(m_n = M|x_n, z_n) - \sum_{n=1}^{N} P(m_n = M|x_n, z_n)\mathbb{1}(d_n = R)\right]\right\}\\ &= \sum_{n=1}^{N}\left[P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right]P(m_n = M|x_n, z_n)\\ &\quad + \lambda\left[N_M - \sum_{n=1}^{N} P(m_n = M|x_n, z_n)\right].\end{aligned} \tag{22}$$

From the Lagrangian, it is clear that if there is a cutoff rule solution, the solution is on the success probabilities alone. While applicants with higher matriculation probabilities are more valuable in terms of the objective, they are also more expensive in terms of the capacity constraint on the number of matriculants. Intuitively, if the committee were to prioritize applicants with higher matriculation probabilities, it would not be able to give as many acceptances.

Moreover, the only way to get a cutoff rule solution that incorporates matriculation probabilities would be to have the constraint be on the number of acceptances. That is, the constraint would be $\sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A$, the Lagrangian would equal $\sum_{n=1}^{N} \left[P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)\mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)\right] + \lambda(N_A - N)$, the applicants would be ranked relative based on $P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)$, and likely matriculants would receive more acceptances.

Crucially, however, because not every accepted applicant will matriculate, this exact problem does not have a cutoff rule solution. For example, suppose $N = 2$, $N_M = 1$, $P(y_1 = S|w_1) = 0.9$, $P(m_1 = M|w_1) = 0.5$, $P(y_2 = S|w_2) = 0.89$, and $P(m_2 = M|w_2) = 0.89$. Since $0.5 + 0.89 > 1$, the committee can accept only one applicant. The cutoff rule solution would be to accept Applicant 1, but Applicant 2 would yield a higher value of the objective since $0.89 \times 0.89 > 0.9 \times 0.5$. But fortunately, any exceptions to the cutoff rule solution disappear once N becomes sufficiently large; that is, when there is a continuum of applicants as in Bai et al. (2022)'s setup.

Theorem 3. *For large N, the committee's portfolio choice problem with matriculation probabilities has a cutoff rule solution on success probabilities.*

Proof. See Appendix F.

Given this result, it appears that committees should not factor in matriculation probabilities when making admissions decisions. In a one-round model, that is the case. However, in a two-round portfolio choice model with matriculation, the cutoff rule is such that the committee should consider matriculation probabilities in Round 1. This rule holds in particular for lower-ranked programs. For example, if a lower-ranked program's committee chooses to read only the files of applicants with perfect GRE scores, then it may have trouble filling its cohort in Round 2, considering that the best applicants are unlikely to matriculate. And even if the committee can fill its cohort, that cohort is likely to be filled with applicants who have weak subjective variables. In contrast, if the committee had considered matriculation probabilities in Round 1, it may have transitioned applicants with lower GRE scores but strong subjective variables who will actually matriculate. I provide the full derivation in Appendix E.2, but I present the dynamic portfolio choice problem and cutoff rule solution here. The Round 2 problem is:

$$\max_{(d_{2,1},\ldots,d_{2,N_T}) \in \{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) \leq N_M. \tag{23}$$

Edge cases aside, its solution with $\lambda_2^* = P(y_2^* = S)$ is:

$$\forall n \in \{1,\ldots,N_T\},\, d_{2,n}(w_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A,R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^*. \end{cases} \tag{24}$$

Let $EV_{2,n}(x_n) = \int_{\widetilde{z}_n} \max\{P(y_n = S|x_n, \widetilde{z}_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|x_n, \widetilde{z}_n) dF_{z|x}(\widetilde{z}_n|x_n)$, where is $\overline{\lambda_2^*}$ is the collapsed interval for λ_2^*. Then the Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N}) \in \{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T. \tag{25}$$

Edge cases aside, its solution with $\lambda_1^* = L(y_1^* = S, m_1^* = M)$ is:

$$\forall n \in \{1,\ldots,N\},\, d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{26}$$

As I show in Appendix E.2, $EV_{2,n}(x_n)$ puts considerably more weight on success than matriculation. Therefore, the matriculation-dynamic and dynamic models have similar decision rules, and the best static approximation for the matriculation-dynamic model is the static model.

4.2.1 Matriculation Models

Because matriculation probabilities do not matter in Round 2, the matriculation-dynamic and static Round 2 value functions are the same as the previous section's Round 2 value functions:

$$v_2(x,z,t,\bar{k},\varepsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x,z,t,\bar{k},d_2) + \varepsilon_2(d_2), \text{ such that:} \tag{27}$$

$$u_2(x,z,t,\bar{k},d_2) = P(y = S|x,z,t,\bar{k}) \mathbb{1}(d_2 = A) + P(y_2^* = S|t) \mathbb{1}(d_2 = R).$$

However, the Round 1 value function contains matriculation probabilities:

$$v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x,t,\bar{k},d_1) + \varepsilon_1(d_1), \text{ such that:} \tag{28}$$

$$u_1(x,t,\bar{k},d_1) = \mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] \mathbb{1}(d_1 = T) + L(y_1^* = S, m_1^* = M|t) \mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\widetilde{z}} \sigma \log\left(e^{[P(y=S|x,\widetilde{z},t,\bar{k}) - P(y_2^*=S|t)]/\sigma} + 1\right) P(m = M|x,\widetilde{z},t) dF(\widetilde{z}|x,t).$$

The derivation of the Round 1 value function is in Appendix E.2. The object I denote by $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]$ is the Round 1 transition payoff, as opposed to the literal conditional expectation of v_2. As such, the resulting social surplus function contains $e^{[P(y=S|x,\widetilde{z},t,\bar{k}) - P(y_2^*=S|t)]/\sigma} + 1$ instead of the usual $e^{P(y=S|x,\widetilde{z},t,\bar{k})/\sigma} + e^{P(y_2^*=S|t)/\sigma}$. Due to the matriculation term $P(m = M|x,\widetilde{z},t)$, the incorrect latter expression does not result in the same ordering of applicants as the correct former expression. The restricted CCPs are the same as in the dynamic and static models:

$$P(d_r|x,(z),t,\bar{k}) = \frac{\exp(u_r(x,(z),t,\bar{k},d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(x,(z),t,\bar{k},\widetilde{d}_r)/\sigma)}. \tag{29}$$

Rather than write out the specifications for the unrestricted CCPs, success probability, and cutoff probabilities, I reference Equations (11)–(13). But the specification for the matriculation probability is: $P_m = P\big(m = M \big| x,z,t;\theta_M\big) = \frac{\exp(f_M(x,z,t)\theta_M)}{1+\exp(f_M(x,z,t)\theta_M)}$. The unrestricted likelihood is then:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)}(1-P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)} \prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_F\big) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}$$

$$(1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} \prod_{n=N_T+1}^{N} \big(1-P_{d_{1,n}}(\theta_T)\big) \prod_{n=1}^{N} f\big(z_n^{\neg c}|x_n,t_n;\theta_F\big) P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \tag{30}$$

As with the restricted dynamic likelihood, I use (θ_F^U,θ_S^U) as an initial guess of (θ_F^R,θ_S^R) for the restricted matriculation-dynamic likelihood. Also, a Heckman correction can be implemented for $f_{z|x,t}$ because *Committee Score* is missing for the applicants rejected in Round 1, but due to the lack of an available Heckman instrument, I do not implement one in my preferred specification.

The matriculation-dynamic model's estimated cutoffs, $\hat{\sigma}$, and AIC are in Table 5, and the unrestricted and restricted estimates are in Table 17 and 18 of Appendix C (with simulated estimates in Tables 26 and 26 of Appendix D). The Round 1 cutoffs are much lower in the matriculation-dynamic model than in the dynamic model, but that is because of the expression for $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]$. Both rounds' cutoffs and σ are well-identified in all five of the years.

Table 5: Restricted Matriculation-Dynamic Cutoffs

	R1 Cutoff Level	R2 Cutoff Probability	Sigma
2015	0.0071	36.35%	
	(0.0026, 0.0194)	(29.47%, 43.84%)	
2016	0.0071	38.25%	
	(0.0028, 0.0180)	(31.05%, 46.00%)	
2017	0.0105	40.65%	
	(0.0041, 0.0267)	(35.15%, 46.39%)	
2018	0.0095	36.59%	
	(0.0038, 0.0240)	(30.01%, 43.71%)	
2019	0.0095	37.41%	
	(0.0035, 0.0255)	(31.04%, 44.26%)	
Sigma			0.0020**
			(0.0010)

Nevertheless, discussions with members of this university's committee indicate that the committee does not consider matriculation probabilities in Round 1 because this university is ranked highly enough that in general, it can fill its cohort with applicants who have strong objective and subjective variables. However, this university has at times considered matriculation probabilities when making admissions decisions; it would have ignored matriculation probabilities entirely only in a world of perfect information about each applicant's intention. Suppose, for example, that every time the committee accepts an applicant, it immediately hears back from that applicant whether the applicant will matriculate. Then in Round 2, the committee would start with the applicant with the highest success probability, immediately hear back, proceed to the applicant with the second-highest success probability, and continue until the cohort is filled.

But in the real world, committees do not immediately hear back from applicants. Therefore, if a committee does not consider matriculation probabilities when giving early non-waitlisted acceptances, by the time it hears back from the accepted applicants (most of which will not matriculate since success and matriculation probabilities are negatively correlated), relatively good applicants who would have matriculated will have moved on to other universities. As a result, the committee may have to settle for weaker applicants than if it had considered matriculation probabilities. In addition, the committee can adjust the number of late acceptances it gives based on how many early matriculants it receives to help it attain the target number of matriculants. These dynamics motivate models of the waitlist process that I present in Online Appendix B.

5 Optimality Test

Suppose the committee were to have chosen the acceptance or matriculation set that was chosen by the dynamic, static, myopic or matriculation-dynamic model. Would the average success probability of the committee's set have increased, and if so, by how much? To answer these questions, I solve the model with $\sigma = 0$ in the conditional choice probabilities (CCPs), compute the average success probabilities of the model's acceptance or matriculation set, and compare those probabilities to the probabilities of the committee's acceptance or matriculation set. When I solve the model, I plug in the unrestricted success probability parameter estimates, not the restricted estimates. The reason is that the restricted estimates may be biased to rationalize suboptimal behavior (see Anderson et al. 2025). For example, if the committee puts too much weight on the quantitative GRE score, then its restricted parameter estimate will be biased up since the restricted CCPs assume that the committee is behaving optimally.

For the dynamic model, I rank the applicants in each year by the expected Round 2 social surplus function $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log\left(e^{P(y=S|x,\tilde{z},t,\bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma}\right) dF(\tilde{z}|x,t)$,[13] transition the number of transitioned applicants in the data, rank the survivors by $P(y=S|x,z,t,\bar{k})$, and accept the number of accepted applicants in the data. In contrast, for the static model, I rank the applicants in Round 1 by $P(y=S|x,t,\bar{k})$. After that, the algorithm is the same as the dynamic model's. Once I have determined which applicants each model accepts, I compute their average

[13]For the myopic model, the expected social surplus is $\int_{\tilde{z}} \sigma \log\left(e^{P(y=S|x,\tilde{z},t,\bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma}\right) dF(\tilde{z}|t)$.

success probability, pooled across years, and compare it to the average success probability of the applicants that the committee's rule selects. Determining which applicants the committee's decision rule selects is straightforward. In Round 1, I rank the applicants by $P(d_1 = T|x,t)$, and in Round 2, I rank the survivors by $P(d_2 = A|x,z,t)$. The number of acceptances is the same.

For the matriculation model, the algorithm is more complicated because the purpose of these models is to evaluate the committee's performance based on the applicants who matriculate at the university. Accepting strong applicants who do not matriculate due to getting offers from higher-ranked university is not accomplishing anything. For the matriculation-dynamic model, I rank the applicants in each year by $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log \left(e^{[P(y=S|x,\tilde{z},t,\bar{k}) - P(y_2^*=S)]/\sigma} + 1 \right) P(m=M|x,\tilde{z},t) dF(\tilde{z}|x,t)$, transition the number of transitioned applicants in the data, rank the surviving applicants by $P(y=S|x,z,t,\bar{k})$, and accept them in descending order, each matriculating with probability $P(m=M|x,z,t)$ until the number of matriculants in the data matriculate. I then compute the average success probability of the matriculation set across matriculants and across 500 simulations, and compare it to that of the committee's set. As I explain in Section 4, the best static approximation of the matriculation-dynamic model is the static model, so I also run the matriculation set's algorithm for the static model's optimal decision rule.

Suppose I find deviation gains. Could committees have incorrect beliefs about which applicants are most likely to succeed? Also, if so, can they do a better job admitting applicants by changing the weights they put on some of the variables? If there are gains in the dynamic, static, or myopic models, the matriculation rate of the model's (stronger) acceptance set may be less than that of the committee's (weaker) acceptance set. In that case, the estimated gains would be an upper bound. However, the matriculation model accepts those liklier to matriculate to prevent losing good applicants with weaker observables by rejecting them before reading their files.

In addition, the test can credit the committee only for information I observe. For example, the committee may personally know a recommendation letter writer, or the letter may contain information that algorithms cannot detect, such as exactly how rigorous an applicant's masters program was. The test evaluates both the committee's and the model's acceptance or matriculation set with the same measure, the unrestricted success probability $P(y = S|x,z,t,\bar{k})$, which is estimated from the observables. If the committee acts on a success-predicting variable I do not observe, the measure cannot detect that variable, so any edge the committee's private information provides goes unmeasured. The end result would be an overstated deviation gain.

The protection against this limitation is the comprehensiveness of the data. The application files I observe are the same files the committee read, and from those files, I construct (in addition to the objective variables x) measures of mathematical preparation, the substance of the application essay, the content and tone of the recommendation letters, and the standing of the recommenders. Most importantly, because z includes the committee score, conditioning the success probability on z absorbs much of the residual private information.[14] In this way, I effectively merge human and data-driven approaches. While humans have access to more information and can use their intuition to evaluate less quantifiable characteristics, they may also fall prey to human biases that models do not. Nonetheless, no set of observables is guaranteed to span the committee's information set, so the deviation gains test is best read as conditional on the observed variables closely reconstructing what the committee saw. That said, the realized success rate of the committee's observed matriculation set is no higher than the average success probability of the committee's decision rule's matriculation set, meaning that any private information the committee had did not measurably help the committee attain a better matriculation set.

A separate consideration is sampling error. There are over 30 success probability parameters, so it is possible for the models to overfit the data; that is, what Andrews et al. (2023) call the "winner's curse" in optimization. Ideally, I would test the performance of each model's decision rule on a holdout set. For example, I could use 2015–2018 success probability parameter estimates to select the model's acceptance or matriculation set for 2019, compute the average success probability of that set with 2019 success probability parameter estimates, and repeat this procedure holding out each of 2015, 2016, 2017, and 2018. However, there are too few observations in any single year (for example, 2019) to reliably estimate the success probability parameters. Even sample sizes from three years of training, two years of testing splits are too small. For now, I follow Anderson et al. (2025) by using a randomization test that works effectively on a tennis dataset where the sample sizes for each player pair are large enough for the estimates to be close to the true parameter values. In a future version of this paper, I will present results from a version of the test I am currently developing that is robust against false positives from the winner's curse.

If I assume for simplicity that the model is a two-round model that chooses the same transition set of applicants as the committee, then the following inequality holds:

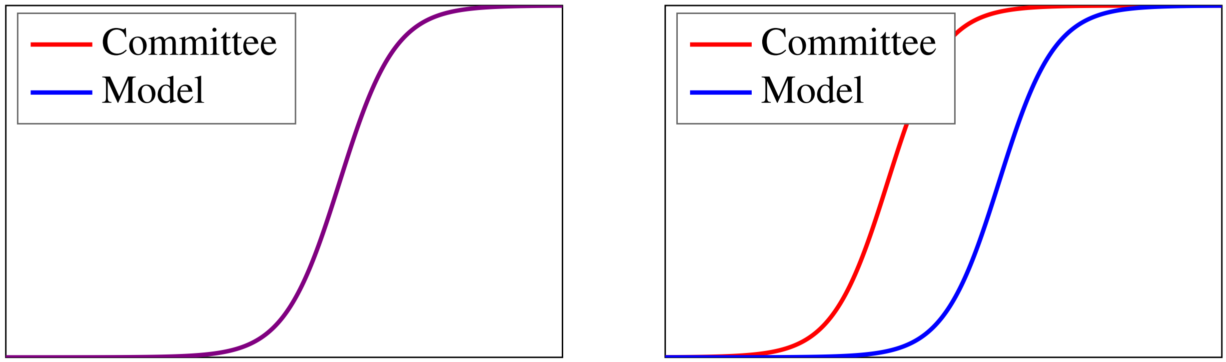
Theorem 4. *Let d^R be the model's decision rule with $\sigma = 0$ in the CCPs, and let d^U be the committee's decision rule. By optimality:*

$$\begin{aligned}
&\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\} \\
&\qquad \geq \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}. \quad (31)
\end{aligned}$$

Proof. See Appendix F.

In this inequality, I am careful to assume that the committee's Round 1 decision rule is the same as the model's. Without this assumption, it is possible in rare cases for the sign to flip if, for

[14]In Table 10 of Appendix B, *Missing Committee Score* survives the lasso with a negative sign (and has an insignificant AME of -0.0277 in Table 2b of Section 3). As such, application variables equal, applicants who were filtered out in Round 1 might have been less likely to succeed. In Round 1, the committee could have observed each applicant's country of origin and undergraduate institution, as well as their letter writers' names. Aside from that, it did not observe anything not in my dataset, at least not until Round 2, where my dataset contains *Committee Score*.



Optimal Committee | Suboptimal Committee

Figure 2: Deviation Gains CDFs

example, the committee favors high-upside applicants in Round 1, and it turns out that favoring such applicants ultimately helps it attain a better set of accepted applicants (see Appendix F).

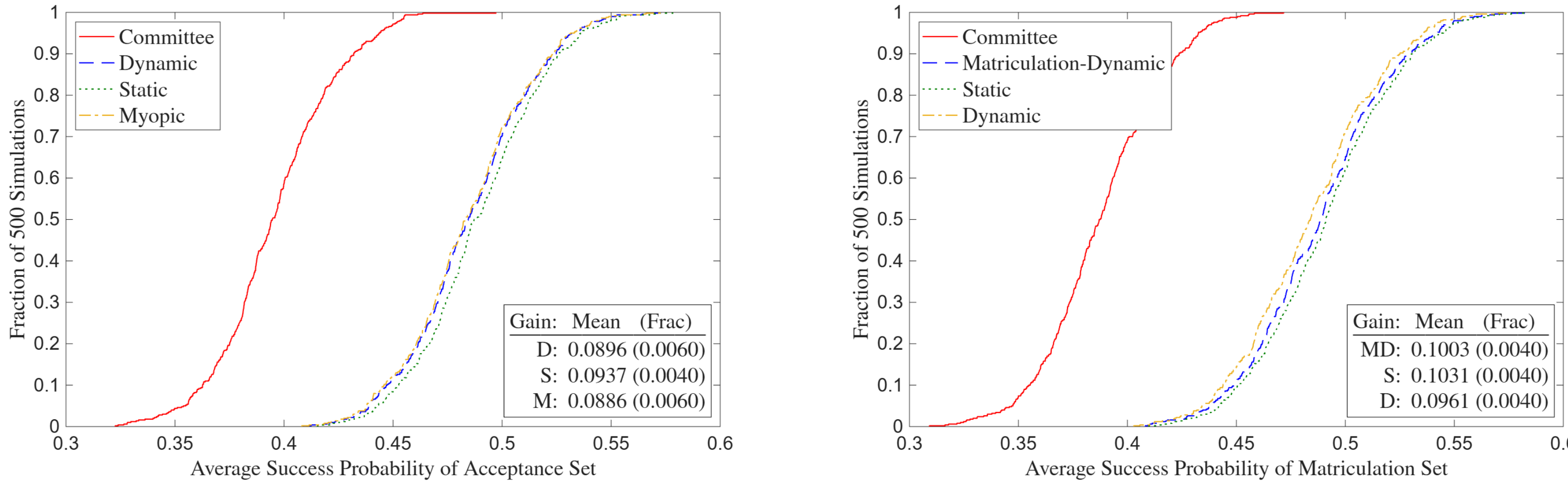
Regardless, for a perturbation $\widetilde{\theta^U_{S,t}}$ about $\hat{\theta}^U_S$'s asymptotic distribution, the sign can flip:

$$\begin{aligned}
&\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \widetilde{\theta^U_{S,t}}\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d^R_{2,n}\left[P_{y_n}\left(\hat{\theta}^U_S\right)\right] = A\right\} \\
&\quad < \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \widetilde{\theta^U_{S,t}}\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d^U_{2,n}\left[P_{y_n}\left(\hat{\theta}^U_A\right)\right] = A\right\}. \quad (32)
\end{aligned}$$

Given that the sign can flip if the committee optimizes or comes close to it, I can use such perturbations of the success parameters to test whether the deviation gains are robust to sampling error. Following Krinsky and Robb (1986), across $B = 500$ perturbations, I plot the CDFs of the average success probabilities of the acceptance and matriculation sets of the dynamic, static, myopic, and matriculation-dynamic models, as well as those of the sets implied by the committee's decision rule. While the acceptance and matriculation sets are selected with the actual parameter estimates, the average success probabilities are computed with the perturbed estimates.

Figure 2 illustrates what the CDFs look like in general. When a committee switches from a suboptimal decision rule to the optimal decision rule, the actual decision rule's CDF shifts right to match the model's CDF. I provide a demonstration on the simulated data, where per Figures 7 and 8 in Appendix D, in the dataset where the committee is nearly optimal, the models' CDFs are very close to the committee's. However, in the dataset where the committee is not optimizing, the models' CDFs clearly stochastically dominate the committee's CDF.

In addition, I prove that with a large number of perturbations, the fraction of perturbations in which the committee's set has a weakly higher average success probability than the model's set is a p-value for the null hypothesis that the committee's set's average success probability is at least as high as the model's. This proof defines a p-value in the standard way; if the null hypothesis is true, the p-value is the probability of getting a positive deviation gain at least as large as the



(a) Acceptance Set

(b) Matriculaton Set

Figure 3: Deviation Gains

one I observe. As the test should, it does not strongly reject equality of the models' CDFs vs. the committee's CDF in the near-optimal simulated dataset, but it always rejects equality of the CDFs in the suboptimal dataset. Formally, the theorem that the fraction of perturbations is a p-value is:

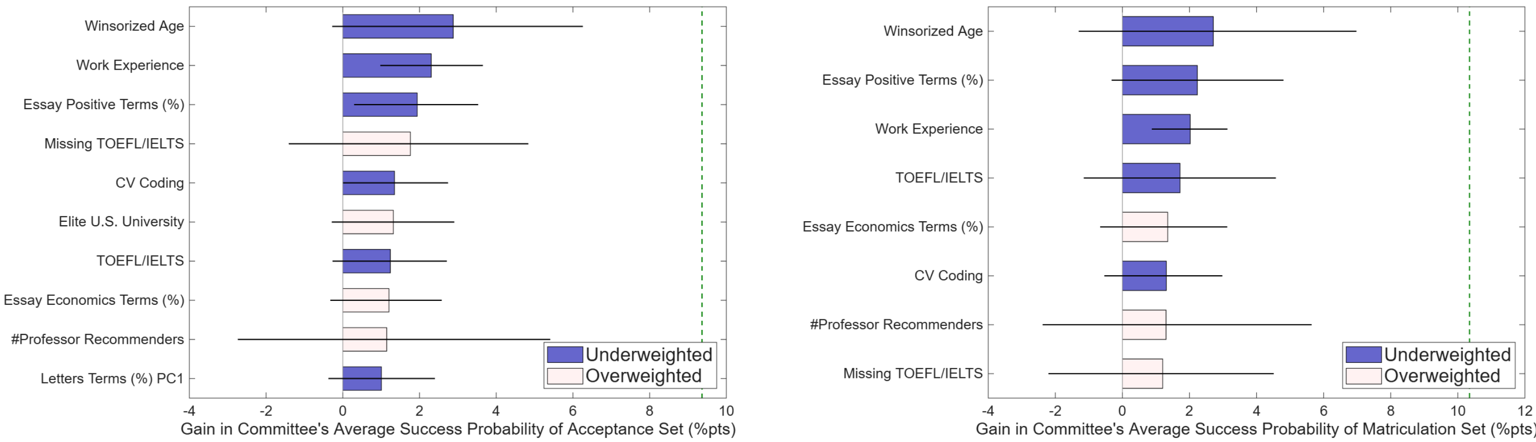
Theorem 5. *For large B, the fraction of* $\tilde{\theta}^U_{S,b}$ *with gain* $g(\tilde{\theta}^U_{S,b}) \leq 0$ *is a p-value for* $H_0\colon g_0 \leq 0$.

Proof. See Appendix F.

Figure 3 reveals that the average success probability of the model's acceptance set is about 9 percentage points higher than that of the committee's, and the average success probability of the model's matriculation set is about 10 percentage points higher. There is little variation across models, though as expected, the myopic model is slightly worse than the dynamic model for the acceptance set, and the dynamic model is slightly worse than the matriculation-dynamic model for the matriculation set. The p-values are also statistically significant at the 1% level.

Next, I determine which variables may be responsible for the gains. For simplicity, I use the static models for this attribution exercise. I start with the Rounds 1 and 2 conditional choice probability parameter weights for each variable. One variable at a time, I hold all the other variables' weights fixed and choose the Rounds 1 and 2 weights that maximize the average success probability of the acceptance or matriculation set. Once those weights are chosen, I recalculate the average success probability using the 500 perturbations and plot the results in Figure 4. The bar shows the average of those average success probabilities across the perturbations, and the whiskers extend to the 5th and 95th percentiles.[15] Also, the bars are dark blue for previously-underweighted variables where the reweighting increases their average in the acceptance or ma-

[15]Unlike a 95% confidence interval, the whiskers do not extend to the 2.5th and 97.5th percentiles, as the null hypothesis of the committee's average success probability being at least as high as the model's is one-sided.



(a) Acceptance Set

(b) Matriculaton Set

Figure 4: Deviation Gains Attribution

triculation set. In contrast, the bars are pink for previously-overweighted variables where the reweighting decreases their average. An objective or subjective variable qualifies for inclusion in the figure if its bar reaches at least one percentage point. For each of the variables in Figure‘4, Table 19 of Appendix 4 contains their average values for the full sample, the committee's acceptance or matriculation set, the single-variable optimized set, and the joint optimized set.

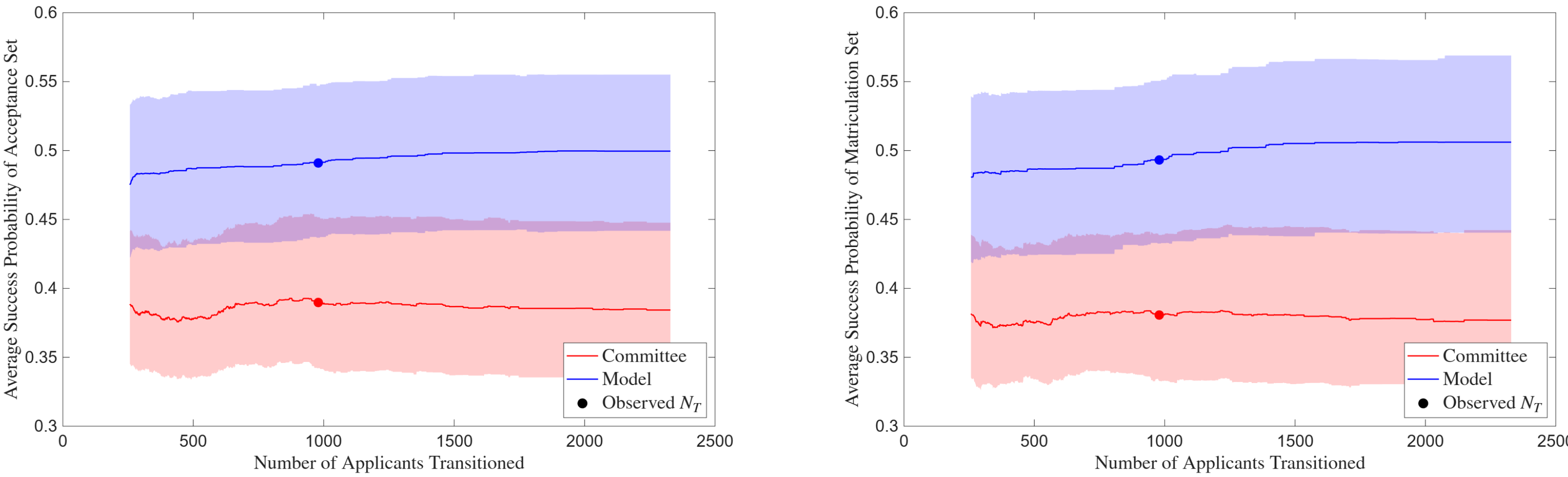
The bars are similar for both the acceptance and matriculation sets; all bars for the matriculation set are present for the acceptance set, only two bars are present for the acceptance set only, and no bars change colors across sets. Starting with the matriculation set, the bars fall into four categories. The first and most pronounced category is experience; older applicants and those with work experience are more likely to succeed, and the committee's set does not give them enough credit. The second category is English proficiency, as applicants who score low on the TOEFL/IELTS are less likely to succeed. The third category is the subtext of the application essays. Applicants with a high percentage of positive words in their essays are more likely to succeed, and those with a high percentage of economics words (i.e. words like microeconomics, macroeconomics, econometrics, etc.) are less likely to succeed. Finally, the fourth category relates to applicants with more of an industry background. In particular, applicants with a high number of coding languages on their CVs are less likely to be accepted despite not being less likely to succeed. Finally, *Elite U.S. University* shows up for the acceptance set but not the matriculation set. Applicants from an elite U.S. university are more likely to be accepted and not more likely to succeed.[16] However, they are also less likely to matriculate if accepted, so accepting them in greater numbers does not hurt the matriculation set's average success probability.

[16] Bai et al. (2022) find that attending an Ivy Plus undergraduate institution predicts success. However, unlike their university, this paper's university is less likely to receive applications from the strongest Ivy Plus students.

There are two additional points worth making. First, as expected, there is substantial overlap between the variables that survive the lasso in the sixth column of Table 10 in Appendix B and the variables that show up in Figure 4. Second, the AMEs in Table 2b of Section 3 hint at the attribution results. For example, *Work Experience* has an AME of 0.0140 for *Accept* vs. 0.0358 for *Success*, and Figure 4 shows that it is underweighted. However, there are other variables in Table 2b whose *Accept* AMEs do not line up with their *Success* AMEs, yet are neither under nor overweighted. For example, *Female* has a positive *Accept* AME and *International Asian* has a negative *Accept* AME, both variables have insignificant *Success* AMEs, and neither variable shows up in Figure 4. The reason is that there are enough strong applicants with *Female* $= 1$ and *International Asian* $= 0$ that the committee's decision rule does not affect the average success probability of the acceptance or matriculation sets. So if these two *Accept* AMEs resulted from a diversity objective, that objective was nonetheless also consistent with a success probability-maximizing objective. In contrast, *CV Coding* has a negative *Accept* AME, an insignificant *Success* AME, and it shows up as underweighted in Figure 4. There is no way to know from reading Table 2b that it would work out that way, which demonstrates why a model is necessary to relate acceptance AMEs to success AMEs when drawing any conclusions about optimality.

Table 2b also shows that *GRE Quantitative Percentile*'s AME is positive and significant until I control for *Program Rank* and *Missing Program Rank*, at which point it is no longer significant. Controlling for *Program Rank* is necessary since the quantitative GRE may only be positively correlated with success because a high percentile is often required to get into a good PhD program, which in turn is often required to attain a good placement. In that case, it would be incorrect to recommend selecting based on the quantitative GRE. However, *GRE Quantitative Percentile*'s insignificant AME does not necessarily mean that selecting on it is incorrect. For instance, this university filters out applicants with low percentiles. It is possible that despite the insignificant AME, such applicants would not do well at this university. The reason is that there are very few applicants at top PhD programs with low percentiles, so the data cannot identify well what would happen if someone with a low percentile attended a top program. Also, Figure 4 shows that there is no evidence that the committee is overweighting *GRE Quantitative Percentile*.

Finally, in Figure 5, I present the results from a counterfactual in which I allow N_T, which is the number of files the committee chooses to transition to Round 2, to vary within each year. Assuming the committee reads every accepted applicant's file, the minimum value of N_T is N_A. And the maximum value of N_T is the total number of applicants N. This counterfactual seeks to answer the question of whether reading more files would have helped the committee attain an acceptance or matriculation set with a higher average success probability. I find little evidence that reading more files would have helped. The committee's line is effectively flat across files, and while the model's line is slightly increasing and concave (indicating positive but diminishing returns to reading more files), the increase in files read all the way from N_A to N_T only slightly increases the average success probabilities. Therefore, reading more files is less effective than reweighting the information attained from the objective variables and the files already read.



(a) Acceptance Set

(b) Matriculaton Set

Figure 5: Benefit of Reading More Files

6 Conclusion

This paper studies how decision-makers select among options whose true quality is uncertain and costly to evaluate. Graduate admissions is a natural setting for such a study, as it features a low-cost filtering process followed by a high-cost review of the application files that survive the filter. Because not all accepted applicants matriculate, the committee must also consider matriculation probabilities when making admissions decisions. By structurally modeling the committee's decision problem, I develop the first framework in the economics graduate admissions literature that links application information to admissions decisions and post-program success. This framework makes optimality testable, and gains from deviating to the optimal decision rule quantifiable.

Before estimating structural modes, I start with regressions of acceptance, matriculation, and success dependent variables on information in application files from a research university's economics PhD program in 2015–2019. I find that different independent variables best predict each dependent variable. The 0 to 10 score the committee gives every applicant whose file it read best predicts acceptance, and attending the university's open house best predicts matriculation if accepted. I use program completion and placement rank to measure success, and my primary success measure is a binary variable that equals 1 if the applicant completed a PhD program and attained a top 100 or top 100-equivalent placement. The best predictor of success is the rank of the PhD program the applicant attended, though other variables like age, work experience, and TOEFL/IELTS scores also predict success. To avoid selection bias, I use publicly-available information from the Internet to track PhD program and placement oucomes for all applicants, not just those who matriculated at this university. I also regress acceptance on application file variables using a second dataset of this university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023, and I find that economics admissions is more data-driven, based on variables such as GRE scores, than admissions in the other programs.

There are papers in the graduate admissions literature dating back to Krueger and Wu (2000) that qualitatively compare the determinants of acceptance to those of success, but I show that while such comparisons can provide useful information, they are insufficient to determine whether an admissions committee is optimizing. If I were to test whether the average marginal effects (AMEs) of variables in application files, such as GRE scores, are the same in an acceptance regression vs. a success regression, I would not be testing optimality. As I show at the end of Section 3, equal AMEs are neither necessary nor sufficient for optimization. Therefore, any optimality tests must be based on structural models of the committee's selection problem that account for evaluation costs and multi-stage decision-making. For this reason, I develop and estimate such models of the committee's problem in order of increasing complexity.

I start with two-round dynamic, static, and myopic models based on cutoff rules. In these models, there is a Round 1 filter based on only objective variables like GRE scores, followed by a Round 2 final decision based on objective and subjective variables, such as those constructed by textual analysis of recommendation letters. I then develop a two-round matriculation-dynamic model, in which not every accepted applicant matriculates, and the committee considers matriculation probabilities in Round 1. The reason for doing so is to ensure that the committee can fill its cohort with applicants who do not have particularly weak subjective variables. I derive all these models from dynamic portfolio choice problems that the committee faces, and in Section 4 and Appendix E, I solve for the cutoff rule solutions to those problems. I estimate the models on this university's data and back out the committee's Rounds 1 and 2 cutoffs in each year.

After estimating the structural models, I use the estimates to compare each two-round model's chosen set to the committee's chosen set. For each model, I simulate the acceptance and matriculation processes, the latter based on estimated matriculation probabilities, to attain two acceptance sets and two matriculation sets: one each for the committee and one each for that particular model. I select the model's acceptance and matriculation sets using the unrestricted success probability parameter estimates, but I calculate the average success probability of each set using perturbations of the estimates about their asymptotic distribution. I also prove that the fraction of perturbations where the model's set's average success probability does not exceed the committee's is an asymptotically valid p-value for the null hypothesis that the model's acceptance or matriculation set's average success probability is no greater than that of the committee.

I find that the model's acceptance and matriculation sets exhibit deviation gains of 9–10 percentage points relative to the committee's sets. For example, the average success probability of the committee's matriculation set is 37%, whereas that of the model's set is 47%. These gains may be attributed to experience (in the form of *Age* and *Work Experience*), English proficiency (*TOEFL/IELTS* scores), application essay subtext (a higher rate of positive terms predicts success, and a higher rate of economics jargon terms predicts failure), and industry backgrounds (*CV Coding* predicts rejection but not failure). But it is important to note that a 9–10 percentage point gain would be unlikely in practice. The determinants of success may have changed since 2015–2019, and the randomization test is not fully robust in terms of out-of-sample performance, a phenomenon that Andrews et al. (2023) denote the "winner's curse." I am developing a version of this test robust to the winner's curse and will implement it in a future version of this paper.

I also run a counterfactual in which I allow the number of files n the committee transitions to Round 2 to vary. The goal is to determine whether reading more files would increase the average success probability of the acceptance or matriculation set. I find no evidence that it would, as the committee's curve is essentially flat in n. However, in a setting where the curve is increasing and concave, the same counterfactual would support a revealed-preference estimate of the dollar value the committee places on success, in the spirit of the value of a statistical life literature, which infers the value of a life from the wage premium workers demand for bearing risk.

Let $\overline{P}(n)$ be the average success probability of the acceptance set in a given year when n files are transitioned. Since the number of acceptances N_A is fixed, the expected number of successes is $N_A\overline{P}(n)$. For the matriculation set version of this counterfactual, N_A becomes N_M. One committee member reads each file, so if they spend h hours on it at an hourly wage of w, then transitioning one more file costs hw. If the committee values each success at V dollars (in present value, since success is realized years after admission) and chooses N_T to maximize $VN_A\overline{P}(n) - hwn$, then at an interior optimum, the last file read must just pay for itself. Namely, $VN_A\overline{P}'(N_T) = hw$, or $V = hw/(N_A\overline{P}'(N_T))$. The denominator is the number of additional expected successes the last file bought, so V is the cost the committee was willing to incur per additional expected success. Also, the cost's wage-free counterpart $h/(N_A\overline{P}'(N_T))$ is in hours per additional expected success. Because this is a marginal calculation, I must calculate the slope of the curve at N_T. Since the curve is a step function, I would fit it to a smooth concave function before differentiating. I could also run the calculation in reverse. Given an outside estimate of V, such as the value of an additional top placement to a department's ranking, comparing $VN_A\overline{P}'(N_T)$ to hw tests whether the committee reads too few or too many files without assuming that N_T is optimal.

In addition to its contributions to the graduate admissions literature and to literatures on dynamic selection and costly evaluation, this paper contributes to the literature on synthetic data and privacy protection. Recent economics papers use synthetic data provided by the U.S. Census Bureau, but this paper is the first to design and implement a comprehensive synthetic data protocol from the construction of the synthetic data to the final estimation results. As a result, it serves as a template for researchers who wish to work with confidential data that they are not allowed to access directly, which can help them make discoveries they could not have otherwise.

More broadly, this paper provides a tractable framework for studying selection under costly evaluation. By doing so, it advances the graduate admissions literature from identifying variables that predict acceptance and success to testing whether decisions were optimal conditional on time and effort constraints. This study is based on a single institution, so the specific empirical results reflect that setting. However, the framework applies broadly to environments where due to evaluation costs, decision-makers must employ a multi-stage process to select among their available options. Such environments range from hiring committees to clinical trials of medicines. That said, admissions committees may pursue objectives beyond maximizing success probabilities, such as diversity or research field balance. The framework can be extended to incorporate such objectives directly into the committee's problem. In addition, a worthwhile avenue for future research would be collecting data from multiple universities to model the applicant's decision problem and compute equilibrium admissions outcomes for applicants across universities.

Ultimately, the methodology in this paper can help decision-makers make better selection decisions based on *ex-ante* information when faced with uncertainty and costly evaluation. Crucially, decision-makers may be able to improve selection outcomes without having to increase evaluation effort. Instead, they can use the paper's framework to refine how they use that information, just as data analytics has helped organizations, such as the Oakland Athletics Major League Baseball team as described in the book *Moneyball* (Lewis 2004), with their decision-making. Consequently, they can do the best possible job selecting from their available options.

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Appendices

A Variables, Rankings, and Dictionaries

This table contains a list of the variables in this paper. Variables available only for the economics program are bold, and those available for all but economics are italic. *#Math Courses* and *Advanced Math* became objective variables starting in 2020 due to a change in the university's online application system. PhD program rankings are from U.S. News, and placement rankings are from IDEAS/RePEc. The dictionaries for the subjective independent variables are from Madera, Hebl, and Martin (2009), Young and Soroka (2012), and Bai et al. (2022). These dictionaries list word roots (e.g. diligen*), and I match words against the roots directly. A word counts for a dictionary if it begins with the same letter as, and contains, one of that dictionary's roots, which credits inflected and derived forms (e.g. diligent, diligence) to their root. Each dictionary is counted independently against the full document, a word counts at most once per dictionary, and a word containing roots from several dictionaries counts in each of those dictionaries.

Table 6: Variables

(a) Dependent

Admission	Success
$\mathbb{1}$(Transition)	**$\mathbb{1}$(Complete Program)**
$\mathbb{1}$(Accept)	**$\mathbb{1}$(Academic Placement)**
$\mathbb{1}$(Waitlist)	**$\mathbb{1}$(Government Placement)**
$\mathbb{1}$(Open House)	**$\mathbb{1}$(Consulting Placement)**
$\mathbb{1}$(Matriculate)	**$\mathbb{1}$(Tech Placement)**
$\mathbb{1}$(Attend Program)	**$\mathbb{1}$(Other Placement)**
Program Rank	**$\mathbb{1}$(No Placement)**
	Placement Rank
	$\mathbb{1}$(Top 100 Placement)

(b) Objective Independent

Demographic	Performance
Year	GRE Verbal Percentile
$\mathbb{1}$*(Covid Year)*	GRE Quantitative Percentile
Winsorized Age	GRE Analytic Percentile
$\mathbb{1}$(Female)	Undergraduate GPA
$\mathbb{1}$(U.S.)	TOEFL/IELTS
$\mathbb{1}$*(U.S. African/Hispanic/Native)*	$\mathbb{1}$(Elite U.S. University)
$\mathbb{1}$*(U.S. Asian)*	$\mathbb{1}$(Elite U.S. Liberal Arts College)
$\mathbb{1}$*(U.S. Other)*	$\mathbb{1}$(Other U.S. University/College)
$\mathbb{1}$(International Asian)	$\mathbb{1}$(Elite Foreign University)
$\mathbb{1}$(International Other)	$\mathbb{1}$(Other Foreign University)
	$\mathbb{1}$(Graduate Degree)
	$\mathbb{1}$(Work Experience)
	#Professor Recommenders
	#Prolific Recommenders
	Concentration Dummy Variables

(c) Subjective Independent

Score/Recommendation Letters	Supplement/CV/Essay
Committee Score	**#Math Courses**
Letters Length	**$\mathbb{1}$(Advanced Math)**
Positive Terms (%) (YS 2012)	**CV Length**
Negative Terms (%) (YS 2012)	**CV Coding**
Standout Terms (%) (BEMS 2022)	**CV $\mathbb{1}$(Publication/Working Paper)**
Ability Terms (%) (BEMS 2022)	**Essay Length**
Research Terms (%) (BEMS 2022)	**Essay $\mathbb{1}$(Specific Professor)**
Grindstone Terms (%) (BEMS 2022)	**Essay Economics Terms (%)**
Teaching Terms (%) (BEMS 2022)	**Essay Research Terms (%)**
Communal Terms (%) (BEMS 2022)	**Essay Positive Terms (%)**
Agentic Terms (%) (MHM 2009)	**Essay Negative Terms (%)**
	Field Dummy Variables

Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)

- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

Academia-Equivalent Rankings (Consulting/Technology/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

Dictionaries (Recommendation Letters)

- **Positive/Negative Words:** Young and Soroka (2012)'s Lexicoder Sentiment Dictionary
- **Standout, Ability, Research, Grindstone, Communal Words:** Bai et al. (2022)
- **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera et al. 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
- **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop. (Note: I constructed this dictionary because it is unavailable in Bai et al. 2022.)

Dictionaries (Application Essays/CVs)

- **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
- **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
- **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
- **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

A.1 Academia-Equivalent Prompt and Link to Rankings

I am writing a research paper that entails using data in application files to predict success in economics PhD programs, where success is defined as the quality of the first post-PhD job placement. For academic jobs, I use RePEc rankings. For non-academic jobs, I will follow the same procedure as Krueger and Wu (2000):

"Another issue concerns the ranking of the various job placements. For our analysis, we assumed that one goal of admissions and graduate training in economics is to place students in leading research jobs, although we recognize that different departments have different objectives. In ranking academic institutions, we relied primarily on the ranking of the top 100 economics departments based on faculty publications in elite journals produced by Scott and Mitias (1996). Students who were placed in business school jobs were given a rank equal to their university's economics department ranking plus five. (A lower rank signified a more prestigious job placement.) The World Bank, IMF, and Federal Reserve Board received a ranking equivalent to the 40th best economics department. Consulting jobs (e.g., NERA, Abt, DRI) were given a rank equal to the 120th best economics department. Finally, applicants who could not be found were assigned a rank of 150, the worst rank we gave, and treated as censored observations in much of the analysis."

I am attaching an Excel file with the RePEc rankings, as well as job titles for Government, Consulting, Tech, and Other Non-Academia. In each of the latter four sheets, please fill in the "Academia-Equivalent Ranking" column with the ranking you deem most appropriate. Keep it as consistent as possible with Krueger and Wu, meaning that the best non-academic placements should receive a ranking of 40. Krueger and Wu deem those placements to be the World Bank, IMF, and Federal Reserve Board (that is, the Board of Governors and not the regional banks). However, because the economics profession has changed since 2000, you can give additional placements a ranking of 40. Limit any rankings of 40 to the Government sheet and the "Tech Research Labs" part of the Tech sheet. Be very strict about which, if any, additional placements you give a 40, since Krueger and Wu gave only three 40s, each of which were to very prestigious placements. You can give at most three additional 40s in Government, and you can give at most three 40s in Tech.

Regarding rankings other than 40 that you can give, you can give 80, 120, and 135. The logic behind 80 is that it is halfway between 40 and 120, and the logic behind 135 is that it is halfway between 120 and 150. Do not give any placements a rank of 150; 150 is reserved for applicants who cannot be found. 40 is for placements in the World Bank tier, 80 is for placements between the World Bank and NERA tiers, 120 is for placements in the NERA tier, and 135 is for placements that are below the NERA tier. If you ever give a Consulting or Other Non-Academia placement a ranking better than 120, make sure you have a very sound reason for doing so because Krueger and Wu capped rankings for consulting placements at 120. Please post all your academia-equivalent rankings in the chat as well, as I will save a screenshot of the chat.

A.2 Synthetic Data Example

As a demonstration of synthetic data, I use a Gaussian copula to simulate 1,000 observations of the following independent variables: *GRE Quantitative Percentile* (continuous), *Gender* (binary), *Demographic* (categorical with three demographics), *Advanced Math* (binary), and *Committee Score* (continuous). Then I use the standard logistic function to simulate binary dependent variables for transitioning past the first-round filter, getting waitlisted, getting accepted into the program, matriculating at the program if accepted, and obtaining a successful placement above a threshold ranking. I also generate the (continuous) rank of the PhD program each applicant attends, which is constant for the applicants with $Matriculate = 1$ and equals 150 for applicants who do not attend a PhD program (for these applicants, $Success = 0$). The top frame of Figure 6 contains the first ten observations of this simulated dataset. *Committee Score* is missing for all applicants who did not survive the first-round filter, which is a feature of the actual data.

I then pass this dataset through another Gaussian copula and use that copula to generate a synthetic dataset, the first ten observations of which are in the bottom frame of the figure. The means, variances, and correlations of the variables are similar in both datasets, with only the correlations in the synthetic dataset slightly attenuated. However, none of the rows of the simulated dataset are present in the synthetic dataset. In addition, important statistical properties; such as *Accept* equaling 1 only when *Transition* equals 1, *Demographic 1* and *Demographic 2* never both equaling 1, and *Committee Score* missing when *Transition* equals 0; are preserved.

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
1	1	0	0	0	89	0	0	0	0	5	150
1	0	1	0	1	96	0	1	0	0	6	23
0	0	0	0	0	88	0	1	0	1	·	150
0	0	0	0	0	80	1	1	0	1	·	150
0	0	0	0	1	99	0	1	0	0	·	29
0	0	0	0	0	81	0	0	1	1	·	23
1	0	1	1	1	92	0	1	0	0	3	37
1	1	1	1	1	93	0	0	0	1	3	37
0	0	0	0	0	83	1	0	1	0	·	48
1	1	0	0	0	87	0	1	0	1	5	34

$\Downarrow$

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
0	0	0	0	0	77	0	0	0	1	·	30
0	0	0	0	1	89	0	0	0	0	·	25
0	0	0	0	0	85	1	0	0	0	·	150
0	0	0	0	0	76	0	1	0	1	·	51
1	1	0	0	0	88	0	1	0	0	8	150
1	0	1	0	0	88	0	0	0	1	5	150
1	1	0	0	0	84	0	0	1	0	2	20
0	0	0	0	0	82	0	1	0	0	·	35
0	0	0	0	0	70	1	0	0	0	·	150
1	1	0	0	1	93	0	1	0	0	5	57

Figure 6: Synthetic Data Example

B Additional Regression Results

Table 7: Summary Statistics

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Transition	0.4204	0.4937	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
Waitlist	0.3324	0.4712	0.7686	0.4220	0.0743	0.2628	0.0896	0.2877
Accept	0.1103	0.3134	0.2625	0.4402	1.0000	0.0000	1.0000	0.0000
Matriculate	0.0339	0.1811	0.0807	0.2725	0.3074	0.4623	1.0000	0.0000
Open House	0.0353	0.1847	0.0817	0.2741	0.3267	0.4702	0.5522	0.5010
Attend Program	0.6024	0.4895	0.6752	0.4685	0.7704	0.4214	1.0000	0.0000
Complete Program	0.5578	0.4968	0.6210	0.4854	0.6654	0.4728	0.6709	0.4729
Top 100 Placement	0.1971	0.3979	0.2472	0.4316	0.3074	0.4623	0.3418	0.4773
Program Rank	88.9133	57.4823	78.5844	57.1361	63.3828	54.1845	37.0000	0.0000
Missing Program Rank	0.4199	0.4937	0.3422	0.4747	0.2490	0.4333	0.0000	0.0000
Placement Rank	127.7505	34.9306	123.0222	37.2570	117.2696	40.5746	117.4430	37.5728
Missing Placement Rank	0.4444	0.4970	0.3830	0.4864	0.3346	0.4728	0.3291	0.4729
2016	0.2052	0.4040	0.1696	0.3754	0.1790	0.3841	0.2025	0.4045
2017	0.2014	0.4011	0.1788	0.3833	0.1829	0.3873	0.1772	0.3843
2018	0.1928	0.3946	0.2686	0.4435	0.2062	0.4054	0.2532	0.4376
2019	0.2027	0.4021	0.2084	0.4064	0.2179	0.4136	0.2152	0.4136
Winsorized Age	25.9910	3.1479	26.2635	3.1823	26.0817	3.0652	26.3291	3.2806
Female	0.3912	0.4881	0.4147	0.4929	0.4397	0.4973	0.4177	0.4963
U.S.	0.2357	0.4245	0.2380	0.4261	0.2996	0.4590	0.3038	0.4628
International Asian	0.4809	0.4997	0.5230	0.4997	0.3424	0.4754	0.4304	0.4983
GRE Verbal Percentile	68.0129	22.5869	76.8421	18.3703	79.5703	18.3047	76.9744	19.4041
GRE Quantitative Percentile	88.4550	11.6330	93.1362	5.4135	93.5391	4.4614	92.3077	5.7554
GRE Analytic Percentile	50.0613	27.5744	56.3653	26.5466	60.3398	26.0880	57.4615	26.6481
Undergraduate GPA	3.6307	0.3026	3.6645	0.3003	3.7227	0.2610	3.7068	0.2880
TOEFL/IELTS	7.4376	0.5587	7.5641	0.5834	7.7011	0.6667	7.4464	0.6137
Elite U.S. University	0.1267	0.3327	0.1736	0.3790	0.2257	0.4188	0.1772	0.3843
Elite U.S. Liberal Arts College	0.0296	0.1696	0.0347	0.1832	0.0428	0.2028	0.0633	0.2450
Elite Foreign University	0.0386	0.1928	0.0490	0.2160	0.0350	0.1842	0.0253	0.1581
Other Foreign University	0.4049	0.4910	0.4188	0.4936	0.3619	0.4815	0.4430	0.4999
Missing Undergraduate Institution	0.2203	0.4145	0.1910	0.3933	0.1984	0.3996	0.1266	0.3346
Graduate Degree	0.5067	0.5001	0.5730	0.4949	0.5136	0.5008	0.5190	0.5028
Work Experience	0.5247	0.4995	0.5894	0.4922	0.6770	0.4685	0.7468	0.4376
#Professor Recommenders	2.6230	0.7371	2.6302	0.7165	2.5992	0.7118	2.5696	0.7104
#Prolific Recommenders	1.2602	1.0686	1.3912	1.0840	1.4825	1.0424	1.3038	1.0664
Missing GRE	0.0335	0.1800	0.0102	0.1006	0.0039	0.0624	0.0127	0.1125
Missing Undergraduate GPA	0.7853	0.4107	0.7487	0.4340	0.6770	0.4685	0.6835	0.4681
Missing TOEFL/IELTS	0.6076	0.4884	0.5935	0.4914	0.6615	0.4741	0.6456	0.4814
Missing Winsorized Age	0.0009	0.0293	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Committee Score	6.4070	1.8701	6.4070	1.8701	8.5003	0.8151	8.4854	0.7157
#Math Courses	5.7637	1.8154	5.8173	1.7904	5.6992	1.8640	5.6585	1.6219
Advanced Math	0.4860	0.5004	0.4909	0.5007	0.5680	0.4973	0.4651	0.5047
Letters Length/100	8.3474	3.5749	8.5333	3.8263	10.0514	4.2864	9.7088	3.4876
Letters Positive Terms (%)	11.9770	2.2993	11.9976	2.2077	12.2167	1.9255	12.5110	2.0904
Letters Negative Terms (%)	5.7286	1.3427	5.7216	1.3318	5.7446	1.2313	5.8938	1.2871
Letters Standout Terms (%)	0.6241	0.4020	0.6241	0.4029	0.7151	0.4509	0.7077	0.5109
Letters Ability Terms (%)	1.3326	0.6239	1.3457	0.5964	1.3796	0.5620	1.4334	0.6037
Letters Research Terms (%)	3.4837	1.3319	3.5208	1.3423	3.8548	1.1984	3.8395	1.2121
Letters Grindstone Terms (%)	1.8189	0.7734	1.8674	0.7808	1.9975	0.7588	2.1600	0.8093
Letters Teaching Terms (%)	1.7379	0.7710	1.7489	0.7759	1.7275	0.7536	1.8976	0.7577
Letters Communal Terms (%)	0.5946	0.3890	0.6019	0.3954	0.5494	0.3234	0.5630	0.3875
Letters Agentic Terms (%)	0.4768	0.3150	0.4842	0.3229	0.5039	0.3222	0.5597	0.3314
Letters Terms (%) PC1	-0.0579	1.6183	0.0000	1.5768	0.2176	1.3817	0.5697	1.5386
Letters Terms (%) PC2	-0.0271	1.1177	0.0000	1.1446	0.1002	1.1774	0.1699	1.2823
CV Length/100	4.3532	2.4237	4.3982	2.4758	4.9641	2.7175	4.8261	2.3473
CV Coding	2.8321	2.1135	2.7931	2.1000	2.8618	2.1129	2.5122	1.6450
CV Publication/Working Paper	0.2518	0.4346	0.2571	0.4377	0.3659	0.4836	0.3902	0.4939
Essay Length/100	4.5099	1.5741	4.5173	1.5805	4.7488	1.6373	4.9816	1.7932
Essay Specific Professor	0.4822	0.5003	0.5219	0.5003	0.5691	0.4972	0.6512	0.4822
Essay Economics Terms (%)	9.7284	2.6219	9.7216	2.6635	9.5303	2.7324	9.8941	3.2608
Essay Research Terms (%)	5.7975	2.0501	5.7402	2.0168	5.7313	1.8797	5.5231	1.8432
Essay Positive Terms (%)	11.8293	2.6518	11.7324	2.6650	11.4734	2.4592	11.3169	2.4455
Essay Negative Terms (%)	6.2844	1.9056	6.3233	1.9712	6.2889	2.0860	6.2651	1.9436
Missing Committee Score	0.5844	0.4929	0.0112	0.1055	0.0195	0.1384	0.0633	0.2450
Missing Application Pdf	0.8154	0.3881	0.6650	0.4722	0.5136	0.5008	0.4557	0.5012
Missing #Math Courses	0.8201	0.3842	0.6701	0.4704	0.5214	0.5005	0.4810	0.5028
Missing Letters	0.8175	0.3863	0.6691	0.4708	0.5136	0.5008	0.4557	0.5012
Missing CV	0.8210	0.3835	0.6742	0.4689	0.5214	0.5005	0.4810	0.5028
Missing Essay	0.8192	0.3849	0.6731	0.4693	0.5214	0.5005	0.4557	0.5012
Observations	2329		979		257		79	

Table 8: Correlation Matrix

	Trans	Wait	Acc	Mat	OH	Att	Com	T100	ProR	MProR	PlaR	MPlaR	2016	2017	2018	2019	Age	Fem	US	IntAsn	GREV	GREQ
Transition	1																					
Waitlist	0.8083	1																				
Accept	0.4136	-0.1908	1																			
Matriculate	0.2200	-0.0994	0.5320	1																		
Open House	0.2192	-0.1104	0.5496	0.5400	1																	
Attend Program	0.1266	0.0767	0.1209	0.1522	0.0833	1																
Complete Program	0.1085	0.0889	0.0763	0.0427	0.0154	0.9124	1															
Top 100 Placement	0.1073	0.0778	0.0977	0.0682	0.0498	0.4025	0.4412	1														
Program Rank	-0.1531	-0.0909	-0.1565	-0.1693	-0.0999	-0.8635	-0.7942	-0.4106	1													
Missing Program Rank	-0.1341	-0.0860	-0.1219	-0.1594	-0.0856	-0.9548	-0.8749	-0.4018	0.9044	1												
Placement Rank	-0.1153	-0.0802	-0.1057	-0.0553	-0.0666	-0.5176	-0.5673	-0.9004	0.4999	0.5032	1											
Missing Placement Rank	-0.1051	-0.0835	-0.0778	-0.0435	-0.0164	-0.9084	-0.9957	-0.4431	0.7924	0.8708	0.5698	1										
2016	-0.0752	-0.0883	-0.0229	-0.0013	-0.0126	-0.0194	0.0073	-0.0005	0.0077	0.0178	-0.0069	-0.0073	1									
2017	-0.0480	-0.0732	-0.0162	-0.0113	0.0162	0.0076	0.0138	0.0096	-0.0025	-0.0064	-0.0040	-0.0138	-0.2552	1								
2018	0.1637	0.1695	0.0120	0.0287	0.0009	0.0101	0.0078	0.0096	-0.0118	-0.0233	-0.0076	-0.0034	-0.2483	-0.2454	1							
2019	0.0121	-0.0050	0.0133	0.0058	-0.0045	0.0145	-0.0565	-0.0081	0.0123	0.0017	0.0166	0.0543	-0.2562	-0.2532	-0.2464	1						
Winsorized Age	0.0738	0.0157	0.0102	0.0201	0.0122	-0.0260	-0.0179	0.0398	0.0474	0.0304	-0.0148	0.0229	0.0102	-0.0091	-0.0128	0.0160	1					
Female	0.0411	0.0255	0.0350	0.0102	0.0328	-0.0068	-0.0179	-0.0189	0.0074	0.0168	0.0007	0.0198	0.0066	-0.0163	0.0142	0.0358	-0.1468	1				
U.S.	0.0046	-0.0393	0.0530	0.0301	0.0318	-0.1069	-0.1165	-0.0310	0.0900	0.1075	0.0538	0.1141	0.0083	0.0263	-0.0073	-0.0409	-0.1537	-0.0886	1			
International Asian	0.0717	0.1739	-0.0976	-0.0189	-0.0210	0.1744	0.1736	0.0416	-0.1725	-0.1816	-0.0646	-0.1707	-0.0210	-0.0162	-0.0107	0.0407	-0.0407	0.1108	-0.5345	1		
GRE Verbal Percentile	0.3349	0.2346	0.1825	0.0746	0.1228	0.0524	0.0343	0.0479	-0.0681	-0.0536	-0.0548	-0.0358	-0.0272	-0.0158	0.0260	0.0343	-0.1056	-0.0155	0.2891	-0.1077	1	
GRE Quantitative Percentile	0.3448	0.2791	0.1559	0.0623	0.0605	0.1957	0.1930	0.0980	-0.2102	-0.1965	-0.1287	-0.1899	0.0061	0.0095	0.0116	-0.0255	-0.0622	-0.0398	-0.1080	0.2269	0.2114	1
GRE Analytic Percentile	0.1960	0.1198	0.1330	0.0505	0.0710	-0.0468	-0.0700	-0.0083	0.0359	0.0504	0.0058	0.0695	-0.0542	0.0113	0.0537	0.0379	-0.1533	0.0022	0.3950	-0.2554	0.5519	-0.0386
Undergraduate GPA	0.0517	-0.0095	0.0747	0.0322	0.0440	0.0329	0.0179	-0.0166	-0.0382	-0.0336	0.0056	-0.0179	-0.0272	-0.0058	0.0260	0.0109	-0.0853	0.0693	0.0418	-0.0425	0.0835	0.0793
TOEFL/IELTS	0.1251	0.1060	0.0898	0.0017	0.0413	-0.0200	-0.0127	0.0342	-0.0053	0.0226	-0.0431	0.0108	-0.0018	0.0047	0.0176	-0.0092	-0.1177	0.0969	0.0119	-0.0524	0.2772	0.0337
Elite U.S. University	0.1203	0.0775	0.1048	0.0285	0.0769	-0.0151	-0.0196	-0.0167	-0.0171	0.0108	0.0124	0.0179	-0.0305	-0.0271	0.0070	0.0392	-0.0769	0.0095	0.2082	-0.0410	0.1155	0.0303
Elite U.S. Liberal Arts College	0.0256	-0.0020	0.0274	0.0372	0.0205	-0.0081	-0.0280	-0.0038	-0.0103	0.0104	-0.0012	0.0272	0.0178	0.0133	-0.0405	-0.0125	-0.0639	0.0260	0.1177	-0.0567	0.0805	0.0260
Elite Foreign University	0.0459	0.0691	-0.0066	-0.0130	-0.0062	0.0218	0.0305	0.0071	-0.0445	-0.0261	0.0027	-0.0314	0.0084	0.0215	-0.0246	-0.0290	-0.0129	0.0447	-0.1008	0.1325	0.0335	0.0642
Other Foreign University	0.0241	0.0431	-0.0309	0.0146	-0.0327	0.0571	0.0564	0.0311	-0.0260	-0.0496	-0.0454	-0.0512	-0.0228	-0.0085	-0.0772	0.1586	0.1709	-0.0105	-0.4231	0.2355	-0.1253	0.0709
Missing Undergraduate Institution	-0.0601	-0.0583	-0.0185	-0.0424	-0.0215	0.0169	0.0435	0.0180	-0.0124	-0.0198	-0.0227	-0.0458	0.0300	-0.0008	0.1946	-0.2576	-0.0011	0.0092	-0.0364	-0.0761	-0.0414	-0.0056
Graduate Degree	0.1131	0.1254	0.0049	0.0046	0.0248	0.0529	0.0706	0.0226	-0.0399	-0.0548	-0.0328	-0.0698	0.0443	0.1379	-0.1556	-0.0729	0.3704	-0.0010	-0.2492	0.1402	-0.1043	0.0039
Work Experience	0.1103	0.0043	0.1074	0.0834	0.0484	-0.0723	-0.0789	0.0371	0.0722	0.0851	-0.0148	0.0795	-0.0017	-0.0002	-0.0056	0.0093	0.4141	-0.0229	0.0870	-0.2765	0.0369	-0.1144
#Professor Recommenders	0.0083	0.0374	-0.0114	-0.0136	-0.0257	0.0606	0.0466	-0.0073	-0.0472	-0.0547	0.0069	-0.0514	-0.0791	0.0186	0.0329	0.0390	-0.2398	-0.0114	-0.0330	0.1402	-0.0051	0.1028
#Prolific Recommenders	0.1044	0.0794	0.0733	0.0076	0.0374	0.1042	0.1084	0.0632	-0.1156	-0.1054	-0.0567	-0.1070	-0.0262	0.0330	0.0430	0.0062	0.0200	-0.0017	-0.2072	0.1686	-0.0282	0.1814
Missing GRE	-0.1102	-0.1318	-0.0579	-0.0217	-0.0358	-0.0731	-0.0649	-0.0322	0.0715	0.0786	0.0303	0.0641	0.0118	-0.0042	-0.0123	0.0071	0.0718	0.0122	-0.0303	-0.0502	-0.0081	-0.0129
Missing Undergraduate GPA	-0.0759	-0.0361	-0.0929	-0.0464	-0.0501	0.0261	0.0292	0.0119	-0.0105	-0.0213	-0.0191	-0.0269	-0.0372	-0.0582	-0.0334	0.2064	0.1637	-0.0095	-0.3206	0.1579	-0.2161	0.0599
Missing TOEFL/IELTS	-0.0246	-0.0503	0.0389	0.0146	0.0438	-0.1211	-0.1296	-0.0660	0.1027	0.1244	0.0753	0.1260	-0.0248	-0.0372	-0.0241	0.0552	-0.1342	-0.0207	0.4256	-0.2472	0.0917	-0.1772
Missing Winsorized Age	-0.0250	-0.0231	-0.0103	-0.0055	-0.0063	-0.0361	-0.0329	-0.0145	0.0312	0.0345	0.0187	0.0328	0.0214	0.0218	-0.0143	-0.0148	0.0000	-0.0235	-0.0163	-0.0282	-0.0442	-0.0172
Committee Score	0.0000	-0.3453	0.5999	0.3028	0.3395	0.0873	0.0554	0.0493	-0.1117	-0.0842	-0.0463	-0.0594	-0.0390	-0.0215	0.0691	0.0682	-0.0598	-0.0014	-0.0064	-0.0812	0.0408	0.0378
#Math Courses	0.0107	0.0311	-0.0215	-0.0177	-0.0467	0.0368	0.0367	0.0132	-0.0467	-0.0422	-0.0177	-0.0375	0.0291	0.0473	0.0642	-0.1633	-0.0764	-0.0104	0.0306	0.0164	0.0307	-0.0093
Advanced Math	0.2226	0.1304	0.2298	0.1071	0.1558	0.0463	0.0406	0.0031	-0.0742	-0.0510	-0.0116	-0.0420	-0.0182	-0.0752	0.0408	0.0398	-0.0349	-0.0054	-0.0328	0.0827	0.0181	0.1157
Letters Length/100	0.0189	-0.1069	0.1781	0.0827	0.0792	0.0261	0.0189	0.0146	-0.0368	-0.0238	-0.0114	-0.0197	0.0075	-0.0049	-0.0155	0.0170	0.0376	0.0367	0.0330	-0.0790	0.0531	-0.0110
Letters Positive Terms (%)	0.0161	-0.0760	0.0674	0.0780	0.0058	-0.0500	-0.0268	0.0708	0.0375	0.0280	-0.0454	0.0186	0.0464	-0.0001	-0.0522	0.0536	0.0432	-0.0145	0.1376	-0.0883	0.0349	0.0533
Letters Negative Terms (%)	-0.0093	-0.0545	0.0077	0.0413	-0.0479	-0.0158	-0.0610	-0.0737	0.0245	0.0241	0.0592	0.0574	0.0186	0.0081	0.0824	-0.0272	0.0807	-0.0036	0.1392	-0.0830	0.0764	-0.0665
Letters Standout Terms (%)	0.0003	-0.1493	0.1464	0.0698	-0.0346	0.0222	0.0131	0.0878	-0.0031	-0.0252	-0.0422	-0.0153	0.0733	-0.0587	-0.0863	0.0900	0.0507	-0.0141	0.0120	-0.0394	0.0100	0.0078
Letters Ability Terms (%)	0.0375	-0.0301	0.0487	0.0543	0.0006	-0.1785	-0.1573	-0.0137	0.1340	0.1765	0.0371	0.1502	0.0512	0.0014	0.0153	-0.0218	0.0866	-0.0015	0.1424	-0.1930	0.0423	0.0487
Letters Research Terms (%)	0.0499	-0.0879	0.1800	0.0897	0.0944	0.0439	0.0375	0.0492	-0.0771	-0.0588	-0.0934	-0.0376	-0.0062	-0.0162	0.1051	0.0225	0.0258	0.0245	0.1273	-0.0539	0.1004	-0.0220
Letters Grindstone Terms (%)	0.1125	-0.0320	0.1493	0.1481	0.0630	-0.1067	-0.1156	-0.0128	0.0728	0.1050	0.0347	0.1104	-0.0133	-0.0563	-0.0136	0.0294	0.1012	0.0839	0.1233	-0.1591	0.0209	-0.0701
Letters Teaching Terms (%)	0.0258	-0.0022	-0.0087	0.0696	0.0224	-0.1522	-0.1563	0.0241	0.1552	0.1493	0.0234	0.1579	0.0595	-0.0270	-0.0398	0.0459	0.1921	0.0369	0.0349	-0.1064	-0.0334	-0.0692
Letters Communal Terms (%)	0.0338	0.0517	-0.0749	-0.0272	-0.0216	-0.0020	0.0357	-0.0500	0.0132	0.0135	0.0186	-0.0394	0.0324	0.0711	-0.0603	-0.0627	0.0382	-0.0136	-0.1655	0.0499	-0.0011	0.0538
Letters Agentic Terms (%)	0.0421	0.0589	0.0557	0.0884	0.0913	-0.0509	-0.0716	0.0030	0.0147	0.0225	0.0016	0.0686	0.0927	-0.0091	0.0022	-0.0789	-0.0172	0.1191	0.1148	-0.0943	0.0638	-0.0830
Letters Terms (%) PC1	0.0130	-0.0283	0.0593	0.0870	0.0251	-0.0585	-0.0577	0.0109	0.0481	0.0525	0.0028	0.0547	0.0310	-0.0050	-0.0129	0.0043	0.0514	0.0207	0.0601	-0.0803	0.0219	-0.0120
Letters Terms (%) PC2	0.0088	-0.0033	0.0396	0.0383	0.0466	-0.0062	-0.0170	-0.0190	-0.0053	0.0063	0.0007	0.0174	-0.0204	-0.0007	0.0628	-0.0096	0.0192	0.0150	0.0516	-0.0323	0.0285	-0.0245
CV Length/100	0.0067	-0.0429	0.0959	0.0421	0.0772	0.0199	0.0202	0.0299	-0.0124	-0.0185	-0.0335	-0.0222	0.0320	-0.0248	-0.0129	-0.0118	0.0194	0.0249	0.0341	-0.0336	0.0120	-0.0114
CV Coding	-0.0067	-0.0225	0.0102	-0.0320	-0.0333	0.0280	0.0338	0.0297	-0.0250	-0.0246	-0.0377	-0.0365	-0.0225	-0.0306	0.0089	0.0368	-0.0633	-0.0004	-0.0209	0.0151	-0.0125	0.0021
CV Publication/Working Paper	0.1587	0.0464	0.2207	0.1422	0.1135	0.0666	0.0560	0.0432	-0.0707	-0.0633	-0.0408	-0.0569	-0.0284	-0.0575	0.0197	0.0603	0.0879	-0.0215	-0.0475	0.0269	-0.0141	0.0741
Essay Length/100	0.0017	-0.0422	0.0591	0.0712	0.0195	0.0473	0.0287	0.0026	-0.0530	-0.0452	-0.0053	-0.0274	0.0204	-0.0137	-0.0110	0.0253	0.0388	-0.0133	-0.0100	0.0078	-0.0314	-0.0020
Essay Specific Professor	0.2518	0.1247	0.2312	0.1775	0.1550	0.0489	0.0361	0.0306	-0.0633	-0.0470	-0.0396	-0.0374	-0.0440	-0.0641	-0.0419	0.0752	0.0149	-0.0044	-0.0712	0.0804	-0.0046	0.1188
Essay Economics Terms (%)	-0.0009	0.0317	-0.0293	0.0156	-0.0042	-0.0358	-0.0452	-0.0431	0.0338	0.0270	0.0456	0.0426	-0.0102	0.0036	0.0191	-0.0257	-0.0088	-0.0026	-0.0373	0.0596	-0.0069	0.0022
Essay Research Terms (%)	-0.0101	0.0042	-0.0059	-0.0277	-0.0234	0.0838	0.0766	0.0144	-0.0690	-0.0720	-0.0305	-0.0802	-0.0296	0.0235	-0.0081	0.0445	-0.0767	0.0498	-0.0432	0.0665	-0.0242	0.0132
Essay Positive Terms (%)	-0.0132	0.0211	-0.0442	-0.0405	-0.0379	-0.0035	0.0102	0.0332	0.0043	0.0060	-0.0127	-0.0110	-0.0123	-0.0010	0.0151	0.0029	-0.0450	0.0208	-0.0019	0.0002	-0.0149	0.0057
Essay Negative Terms (%)	0.0074	0.0139	-0.0041	-0.0057	-0.0264	0.0107	0.0005	-0.0066	-0.0026	-0.0143	0.0085	-0.0030	0.0015	0.0056	-0.0294	0.0257	0.0215	-0.0251	-0.0063	0.0323	-0.0046	0.0103
Missing Committee Score	-0.9903	-0.7997	-0.4037	-0.1981	-0.2043	-0.1226	-0.1054	-0.1012	0.1477	0.1297	0.1106	0.1020	0.0748	0.0433	-0.1554	-0.0105	-0.0656	-0.0471	-0.0078	-0.0793	-0.3358	-0.3437
Missing Application Pdf	-0.3301	-0.2162	-0.2739	-0.1737	-0.1677	-0.0565	-0.0516	-0.0230	0.0840	0.0596	0.0243	0.0514	0.0528	0.0955	-0.0508	-0.1152	-0.0056	-0.0041	0.1078	-0.1112	0.0270	-0.1797
Missing #Math Courses	-0.3326	-0.2159	-0.2739	-0.1654	-0.1709	-0.0585	-0.0547	-0.0293	0.0872	0.0610	0.0309	0.0584	0.0498	0.0902	-0.0516	-0.1115	-0.0077	-0.0025	0.1021	-0.1108	0.0188	-0.1817
Missing Letters	-0.3273	-0.2105	-0.2771	-0.1755	-0.1700	-0.0545	-0.0491	-0.0230	0.0815	0.0574	0.0229	0.0489	0.0502	0.0986	-0.0509	-0.1130	-0.0081	-0.0063	0.1052	-0.1060	0.0263	-0.1765
Missing CV	-0.3261	-0.2095	-0.2752	-0.1661	-0.1568	-0.0590	-0.0528	-0.0248	0.0842	0.0615	0.0230	0.0525	0.0571	0.0893	-0.0528	-0.1156	-0.0142	-0.0020	0.1142	-0.1198	0.0255	-0.1806
Missing Essay	-0.3233	-0.2066	-0.2726	-0.1770	-0.1718	-0.0533	-0.0476	-0.0253	0.0819	0.0560	0.0253	0.0474	0.0508	0.0912	-0.0448	-0.1101	-0.0035	-0.0099	0.1058	-0.1106	0.0263	-0.1746
Observations	2329																					

	GREA	GPA	T/I	EUni	ELA	EFor	OFor	MUg	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I	MAge	Score	NMath	AMath	LetN	LPos	LNeg	LSta
GRE Analytic Percentile	1																						
Undergraduate GPA	0.0863	1																					
TOEFL/IELTS	0.2465	0.0031	1																				
Elite U.S. University	0.1320	-0.0917	0.0035	1																			
Elite U.S. Liberal Arts College	0.1074	0.0216	0.0113	-0.0665	1																		
Elite Foreign University	-0.0303	0.0047	0.0366	-0.0764	-0.0350	1																	
Other Foreign University	-0.2141	-0.0002	-0.0057	-0.3141	-0.1441	-0.1654	1																
Missing Undergraduate Institution	-0.0322	0.0301	-0.0164	-0.2024	-0.0929	-0.1066	-0.4384	1															
Graduate Degree	-0.1313	-0.0877	0.0107	-0.1458	-0.0808	0.0597	0.1649	0.0831	1														
Work Experience	0.1031	0.0121	0.0128	0.0083	0.0294	-0.0367	-0.0399	0.0163	0.1219	1													
#Professor Recommenders	-0.0399	0.0557	-0.0180	-0.0592	0.0069	0.0270	0.0136	-0.0023	-0.0631	-0.2477	1												
#Prolific Recommenders	-0.1224	-0.0547	0.0052	-0.0311	-0.0497	0.0492	0.1487	-0.0014	0.1173	-0.0724	0.2047	1											
Missing GRE	0.0026	-0.0117	-0.0034	0.0080	-0.0325	0.0122	-0.0028	-0.0183	-0.0073	0.0434	-0.0375	-0.0163	1										
Missing Undergraduate GPA	-0.3214	0.0000	-0.0211	-0.2096	-0.1183	0.0940	0.3226	0.0180	0.1345	-0.0097	0.0036	0.1459	0.0625	1									
Missing TOEFL/IELTS	0.2285	0.0072	0.0000	0.2876	0.1352	-0.0396	-0.3778	-0.1160	-0.1265	0.0345	-0.0031	-0.2002	0.0714	-0.2810	1								
Missing Winsorized Age	0.0000	0.0000	0.0000	-0.0112	-0.0051	-0.0059	0.0057	0.0198	0.0289	0.0279	0.0150	-0.0071	0.1575	0.0153	0.0236	1							
Committee Score	0.0481	0.0800	0.0726	0.0496	-0.0133	-0.0126	-0.0104	0.0234	-0.0609	0.0246	0.0279	0.0849	-0.0451	-0.0466	0.0188	0.0000	1						
#Math Courses	0.0315	-0.0128	0.0409	0.0339	-0.0032	-0.0111	-0.0406	-0.0053	-0.0392	0.0041	-0.0020	-0.0002	0.0170	-0.0632	0.0423	0.0004	-0.0232	1					
Advanced Math	-0.0207	0.0099	0.0072	0.0701	0.0337	0.0150	0.0042	-0.0763	-0.0478	-0.0561	0.0363	0.1345	-0.0417	-0.0261	-0.0030	-0.0092	0.1388	0.2255	1				
Letters Length/100	0.0579	-0.0050	0.0487	0.0012	0.0635	0.0250	-0.0216	-0.0030	-0.0020	0.1001	-0.0420	0.0333	-0.0082	-0.0084	0.0034	0.0007	0.1631	0.0027	0.0502	1			
Letters Positive Terms (%)	0.1164	-0.0099	-0.0573	-0.0018	0.1339	-0.0416	-0.0931	-0.0256	-0.0638	0.0769	-0.1049	-0.1867	0.1234	-0.0626	-0.0402		0.0587	-0.0070	-0.0916	0.0015	1		
Letters Negative Terms (%)	0.0632	-0.0612	-0.1270	0.0379	0.0199	0.0261	-0.0506	-0.0449	0.0424	0.0943	-0.1391	0.0507	0.0151	-0.0109	0.1318		0.0330	0.0100	-0.0104	0.0546	0.1546	1	
Letters Standout Terms (%)	0.0742	0.0477	0.0066	-0.1116	0.0846	0.0073	0.0514	-0.0690	-0.0211	0.0198	0.0226	0.0132	0.1382	0.0456	-0.0033		0.1475	0.0194	-0.1303	0.0427	0.4324	-0.0327	1
Letters Ability Terms (%)	0.1273	0.0808	-0.0476	-0.0047	0.0583	0.0035	-0.1331	0.0026	0.0081	0.1310	-0.1811	-0.2310	-0.0190	-0.0979	0.0358		0.0943	-0.0408	-0.1419	0.0162	0.5004	0.1896	0.2105
Letters Research Terms (%)	0.1052	-0.0448	0.0235	0.0296	0.0482	-0.0391	-0.0767	0.0477	0.0037	0.1626	-0.2141	-0.0945	-0.0446	0.0039	0.0552		0.1776	0.0127	0.0874	0.2071	0.0820	0.1569	-0.0520
Letters Grindstone Terms (%)	0.0946	0.0185	-0.0524	0.0140	0.1687	-0.0368	-0.1797	0.0350	0.0451	0.2148	-0.1778	-0.1497	0.0905	-0.0876	0.0740		0.1127	-0.0523	-0.0447	0.0523	0.3502	0.1286	0.0678
Letters Teaching Terms (%)	0.0481	0.0218	-0.0055	-0.0565	0.0274	-0.0166	0.0065	-0.0288	0.0670	0.1777	-0.0771	-0.1383	-0.0047	-0.0060	-0.0189		-0.0163	-0.0878	-0.1488	0.1290	0.3650	-0.0016	0.1177
Letters Communal Terms (%)	0.0001	0.0203	-0.1205	-0.0344	0.0260	0.0362	-0.0034	0.0305	0.0770	-0.0646	0.1095	-0.0050	0.1483	-0.0116	-0.1435		-0.1044	-0.0440	-0.0754	-0.1683	0.2329	-0.0224	0.0070
Letters Agentic Terms (%)	0.0517	-0.0681	0.0347	0.0740	0.1425	-0.0270	-0.1247	0.0009	-0.0336	0.0006	-0.1529	-0.0772	0.0548	-0.1457	0.1325		0.0353	0.1087	0.0595	0.0830	0.3284	0.0143	0.1124
Letters Terms (%) PC1	0.0601	0.0043	-0.0258	-0.0053	0.0708	-0.0127	-0.0645	-0.0019	0.0041	0.0764	-0.0732	-0.0963	0.0261	-0.0426	0.0213	0.0004	0.0492	-0.0199	-0.1116	0.0803	0.8114	0.2386	0.3964
Letters Terms (%) PC2	0.0213	-0.0207	-0.0088	0.0366	-0.0004	-0.0110	-0.0449	0.0199	0.0192	0.0790	-0.0774	-0.0236	-0.0271	-0.0068	0.0497	0.0003	0.0365	-0.0202	0.0655	0.1600	-0.2292	0.4519	-0.5572
CV Length/100	0.0448	0.0014	0.0399	-0.0027	0.0241	-0.0241	0.0074	-0.0123	0.0314	0.1054	-0.0738	0.0126	-0.0228	-0.0163	-0.0019	0.0002	0.0962	0.0312	0.0677	0.2051	-0.0026	-0.0135	0.0985
CV Coding	-0.0083	-0.0131	0.0023	0.0122	0.0125	-0.0072	0.0100	-0.0158	0.0065	-0.0100	0.0106	0.0452	-0.0094	0.0346	0.0194	-0.0002	0.0740	0.0970	0.0863	0.1100	-0.0178	0.0284	0.0407
CV Publication/Working Paper	-0.0197	-0.0454	-0.0400	0.0292	0.0109	-0.0006	-0.0358	-0.0256	0.0282	0.0825	-0.0292	0.0846	-0.0289	0.0128	-0.0457	-0.0064	0.1599	-0.0381	0.2648	0.1565	0.1046	0.0451	0.1148
Essay Length/100	-0.0472	-0.0304	-0.0084	-0.0051	-0.0053	0.0103	0.0145	-0.0109	0.0979	0.0273	0.0028	0.0398	-0.0057	0.0249	0.0041	0.0001	0.0474	-0.0164	0.0317	0.1083	-0.0279	0.0331	-0.0042
Essay Specific Professor	-0.0416	0.0123	0.0222	0.0562	-0.0091	-0.0067	0.0521	-0.0577	0.0065	-0.0138	0.0279	0.1057	-0.0406	0.0133	-0.0322	-0.0091	0.1428	0.0022	0.4462	0.0738	-0.0047	0.0772	-0.0022
Essay Economics Terms (%)	-0.0280	0.0013	-0.0115	0.0065	-0.0007	-0.0222	-0.0436	0.0058	0.0017	-0.0049	-0.0217	0.0205	0.0164	-0.0006	0.0160	0.0000	0.0035	0.1185	0.0515	-0.0282	-0.0572	-0.0354	-0.0553
Essay Research Terms (%)	-0.0050	0.0206	-0.0017	0.0157	0.0134	0.0160	-0.0295	-0.0003	-0.0478	-0.0345	0.0073	0.0095	-0.0019	-0.0008	0.0049	-0.0003	0.0376	0.0664	0.0430	0.0499	-0.0016	0.0032	0.0595
Essay Positive Terms (%)	0.0096	0.0287	0.0103	0.0155	-0.0262	-0.0164	-0.0213	-0.0069	-0.0358	-0.0021	0.0141	-0.0244	0.0013	-0.0200	0.0169	-0.0005	-0.0277	0.0540	-0.0383	-0.1339	0.0887	-0.0558	-0.0026
Essay Negative Terms (%)	-0.0007	-0.0584	-0.0316	0.0534	0.0249	0.0123	-0.0148	-0.0414	-0.0002	0.0003	-0.0086	0.0500	-0.0132	0.0141	0.0353	0.0003	0.0270	0.0719	0.0827	-0.0128	0.0174	0.1960	0.0757
Missing Committee Score	-0.1972	-0.0516	-0.1281	-0.1189	-0.0273	-0.0479	-0.0179	0.0551	-0.1108	-0.1049	-0.0094	-0.1045	0.1134	0.0768	0.0216	0.0247	0.0000	-0.0153	-0.2077	-0.0137	-0.0089	0.0003	0.0162
Missing Application Pdf	0.0709	-0.0220	0.0179	-0.0650	-0.0082	-0.0079	-0.0674	0.0874	0.0263	0.0656	-0.0632	-0.1400	0.0578	-0.0143	0.0482	0.0140	-0.1544	0.0263	-0.6598	0.0467			
Missing #Math Courses	0.0657	-0.0200	0.0155	-0.0670	-0.0039	-0.0105	-0.0646	0.0844	0.0275	0.0646	-0.0636	-0.1423	0.0623	-0.0108	0.0517	0.0137	-0.1618	0.0268	-0.6547	0.0402	-0.0241	-0.0133	0.0534
Missing Letters	0.0686	-0.0209	0.0219	-0.0674	-0.0092	-0.0091	-0.0632	0.0848	0.0274	0.0645	-0.0622	-0.1461	0.0571	-0.0142	0.0437	0.0139	-0.1570	0.0336	-0.6607	0.0470			
Missing CV	0.0760	-0.0193	0.0171	-0.0579	-0.0109	-0.0110	-0.0757	0.0815	0.0297	0.0556	-0.0596	-0.1431	0.0558	-0.0205	0.0536	0.0137	-0.1568	0.0332	-0.6449	0.0394	-0.0109	-0.0393	-0.0461
Missing Essay	0.0683	-0.0222	0.0160	-0.0660	-0.0101	-0.0042	-0.0649	0.0881	0.0230	0.0668	-0.0601	-0.1415	0.0564	-0.0119	0.0429	0.0138	-0.1489	0.0146	-0.6606	0.0474	-0.0201	-0.0545	-0.0171

	LAbi	LRes	LGri	LTea	LCom	LAge	LPC1	LPC2	CVN	CVCod	CVPap	EssN	EssProf	EssEcon	EssRes	EssPos	EssNeg	MScore	MPDF	MNMath	ML	MCV	MEss
Letters Ability Terms (%)	1																						
Letters Research Terms (%)	0.0863	1																					
Letters Grindstone Terms (%)	0.3655	0.2281	1																				
Letters Teaching Terms (%)	0.3161	0.0976	0.2383	1																			
Letters Communal Terms (%)	0.0719	-0.1124	0.1381	0.0216	1																		
Letters Agentic Terms (%)	0.3027	0.0256	0.2276	0.2610	0.0207	1																	
Letters Terms (%) PC1	0.7486	0.2082	0.6157	0.5816	0.2083	0.5580	1																
Letters Terms (%) PC2	-0.0082	0.7168	0.3168	0.0333	-0.2524	-0.1306	-0.0206	1															
CV Length/100	0.0481	0.1497	0.0613	0.0839	-0.1494	0.1032	0.0820	0.0721	1														
CV Coding	-0.0115	0.2123	0.0344	-0.0499	-0.0274	0.0278	0.0147	0.1218	0.2274	1													
CV Publication/Working Paper	0.0639	0.2120	0.1178	0.1343	-0.0182	0.0086	0.1158	0.0865	0.2482	0.1433	1												
Essay Length/100	-0.0206	0.0524	-0.0092	-0.0672	-0.0119	-0.0283	-0.0332	0.0460	0.0706	0.1168	0.0772	1											
Essay Specific Professor	-0.0215	0.0483	0.0270	0.0159	-0.0317	0.0294	-0.0084	0.0360	0.0795	0.1038	0.2923	0.0701	1										
Essay Economics Terms (%)	0.0057	0.1233	0.0995	0.0289	0.0579	0.0002	0.0166	0.1110	0.0016	0.0578	0.0391	-0.0593	0.0049	1									
Essay Research Terms (%)	-0.0279	-0.2357	-0.0866	-0.0744	-0.0378	-0.0300	-0.0307	0.0886	0.0185	0.1414	0.0621	0.0651	-0.0097	0.1593	1								
Essay Positive Terms (%)	-0.0436	-0.0223	0.0400	-0.0516	0.0216	0.0132	-0.0494	-0.0413	-0.0248	-0.0291	-0.1956	-0.0535	-0.0689	-0.0250	[illegible]	1							
Essay Negative Terms (%)	-0.0206	-0.0427	-0.0338	-0.0425	-0.0131	0.0363	0.0140	-0.0083	0.0279	0.0566	0.0102	0.0716	0.0666	-0.1217	-0.1206	-0.0714	1						
Missing Committee Score	-0.0200	-0.0288	-0.0786	0.0211	-0.0473	-0.0157	-0.0015	-0.0054	-0.0070	0.0098	-0.1443	0.0100	-0.2336	-0.0045	0.0027	0.0134	-0.0106	1					
Missing Application Pdf							0.0322	0.0218	0.0165	-0.0164	-0.4566	0.0042	-0.6494	-0.0023	-0.0250	-0.0327	0.0183	0.3105	1				
Missing #Math Courses	-0.0279	-0.0512	0.0813	0.0401	-0.0056	0.0053	0.0343	0.0164	0.0076	-0.0239	-0.4531	0.0007	-0.6399	0.0044	-0.0279	-0.0345	0.0283	0.3149	0.9843	1			
Missing Letters							0.0324	0.0219	0.0199	-0.0202	-0.4599	0.0066	-0.6462	0.0039	-0.0208	-0.0261	0.0113	0.3076	0.9929	0.9769	1		
Missing CV	0.0037	-0.0206	0.0649	0.0421	0.0770	0.1614	0.0419	0.0187	0.0168	-0.0168	-0.4653	0.0026	-0.6299	-0.0129	-0.0385	-0.0266	0.0242	0.3061	0.9814	0.9650	0.9740	1	
Missing Essay	0.0164	-0.0382	0.0205	-0.0490	-0.0453	-0.0824	0.0253	0.0200	0.0134	-0.0246	-0.4464	0.0042	-0.6578	-0.0023	-0.0253	-0.0331	0.0185	0.3034	0.9871	0.9710	0.9798	0.9709	1

Table 9a: Logistic Regression Results (Synthetic)

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.2091	-0.1311	0.1291	0.0997	0.0892	0.1020	-0.8042	-0.5831	-1.3927
	(0.2288)	(0.2577)	(0.6478)	(0.1812)	(0.1814)	(0.1970)	(4.5726)	(4.5702)	(4.1436)
2017	-0.3355	-0.5461**	0.6767	0.2827	0.2903	0.1565	-12.4495***	-12.2002***	-6.3639
	(0.2390)	(0.2627)	(0.6543)	(0.1761)	(0.1767)	(0.1925)	(4.4305)	(4.4347)	(3.9262)
2018	0.2889	-0.2993	0.2947	0.1627	0.2192	0.1272	-9.3709**	-9.2465**	-6.0098
	(0.2056)	(0.2212)	(0.5517)	(0.1806)	(0.1827)	(0.1972)	(4.4278)	(4.4749)	(3.9975)
2019	-0.1933	-0.3208	-0.0662	0.0116	0.0023	-0.1488	-5.3768	-4.7378	1.0400
	(0.2227)	(0.2490)	(0.6069)	(0.1927)	(0.1933)	(0.2061)	(4.4927)	(4.5026)	(3.9171)
Winsorized Age	-0.0508**	-0.0414*	0.0065	-0.0141	-0.0150	-0.0094	0.4466	0.4383	0.2496
	(0.0220)	(0.0243)	(0.0600)	(0.0170)	(0.0170)	(0.0185)	(0.4147)	(0.4141)	(0.3766)
Female	-0.1826	-0.1535	0.0691	0.1351	0.1379	0.2109*	-1.5097	-1.6305	-4.7144*
	(0.1446)	(0.1590)	(0.4006)	(0.1167)	(0.1168)	(0.1259)	(2.9357)	(2.9318)	(2.6789)
U.S.	0.0002	-0.0549	0.0998	0.0198	0.0241	-0.0123	2.8625	2.5697	4.2450
	(0.2240)	(0.2468)	(0.6663)	(0.1794)	(0.1801)	(0.1997)	(4.4691)	(4.4669)	(4.1868)
International Asian	-0.1230	-0.0646	-0.1205	0.2014	0.1972	0.1700	-0.7475	-0.9433	1.1783
	(0.1688)	(0.1838)	(0.5028)	(0.1394)	(0.1392)	(0.1541)	(3.3754)	(3.3732)	(3.1231)
GRE Verbal Percentile	-0.0025	-0.0001	0.0008	0.0004	0.0001	-0.0021	-0.0668	-0.0623	0.0181
	(0.0034)	(0.0036)	(0.0089)	(0.0026)	(0.0026)	(0.0029)	(0.0661)	(0.0663)	(0.0595)
GRE Quantitative Percentile	-0.0028	-0.0005	-0.0125	0.0145**	0.0143**	-0.0003	-0.5982***	-0.5916***	-0.0184
	(0.0075)	(0.0080)	(0.0195)	(0.0065)	(0.0065)	(0.0069)	(0.1570)	(0.1569)	(0.1376)
GRE Analytic Percentile	0.0041	0.0019	-0.0010	0.0009	0.0011	0.0019	0.0181	0.0156	-0.0008
	(0.0031)	(0.0033)	(0.0074)	(0.0024)	(0.0024)	(0.0026)	(0.0587)	(0.0587)	(0.0529)
Undergraduate GPA	-0.4230	-0.3157	0.9077	0.4075	0.3986	0.4627	-2.7870	-2.7400	-5.8113
	(0.3862)	(0.4536)	(1.8572)	(0.3984)	(0.3990)	(0.4326)	(9.4800)	(9.4624)	(8.8952)
TOEFL/IELTS	0.2329	0.2776	-0.2073	-0.0682	-0.0625	-0.0142	2.4237	2.2404	-0.4376
	(0.2154)	(0.2202)	(0.4100)	(0.1509)	(0.1507)	(0.1670)	(3.7070)	(3.7021)	(3.1595)
Elite U.S. University	-0.0473	-0.1157	-0.2546	0.3145	0.3119	0.3760*	-6.7562	-6.6352	-7.1987
	(0.2618)	(0.2943)	(0.7160)	(0.2054)	(0.2062)	(0.2230)	(5.3790)	(5.3875)	(4.8870)
Elite U.S. Liberal Arts College	-0.6642	-0.6053		0.0069	-0.0044	-0.0958	-2.2643	-2.2283	3.3486
	(0.5448)	(0.5744)		(0.3656)	(0.3659)	(0.3964)	(8.9098)	(8.8713)	(7.5578)
Elite Foreign University	-0.1797	-0.1798	0.5156	-0.0538	-0.0692	-0.0298	-1.0302	-0.9315	-0.2765
	(0.3900)	(0.4341)	(1.2147)	(0.3273)	(0.3294)	(0.3495)	(8.2453)	(8.2793)	(7.1254)
Other Foreign University	-0.0406	-0.1089	0.6613	0.1055	0.1125	0.0046	-6.0523	-6.2650	-0.6409
	(0.2078)	(0.2262)	(0.5235)	(0.1688)	(0.1689)	(0.1840)	(4.2123)	(4.2076)	(3.8417)
Missing Undergraduate Institution	-0.0012	0.0377	-0.6367	0.1037	0.0966	0.0297	-5.9887	-5.9936	-2.3404
	(0.2217)	(0.2413)	(0.5829)	(0.1844)	(0.1842)	(0.1977)	(4.5947)	(4.5865)	(4.1622)
Graduate Degree	-0.0846	-0.1354	-0.0100	0.1515	0.1611	0.1906	-1.4756	-1.6341	-2.0725
	(0.1458)	(0.1626)	(0.3834)	(0.1166)	(0.1165)	(0.1252)	(2.8643)	(2.8619)	(2.5933)
Work Experience	0.2547*	0.2149	-0.2181	0.0750	0.0805	0.1443	0.4084	0.3542	-2.7442
	(0.1463)	(0.1588)	(0.3573)	(0.1161)	(0.1160)	(0.1250)	(2.8682)	(2.8651)	(2.6268)
#Professor Recommenders	0.0194	-0.0565	0.1806	0.0038	0.0074	-0.0302	-0.2230	-0.1761	1.2584
	(0.0950)	(0.1021)	(0.2725)	(0.0759)	(0.0761)	(0.0794)	(1.9087)	(1.9104)	(1.7006)
#Prolific Recommenders	-0.0388	-0.0071	0.0168	0.0027	-0.0000	0.0325	0.3327	0.3170	-1.1079
	(0.0671)	(0.0713)	(0.1685)	(0.0521)	(0.0522)	(0.0575)	(1.3245)	(1.3242)	(1.2261)
Missing GRE	0.3002	0.0736		0.3136	0.3314	0.2674	-12.0035	-11.9570	-11.5860
	(0.3757)	(0.3872)		(0.2858)	(0.2857)	(0.3297)	(8.1036)	(8.0998)	(8.0641)
Missing Undergraduate GPA	0.1997	0.1720	0.2910	-0.1321	-0.1290	-0.0816	5.2305	5.1674	4.3351
	(0.1863)	(0.2016)	(0.5443)	(0.1391)	(0.1389)	(0.1534)	(3.6167)	(3.6124)	(3.3094)
Missing TOEFL/IELTS	-0.1740	-0.0792	-0.3009	0.1948	0.1815	0.2286*	-1.5099	-1.4121	-3.1183
	(0.1609)	(0.1741)	(0.4256)	(0.1280)	(0.1281)	(0.1380)	(3.1162)	(3.1192)	(2.7616)
Committee Score		0.0372	-0.1004		0.0465	0.0330		-1.6627	-0.4906
		(0.0423)	(0.1060)		(0.0505)	(0.0558)		(1.1655)	(1.1645)
#Math Courses	0.0632	0.0756	-0.1683	-0.1269	-0.1305*	-0.1764**	1.1826	1.2073	3.0509**
	(0.0687)	(0.0733)	(0.1601)	(0.0786)	(0.0780)	(0.0863)	(1.7287)	(1.7209)	(1.5010)
Advanced Math	0.0222	-0.0336	0.9873	-0.1639	-0.1630	-0.2842	1.1644	1.2290	6.1671
	(0.2590)	(0.2747)	(0.7209)	(0.3004)	(0.3013)	(0.3256)	(6.4828)	(6.4875)	(5.6351)
Letters Length/100	-0.0254	-0.0128	-0.0575	-0.1049***	-0.1057***	-0.1218***	1.6052**	1.6308**	2.1010***
	(0.0333)	(0.0361)	(0.0885)	(0.0381)	(0.0376)	(0.0428)	(0.7873)	(0.7869)	(0.6934)
Letters Terms (%) PC1	0.0280	-0.0073	0.0381	0.0574	0.0643	0.0566	1.1364	1.1816	1.0356
	(0.0779)	(0.0821)	(0.2097)	(0.0828)	(0.0834)	(0.0930)	(1.8933)	(1.9000)	(1.6894)
Letters Terms (%) PC2	-0.2328*	-0.2369*	-0.4921	0.0802	0.0718	0.1318	-2.8589	-2.7586	-5.0669**
	(0.1226)	(0.1255)	(0.3028)	(0.1205)	(0.1211)	(0.1250)	(2.6915)	(2.7005)	(2.1872)
CV Length/100	0.0971**	0.0795	0.1022	-0.0358	-0.0336	-0.0095	2.4879**	2.5172**	0.5354
	(0.0492)	(0.0508)	(0.1340)	(0.0548)	(0.0551)	(0.0641)	(1.2406)	(1.2458)	(1.0685)
CV Coding	-0.0198	-0.0473	-0.0834	0.0688	0.0720*	0.0795*	-1.6529	-1.6778	-1.0250
	(0.0411)	(0.0448)	(0.1069)	(0.0434)	(0.0434)	(0.0474)	(1.0435)	(1.0451)	(0.8904)
CV Publication/Working Paper	0.3079	0.3005	1.3491*	0.1111	0.1183	0.1111	2.7524	2.4246	5.5955
	(0.2852)	(0.2962)	(0.7475)	(0.3264)	(0.3272)	(0.3482)	(7.3505)	(7.3749)	(6.2211)
Essay Length/100	0.0679	0.0790	-0.0558	-0.0667	-0.0636	-0.0694	0.5523	0.4398	0.3231
	(0.0770)	(0.0831)	(0.1489)	(0.0780)	(0.0794)	(0.0871)	(1.7254)	(1.7321)	(1.3973)
Essay Specific Professor	-0.1756	-0.2494	-0.1115	-0.0547	-0.0524	0.0009	1.2639	1.2219	-3.1507
	(0.2564)	(0.2714)	(0.6302)	(0.2774)	(0.2798)	(0.3022)	(6.3588)	(6.3580)	(5.5747)
Essay Economics Terms (%)	-0.0123	-0.0094	-0.0406	-0.0091	-0.0082	-0.0222	0.2276	0.2960	0.6769
	(0.0542)	(0.0572)	(0.1334)	(0.0539)	(0.0547)	(0.0624)	(1.3179)	(1.3189)	(1.1799)
Essay Research Terms (%)	0.0303	0.0066	-0.0464	0.1042	0.1084	0.1176	-1.8679	-1.9069	-1.8170
	(0.0656)	(0.0702)	(0.1578)	(0.0703)	(0.0705)	(0.0723)	(1.5218)	(1.5245)	(1.2596)
Essay Positive Terms (%)	0.0180	0.0074	0.3699***	0.0871*	0.0868*	0.0947	-1.5542	-1.4775	-1.1629
	(0.0474)	(0.0484)	(0.1213)	(0.0511)	(0.0517)	(0.0609)	(1.1518)	(1.1568)	(1.1006)
Essay Negative Terms (%)	0.0333	0.0312	-0.0731	-0.0304	-0.0280	-0.0168	0.5011	0.4716	0.2911
	(0.0314)	(0.0351)	(0.0769)	(0.0359)	(0.0359)	(0.0382)	(0.8608)	(0.8614)	(0.7100)
Missing Committee Score		-4.3141***			0.2635**	0.4129***		-0.9440	-6.1957**
		(0.4312)			(0.1267)	(0.1388)		(3.0258)	(2.8364)
Missing Application Pdf	-0.9321***	-0.0588	0.9473	0.0574	-0.0560	-0.1100	4.5136	4.8399	5.6614
	(0.2534)	(0.2753)	(0.6793)	(0.2628)	(0.2678)	(0.2943)	(5.9238)	(6.0503)	(5.4160)
Open House			1.0643						
			(0.8268)						
Program Rank						-0.0049**			0.1176***
						(0.0019)			(0.0386)
Missing Program Rank						-2.5925***			94.5255***
						(0.3273)			(5.6183)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	237	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.0552	0.2804	0.1402	0.0221	0.0245	0.1935	0.0040	0.0042	0.0953

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 9b: Logistic Regression AMEs (Synthetic)

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.0185	-0.0100	0.0201	0.0136	0.0122	0.0120	-0.3764	-0.2729	-0.6112
	(0.0203)	(0.0197)	(0.1005)	(0.0248)	(0.0248)	(0.0233)	(2.1401)	(2.1390)	(1.8189)
2017	-0.0297	-0.0417**	0.1052	0.0387	0.0396	0.0185	-5.8265***	-5.7099***	-2.7929
	(0.0212)	(0.0199)	(0.1005)	(0.0241)	(0.0241)	(0.0227)	(2.0748)	(2.0770)	(1.7228)
2018	0.0256	-0.0229	0.0458	0.0223	0.0299	0.0150	-4.3857**	-4.3275**	-2.6375
	(0.0182)	(0.0168)	(0.0856)	(0.0247)	(0.0250)	(0.0233)	(2.0732)	(2.0958)	(1.7555)
2019	-0.0171	-0.0245	-0.0103	0.0016	0.0003	-0.0176	-2.5164	-2.2174	0.4564
	(0.0197)	(0.0189)	(0.0944)	(0.0264)	(0.0264)	(0.0243)	(2.1013)	(2.1061)	(1.7192)
Winsorized Age	-0.0045**	-0.0032*	0.0010	-0.0019	-0.0020	-0.0011	0.2090	0.2051	0.1095
	(0.0020)	(0.0019)	(0.0093)	(0.0023)	(0.0023)	(0.0022)	(0.1941)	(0.1938)	(0.1654)
Female	-0.0162	-0.0117	0.0107	0.0185	0.0188	0.0249*	-0.7066	-0.7631	-2.0690*
	(0.0128)	(0.0121)	(0.0621)	(0.0160)	(0.0159)	(0.0149)	(1.3745)	(1.3728)	(1.1767)
U.S.	0.0000	-0.0042	0.0155	0.0027	0.0033	-0.0015	1.3397	1.2026	1.8630
	(0.0198)	(0.0189)	(0.1035)	(0.0246)	(0.0246)	(0.0236)	(2.0914)	(2.0905)	(1.8379)
International Asian	-0.0109	-0.0049	-0.0187	0.0276	0.0269	0.0201	-0.3498	-0.4415	0.5171
	(0.0150)	(0.0140)	(0.0782)	(0.0191)	(0.0190)	(0.0182)	(1.5798)	(1.5789)	(1.3706)
GRE Verbal Percentile	-0.0002	-0.0000	0.0001	0.0001	0.0000	-0.0002	-0.0313	-0.0291	0.0080
	(0.0003)	(0.0003)	(0.0014)	(0.0004)	(0.0004)	(0.0003)	(0.0309)	(0.0310)	(0.0261)
GRE Quantitative Percentile	-0.0002	-0.0000	-0.0019	0.0020**	0.0020**	-0.0000	-0.2800***	-0.2769***	-0.0081
	(0.0007)	(0.0006)	(0.0030)	(0.0009)	(0.0009)	(0.0008)	(0.0734)	(0.0734)	(0.0604)
GRE Analytic Percentile	0.0004	0.0001	-0.0002	0.0001	0.0002	0.0002	0.0085	0.0073	-0.0004
	(0.0003)	(0.0003)	(0.0011)	(0.0003)	(0.0003)	(0.0003)	(0.0275)	(0.0275)	(0.0232)
Undergraduate GPA	-0.0375	-0.0241	0.1411	0.0558	0.0544	0.0546	-1.3043	-1.2824	-2.5504
	(0.0342)	(0.0346)	(0.2875)	(0.0545)	(0.0545)	(0.0510)	(4.4373)	(4.4292)	(3.9043)
TOEFL/IELTS	0.0206	0.0212	-0.0322	-0.0093	-0.0085	-0.0017	1.1343	1.0486	-0.1921
	(0.0191)	(0.0168)	(0.0637)	(0.0207)	(0.0206)	(0.0197)	(1.7354)	(1.7331)	(1.3866)
Elite U.S. University	-0.0042	-0.0088	-0.0396	0.0431	0.0426	0.0444*	-3.1620	-3.1054	-3.1593
	(0.0232)	(0.0225)	(0.1109)	(0.0281)	(0.0282)	(0.0263)	(2.5179)	(2.5220)	(2.1436)
Elite U.S. Liberal Arts College	-0.0588	-0.0462		0.0009	-0.0006	-0.0113	-1.0597	-1.0429	1.4696
	(0.0483)	(0.0439)		(0.0500)	(0.0500)	(0.0468)	(4.1696)	(4.1517)	(3.3177)
Elite Foreign University	-0.0159	-0.0137	0.0801	-0.0074	-0.0095	-0.0035	-0.4822	-0.4360	-0.1214
	(0.0346)	(0.0332)	(0.1891)	(0.0448)	(0.0450)	(0.0412)	(3.8590)	(3.8749)	(3.1271)
Other Foreign University	-0.0036	-0.0083	0.1028	0.0144	0.0154	0.0005	-2.8325	-2.9321	-0.2813
	(0.0184)	(0.0173)	(0.0813)	(0.0231)	(0.0231)	(0.0217)	(1.9702)	(1.9680)	(1.6858)
Missing Undergraduate Institution	-0.0001	0.0029	-0.0989	0.0142	0.0132	0.0035	-2.8028	-2.8051	-1.0271
	(0.0196)	(0.0184)	(0.0902)	(0.0252)	(0.0251)	(0.0233)	(2.1499)	(2.1460)	(1.8264)
Graduate Degree	-0.0075	-0.0103	-0.0015	0.0207	0.0220	0.0225	-0.6906	-0.7648	-0.9095
	(0.0129)	(0.0124)	(0.0596)	(0.0160)	(0.0159)	(0.0147)	(1.3406)	(1.3395)	(1.1381)
Work Experience	0.0226*	0.0164	-0.0339	0.0103	0.0110	0.0170	0.1911	0.1658	-1.2044
	(0.0129)	(0.0121)	(0.0551)	(0.0159)	(0.0158)	(0.0147)	(1.3422)	(1.3408)	(1.1533)
#Professor Recommenders	0.0017	-0.0043	0.0281	0.0005	0.0010	-0.0036	-0.1044	-0.0824	0.5523
	(0.0084)	(0.0078)	(0.0423)	(0.0104)	(0.0104)	(0.0094)	(0.8933)	(0.8941)	(0.7465)
#Prolific Recommenders	-0.0034	-0.0005	0.0026	0.0004	-0.0000	0.0038	0.1557	0.1484	-0.4862
	(0.0059)	(0.0054)	(0.0262)	(0.0071)	(0.0071)	(0.0068)	(0.6197)	(0.6196)	(0.5384)
Missing GRE	0.0266	0.0056		0.0429	0.0453	0.0316	-5.6177	-5.5961	-5.0848
	(0.0333)	(0.0296)		(0.0391)	(0.0390)	(0.0389)	(3.7950)	(3.7934)	(3.5412)
Missing Undergraduate GPA	0.0177	0.0131	0.0452	-0.0181	-0.0176	-0.0096	2.4479	2.4184	1.9026
	(0.0165)	(0.0154)	(0.0847)	(0.0190)	(0.0190)	(0.0181)	(1.6942)	(1.6921)	(1.4526)
Missing TOEFL/IELTS	-0.0154	-0.0060	-0.0468	0.0267	0.0248	0.0270*	-0.7067	-0.6609	-1.3685
	(0.0143)	(0.0133)	(0.0659)	(0.0175)	(0.0175)	(0.0162)	(1.4590)	(1.4604)	(1.2133)
Committee Score		0.0028	-0.0156		0.0064	0.0039		-0.7782	-0.2153
		(0.0032)	(0.0165)		(0.0069)	(0.0066)		(0.5455)	(0.5111)
#Math Courses	0.0056	0.0058	-0.0262	-0.0174	-0.0178*	-0.0208**	0.5535	0.5650	1.3390**
	(0.0061)	(0.0056)	(0.0248)	(0.0107)	(0.0106)	(0.0101)	(0.8095)	(0.8059)	(0.6591)
Advanced Math	0.0020	-0.0026	0.1534	-0.0224	-0.0223	-0.0335	0.5450	0.5752	2.7066
	(0.0229)	(0.0210)	(0.1122)	(0.0411)	(0.0411)	(0.0384)	(3.0339)	(3.0362)	(2.4720)
Letters Length/100	-0.0022	-0.0010	-0.0089	-0.0144***	-0.0144***	-0.0144***	0.7512**	0.7632**	0.9221***
	(0.0029)	(0.0028)	(0.0137)	(0.0052)	(0.0051)	(0.0050)	(0.3694)	(0.3693)	(0.3050)
Letters Terms (%) PC1	0.0025	-0.0006	0.0059	0.0079	0.0088	0.0067	0.5318	0.5530	0.4545
	(0.0069)	(0.0063)	(0.0325)	(0.0113)	(0.0114)	(0.0110)	(0.8858)	(0.8889)	(0.7413)
Letters Terms (%) PC2	-0.0206*	-0.0181*	-0.0765*	0.0110	0.0098	0.0156	-1.3380	-1.2911	-2.2237**
	(0.0108)	(0.0095)	(0.0462)	(0.0165)	(0.0165)	(0.0147)	(1.2609)	(1.2651)	(0.9607)
CV Length/100	0.0086**	0.0061	0.0159	-0.0049	-0.0046	-0.0011	1.1643**	1.1781**	0.2350
	(0.0043)	(0.0039)	(0.0210)	(0.0075)	(0.0075)	(0.0076)	(0.5802)	(0.5825)	(0.4688)
CV Coding	-0.0018	-0.0036	-0.0130	0.0094	0.0098*	0.0094*	-0.7736	-0.7852	-0.4499
	(0.0036)	(0.0034)	(0.0167)	(0.0059)	(0.0059)	(0.0056)	(0.4887)	(0.4894)	(0.3909)
CV Publication/Working Paper	0.0273	0.0230	0.2097*	0.0152	0.0162	0.0131	1.2881	1.1348	2.4557
	(0.0252)	(0.0226)	(0.1128)	(0.0447)	(0.0447)	(0.0411)	(3.4400)	(3.4515)	(2.7307)
Essay Length/100	0.0060	0.0060	-0.0087	-0.0091	-0.0087	-0.0082	0.2585	0.2058	0.1418
	(0.0068)	(0.0063)	(0.0231)	(0.0107)	(0.0108)	(0.0103)	(0.8075)	(0.8107)	(0.6133)
Essay Specific Professor	-0.0156	-0.0190	-0.0173	-0.0075	-0.0072	0.0001	0.5915	0.5719	-1.3828
	(0.0227)	(0.0207)	(0.0978)	(0.0380)	(0.0382)	(0.0357)	(2.9760)	(2.9757)	(2.4463)
Essay Economics Terms (%)	-0.0011	-0.0007	-0.0063	-0.0012	-0.0011	-0.0026	0.1065	0.1385	0.2971
	(0.0048)	(0.0044)	(0.0207)	(0.0074)	(0.0075)	(0.0074)	(0.6168)	(0.6173)	(0.5179)
Essay Research Terms (%)	0.0027	0.0005	-0.0072	0.0143	0.0148	0.0139	-0.8742	-0.8925	-0.7975
	(0.0058)	(0.0054)	(0.0245)	(0.0096)	(0.0096)	(0.0085)	(0.7127)	(0.7141)	(0.5530)
Essay Positive Terms (%)	0.0016	0.0006	0.0575***	0.0119*	0.0118*	0.0112	-0.7274	-0.6915	-0.5104
	(0.0042)	(0.0037)	(0.0182)	(0.0070)	(0.0071)	(0.0072)	(0.5394)	(0.5418)	(0.4832)
Essay Negative Terms (%)	0.0030	0.0024	-0.0114	-0.0042	-0.0038	-0.0020	0.2345	0.2207	0.1278
	(0.0028)	(0.0027)	(0.0120)	(0.0049)	(0.0049)	(0.0045)	(0.4029)	(0.4032)	(0.3117)
Missing Committee Score		-0.3296***			0.0360**	0.0487***		-0.4418	-2.7191**
		(0.0330)			(0.0173)	(0.0162)		(1.4166)	(1.2447)
Missing Application Pdf	-0.0826***	-0.0045	0.1472	0.0079	-0.0076	-0.0130	2.1124	2.2652	2.4846
	(0.0225)	(0.0210)	(0.1064)	(0.0360)	(0.0366)	(0.0347)	(2.7720)	(2.8317)	(2.3765)
Open House			0.1654						
			(0.1278)						
Program Rank						-0.0006**			0.0516***
						(0.0002)			(0.0169)
Missing Program Rank						-0.3060***			41.4849***
						(0.0381)			(2.4608)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	237	2329	2329	2329	2329	2329	2329

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 10: Lasso Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.1107								
2017		-0.4928							
2018		-2.4456	0.0186						
2019		-1.7752							
Winsorized Age				0.0084	0.0047	0.0184			
Female	0.2674	0.5589							
U.S.	-0.1959							0.0477	
International Asian	-0.8686	-0.8651	0.1271				-0.5295	-0.7927	
GRE Verbal Percentile	0.0227	0.0047	-0.0015	0.0015			-0.0085		
GRE Quantitative Percentile	0.1027	0.0174	-0.0273	0.0179	0.0125		-0.2706	-0.2243	-0.0070
GRE Analytic Percentile	0.0056	0.0035							
Undergraduate GPA	0.9812	0.7135				-0.1316			0.2561
TOEFL/IELTS	0.3055		-0.2943	0.0300		0.0934	-0.6903	-0.5850	-2.2692
Elite U.S. University	0.4024								
Elite U.S. Liberal Arts College		0.1112							
Elite Foreign University		-0.4025							
Other Foreign University	0.0790	0.0940	0.1998						-0.2265
Missing Undergraduate Institution			-0.0036						
Graduate Degree	0.1178								
Work Experience	0.6006	0.4068	0.0404	0.0961	0.0470	0.1976			-2.0809
#Professor Recommenders									0.0922
#Prolific Recommenders	0.1534	0.0301	-0.0870	0.0526	0.0363		-0.0789	-0.0358	
Missing GRE	-0.6534			-0.0697					
Missing Undergraduate GPA	-0.2348	-0.1148							
Missing TOEFL/IELTS	0.1447			-0.1243	-0.1295		1.8402	2.2548	
Committee Score		3.4243			0.0353			-0.4899	
#Math Courses	-0.0553	-0.0354							
Advanced Math	0.4302	0.7352							
Letters Length/100	0.1196	0.0587							
Letters Terms (%) PC1	0.0746	0.1485	0.1258			0.0102			
Letters Terms (%) PC2	-0.0162	-0.2901							
CV Length/100	0.0219	0.0298		0.0014			-0.0286	-0.1179	
CV Coding	-0.0038	-0.1754		0.0165	0.0076		-0.2540	-0.4154	-0.0998
CV Publication/Working Paper	0.5563	0.6436		0.0892			-0.0987		
Essay Length/100	0.1202	0.0295							
Essay Specific Professor	0.3403	0.3958	0.0674						
Essay Economics Terms (%)	-0.0016			-0.0405	-0.0325	-0.0104	0.4473	0.6570	0.2117
Essay Research Terms (%)		-0.0278						-0.1506	
Essay Positive Terms (%)	-0.0049	-0.0382		0.0192	0.0140	0.0267			
Essay Negative Terms (%)		-0.0417							
Missing Committee Score		-0.8416			-0.2615	-0.0714		4.0146	0.7971
Missing Application Pdf	-1.0365								
Open House			1.0982						
Program Rank						-0.0082			0.1385
Missing Program Rank						-2.2557			19.4273
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.2841	0.8007	0.1281	0.0196	0.0213	0.2265	0.0206	0.0278	0.2693
CV Pseudo-R^2	0.2441	0.7605	0.0596	0.0082	0.0129	0.2199	0.0121	0.0183	0.2638

C Additional Structural Results

Table 11: Unrestricted Dynamic Estimates

	Committee Score	Transition		Accept		Success	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-2.7966	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.4553**	-0.5635**
	(2.1751)	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(2.1831)	(0.2752)
2016	0.0915	-0.2896	-0.0435	-0.5294	-0.0237	0.0762	0.0096
	(0.1856)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1883)	(0.0238)
2017	0.2354	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1266	0.0160
	(0.2130)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1929)	(0.0244)
2018	0.5999***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0095	0.0012
	(0.1777)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.2011)	(0.0254)
2019	0.6118***	0.2905	0.0437	-3.0765***	-0.1377***	0.0114	0.0014
	(0.1907)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1974)	(0.0250)
Winsorized Age	-0.0621***	0.0628***	0.0094***	-0.0133	-0.0006	0.0428*	0.0054*
	(0.0235)	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0235)	(0.0030)
Female	-0.0543	0.3919***	0.0589***	0.8909**	0.0399***	-0.0422	-0.0053
	(0.1125)	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1246)	(0.0158)
U.S.	-0.5071**	-0.1124	-0.0169	0.0293	0.0013	0.1597	0.0202
	(0.2043)	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2219)	(0.0280)
International Asian	-0.7530***	0.0087	0.0013	-1.1484***	-0.0514***	-0.0921	-0.0116
	(0.1452)	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1540)	(0.0195)
GRE Verbal Percentile	0.0066*	0.0326***	0.0049***	0.0039	0.0002	0.0006	0.0001
	(0.0036)	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0036)	(0.0005)
GRE Quantitative Percentile	0.0447***	0.1829***	0.0275***	0.0402	0.0018	0.0028	0.0004
	(0.0111)	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0080)	(0.0010)
GRE Analytic Percentile	0.0029	0.0152***	0.0023***	0.0078	0.0004	-0.0022	-0.0003
	(0.0027)	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0029)	(0.0004)
TOEFL/IELTS	0.4407***	-0.0097	-0.0015	-0.1286	-0.0058	0.3375*	0.0427*
	(0.1696)	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1752)	(0.0221)
Elite U.S. University	0.5507***	0.9066***	0.1363***	-0.5317	-0.0238	-0.0941	-0.0119
	(0.1994)	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2319)	(0.0293)
Elite U.S. Liberal Arts College	-0.0711	0.3268	0.0491	0.3253	0.0146	-0.0866	-0.0110
	(0.3683)	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3922)	(0.0496)
Elite Foreign University	0.2902	0.0518	0.0078	-0.9107	-0.0408	0.0958	0.0121
	(0.2991)	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3439)	(0.0435)
Other Foreign University	0.4797**	0.1161	0.0174	0.2293	0.0103	0.1627	0.0206
	(0.2022)	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2168)	(0.0274)
Missing Undergraduate Institution	0.5097**	-0.4337**	-0.0652**	0.0505	0.0023	0.1768	0.0224
	(0.2157)	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2321)	(0.0294)
Graduate Degree	-0.0757	1.0229***	0.1537***	-0.3419	-0.0153	-0.1684	-0.0213
	(0.1279)	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1367)	(0.0172)
Work Experience	0.3458**	0.7870***	0.1183***	0.5914*	0.0265*	0.2740*	0.0347*
	(0.1345)	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1424)	(0.0179)
#Professor Recommenders	0.0618	-0.0211	-0.0032	-0.1896	-0.0085	-0.0622	-0.0079
	(0.0769)	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0859)	(0.0109)
#Prolific Recommenders	0.1667***	0.1325***	0.0199***	0.1903	0.0085	0.0667	0.0084
	(0.0548)	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0582)	(0.0073)
Missing GRE	-1.6087*	-1.8142***	-0.2727***	1.6533	0.0740	0.0247	0.0031
	(0.9422)	(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3772)	(0.0477)
Missing TOEFL/IELTS	0.3199**	0.0947	0.0142	0.4511	0.0202	-0.0036	-0.0005
	(0.1319)	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1435)	(0.0181)
Committee Score				3.7089***	0.1660***	-0.0006	-0.0001
				(0.3984)	(0.0065)	(0.0486)	(0.0062)
#Math Courses	-0.0790			-0.2494	-0.0112	0.0007	0.0001
	(0.0519)			(0.2386)	(0.0108)	(0.0879)	(0.0111)
Advanced Math	0.2979			1.1003	0.0492	-0.2220	-0.0281
	(0.1889)			(0.7916)	(0.0347)	(0.2926)	(0.0370)
Letters Length/100	0.0858***			0.0770	0.0034	-0.0135	-0.0017
	(0.0245)			(0.1080)	(0.0048)	(0.0385)	(0.0049)
Letters Terms (%) PC1	0.1027*			0.2451	0.0110	0.1278	0.0162
	(0.0551)			(0.2489)	(0.0112)	(0.0911)	(0.0115)
Letters Terms (%) PC2	-0.0208			-0.3767	-0.0169	-0.1364	-0.0172
	(0.0823)			(0.3450)	(0.0153)	(0.1257)	(0.0159)
CV Length/100	0.0513			0.0391	0.0017	0.0347	0.0044
	(0.0329)			(0.1060)	(0.0047)	(0.0592)	(0.0075)
CV Coding	0.0381			-0.2228*	-0.0100*	0.1024	0.0130
	(0.0424)			(0.1301)	(0.0060)	(0.0651)	(0.0082)
CV Publication/Working Paper	0.5482***			0.7611	0.0341	0.1440	0.0182
	(0.1912)			(0.8128)	(0.0364)	(0.3359)	(0.0425)
Essay Length/100	0.0927			0.2047	0.0092	-0.0718	-0.0091
	(0.0597)			(0.3483)	(0.0157)	(0.0818)	(0.0103)
Essay Specific Professor	0.1411			0.4125	0.0185	0.2894	0.0366
	(0.1744)			(0.6138)	(0.0275)	(0.2770)	(0.0350)
Essay Economics Terms (%)	0.0287			0.0154	0.0007	-0.0854	-0.0108
	(0.0312)			(0.0814)	(0.0036)	(0.0614)	(0.0077)
Essay Research Terms (%)	0.0396			-0.0751	-0.0034	0.0146	0.0018
	(0.0426)			(0.1492)	(0.0067)	(0.0672)	(0.0085)
Essay Positive Terms (%)	0.0171			-0.0689	-0.0031	0.0977*	0.0124*
	(0.0345)			(0.0968)	(0.0043)	(0.0539)	(0.0068)
Essay Negative Terms (%)	0.0441			-0.0863	-0.0039	-0.0441	-0.0056
	(0.0443)			(0.1368)	(0.0061)	(0.0745)	(0.0094)
Missing Committee Score						-0.2179	-0.0276
						(0.1515)	(0.0191)
Missing Application Pdf	-0.5376***			0.2614	0.0117	0.2498	0.0316
	(0.1841)			(0.7005)	(0.0311)	(0.2756)	(0.0348)
Program Rank						-0.0086***	-0.0011***
						(0.0021)	(0.0003)
Missing Program Rank						-3.1438***	-0.3976***
						(0.4076)	(0.0514)
$\sqrt{\text{MSE}}$	1.6467***						
	(0.0370)						
Obs | Param | LL | AIC	2329	582	-14135.6	29435.1			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 12: Restricted Dynamic Estimates

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-5.6768**	-0.9860***	0.3731	-0.5503***	36.58%	-0.7153*	
	(2.3401)	(0.1000)	(0.3067, 0.4539)	(0.1561)	(29.81%, 43.92%)	(0.3751)	
2016	0.0001	0.0171	0.3795	0.0269	37.21%	0.0155	
	(0.1827)	(0.1265)	(0.3025, 0.4761)	(0.2038)	(29.20%, 45.99%)	(0.2039)	
2017	0.4867**	0.1006	0.4126	0.1561	40.27%	0.1404	
	(0.2144)	(0.1085)	(0.3563, 0.4778)	(0.1786)	(34.52%, 46.30%)	(0.1795)	
2018	0.9329***	0.0277	0.3836	0.0473	37.68%	-0.0004	
	(0.1967)	(0.1208)	(0.3271, 0.4497)	(0.1943)	(31.79%, 43.97%)	(0.2015)	
2019	0.6228***	0.0411	0.3887	0.0667	38.14%	0.0232	
	(0.1963)	(0.1142)	(0.3268, 0.4624)	(0.1852)	(31.89%, 44.81%)	(0.1863)	
Winsorized Age	-0.0454*					-0.0004	
	(0.0246)					(0.0009)	
Female	-0.1026					0.0114	
	(0.1055)					(0.0091)	
U.S.	-0.3132					0.0035	
	(0.2098)					(0.0081)	
International Asian	-0.3624**					-0.0051	
	(0.1577)					(0.0070)	
GRE Verbal Percentile	0.0110***					0.0002	
	(0.0036)					(0.0003)	
GRE Quantitative Percentile	0.0580***					0.0007	
	(0.0107)					(0.0008)	
GRE Analytic Percentile	0.0026					0.0001	
	(0.0028)					(0.0001)	
TOEFL/IELTS	0.3161*					0.0007	
	(0.1703)					(0.0073)	
Elite U.S. University	0.5975***					-0.0108	
	(0.1775)					(0.0108)	
Elite U.S. Liberal Arts College	-0.4500					0.0047	
	(0.3224)					(0.0151)	
Elite Foreign University	0.2329					-0.0086	
	(0.2727)					(0.0136)	
Other Foreign University	0.4356**					0.0048	
	(0.1932)					(0.0112)	
Missing Undergraduate Institution	0.3922*					-0.0001	
	(0.2038)					(0.0098)	
Graduate Degree	-0.0531					-0.0022	
	(0.1498)					(0.0063)	
Work Experience	0.0819					0.0088	
	(0.1415)					(0.0079)	
#Professor Recommenders	0.1025					-0.0005	
	(0.0784)					(0.0037)	
#Prolific Recommenders	0.0536					-0.0002	
	(0.0598)					(0.0026)	
Missing GRE	-1.2274					0.0216	
	(0.9140)					(0.0228)	
Missing TOEFL/IELTS	0.2149*					0.0036	
	(0.1252)					(0.0061)	
Committee Score						0.0514	
						(0.0355)	
#Math Courses	-0.1443					-0.0029	
	(0.0920)					(0.0047)	
Advanced Math	0.4000					-0.0022	
	(0.2955)					(0.0108)	
Letters Length/100	0.0906**					0.0005	
	(0.0439)					(0.0020)	
Letters Terms (%) PC1	0.2131**					0.0053	
	(0.0942)					(0.0056)	
Letters Terms (%) PC2	-0.1480					-0.0045	
	(0.1963)					(0.0067)	
CV Length/100	0.0283					0.0013	
	(0.0724)					(0.0021)	
CV Coding	0.0306					-0.0044	
	(0.0981)					(0.0040)	
CV Publication/Working Paper	1.1465***					0.0006	
	(0.4362)					(0.0134)	
Essay Length/100	0.2719***					0.0080	
	(0.1044)					(0.0090)	
Essay Specific Professor	0.1374					-0.0060	
	(0.2212)					(0.0136)	
Essay Economics Terms (%)	0.0664					0.0004	
	(0.0550)					(0.0015)	
Essay Research Terms (%)	0.1364*					-0.0002	
	(0.0821)					(0.0030)	
Essay Positive Terms (%)	0.0422					-0.0004	
	(0.0718)					(0.0029)	
Essay Negative Terms (%)	0.1003					0.0008	
	(0.0766)					(0.0033)	
Missing Committee Score						-0.2001	
						(0.1292)	
Missing Application Pdf	-0.7069**					-0.0267	
	(0.2763)					(0.0180)	
Program Rank						-0.0080***	
						(0.0020)	
Missing Program Rank						-3.0773***	
						(0.4051)	
$\sqrt{\text{MSE}}$	1.7300***						
	(0.0475)						
Sigma							0.0020
							(0.0013)
Obs | Param | LL | AIC	2329	529	-14131.0	29319.9			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 13: Unrestricted Static Estimates

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.6847***	-0.5998***	-4.4553**	-0.5635**
	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(1.6180)	(0.2054)	(2.1831)	(0.2752)
2016	-0.2896	-0.0435	-0.5294	-0.0237	0.0683	0.0087	0.0762	0.0096
	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1852)	(0.0237)	(0.1883)	(0.0238)
2017	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1166	0.0149	0.1266	0.0160
	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1877)	(0.0240)	(0.1929)	(0.0244)
2018	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0112	0.0014	0.0095	0.0012
	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.1920)	(0.0246)	(0.2011)	(0.0254)
2019	0.2905	0.0437	-3.0765***	-0.1377***	0.0177	0.0023	0.0114	0.0014
	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1913)	(0.0245)	(0.1974)	(0.0250)
Winsorized Age	0.0628***	0.0094***	-0.0133	-0.0006	0.0439*	0.0056*	0.0428*	0.0054*
	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0229)	(0.0029)	(0.0235)	(0.0030)
Female	0.3919***	0.0589***	0.8909**	0.0399***	-0.0342	-0.0044	-0.0422	-0.0053
	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1232)	(0.0158)	(0.1246)	(0.0158)
U.S.	-0.1124	-0.0169	0.0293	0.0013	0.1634	0.0209	0.1597	0.0202
	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2198)	(0.0281)	(0.2219)	(0.0280)
International Asian	0.0087	0.0013	-1.1484***	-0.0514***	-0.0894	-0.0114	-0.0921	-0.0116
	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1489)	(0.0191)	(0.1540)	(0.0195)
GRE Verbal Percentile	0.0326***	0.0049***	0.0039	0.0002	0.0015	0.0002	0.0006	0.0001
	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0034)	(0.0004)	(0.0036)	(0.0005)
GRE Quantitative Percentile	0.1829***	0.0275***	0.0402	0.0018	0.0065	0.0008	0.0028	0.0004
	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0076)	(0.0010)	(0.0080)	(0.0010)
GRE Analytic Percentile	0.0152***	0.0023***	0.0078	0.0004	-0.0011	-0.0001	-0.0022	-0.0003
	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0028)	(0.0004)	(0.0029)	(0.0004)
TOEFL/IELTS	-0.0097	-0.0015	-0.1286	-0.0058	0.3267*	0.0418*	0.3375*	0.0427*
	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1721)	(0.0220)	(0.1752)	(0.0221)
Elite U.S. University	0.9066***	0.1363***	-0.5317	-0.0238	-0.0506	-0.0065	-0.0941	-0.0119
	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2283)	(0.0292)	(0.2319)	(0.0293)
Elite U.S. Liberal Arts College	0.3268	0.0491	0.3253	0.0146	-0.0812	-0.0104	-0.0866	-0.0110
	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3863)	(0.0495)	(0.3922)	(0.0496)
Elite Foreign University	0.0518	0.0078	-0.9107	-0.0408	0.0910	0.0117	0.0958	0.0121
	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3413)	(0.0437)	(0.3439)	(0.0435)
Other Foreign University	0.1161	0.0174	0.2293	0.0103	0.2023	0.0259	0.1627	0.0206
	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2140)	(0.0274)	(0.2168)	(0.0274)
Missing Undergraduate Institution	-0.4337**	-0.0652**	0.0505	0.0023	0.1941	0.0249	0.1768	0.0224
	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2292)	(0.0293)	(0.2321)	(0.0294)
Graduate Degree	1.0229***	0.1537***	-0.3419	-0.0153	-0.1377	-0.0176	-0.1684	-0.0213
	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1312)	(0.0168)	(0.1367)	(0.0172)
Work Experience	0.7870***	0.1183***	0.5914*	0.0265*	0.3144**	0.0403**	0.2740*	0.0347*
	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1361)	(0.0173)	(0.1424)	(0.0179)
#Professor Recommenders	-0.0211	-0.0032	-0.1896	-0.0085	-0.0570	-0.0073	-0.0622	-0.0079
	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0846)	(0.0108)	(0.0859)	(0.0109)
#Prolific Recommenders	0.1325***	0.0199***	0.1903	0.0085	0.0559	0.0072	0.0667	0.0084
	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0574)	(0.0073)	(0.0582)	(0.0073)
Missing GRE	-1.8142***	-0.2727***	1.6533	0.0740	-0.0087	-0.0011	0.0247	0.0031
	(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3716)	(0.0476)	(0.3772)	(0.0477)
Missing TOEFL/IELTS	0.0947	0.0142	0.4511	0.0202	-0.0080	-0.0010	-0.0036	-0.0005
	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1419)	(0.0182)	(0.1435)	(0.0181)
Committee Score			3.7089***	0.1660***			-0.0006	-0.0001
			(0.3984)	(0.0065)			(0.0486)	(0.0062)
#Math Courses			-0.2494	-0.0112			0.0007	0.0001
			(0.2386)	(0.0108)			(0.0879)	(0.0111)
Advanced Math			1.1003	0.0492			-0.2220	-0.0281
			(0.7916)	(0.0347)			(0.2926)	(0.0370)
Letters Length/100			0.0770	0.0034			-0.0135	-0.0017
			(0.1080)	(0.0048)			(0.0385)	(0.0049)
Letters Terms (%) PC1			0.2451	0.0110			0.1278	0.0162
			(0.2489)	(0.0112)			(0.0911)	(0.0115)
Letters Terms (%) PC2			-0.3767	-0.0169			-0.1364	-0.0172
			(0.3450)	(0.0153)			(0.1257)	(0.0159)
CV Length/100			0.0391	0.0017			0.0347	0.0044
			(0.1060)	(0.0047)			(0.0592)	(0.0075)
CV Coding			-0.2228*	-0.0100*			0.1024	0.0130
			(0.1301)	(0.0060)			(0.0651)	(0.0082)
CV Publication/Working Paper			0.7611	0.0341			0.1440	0.0182
			(0.8128)	(0.0364)			(0.3359)	(0.0425)
Essay Length/100			0.2047	0.0092			-0.0718	-0.0091
			(0.3483)	(0.0157)			(0.0818)	(0.0103)
Essay Specific Professor			0.4125	0.0185			0.2894	0.0366
			(0.6138)	(0.0275)			(0.2770)	(0.0350)
Essay Economics Terms (%)			0.0154	0.0007			-0.0854	-0.0108
			(0.0814)	(0.0036)			(0.0614)	(0.0077)
Essay Research Terms (%)			-0.0751	-0.0034			0.0146	0.0018
			(0.1492)	(0.0067)			(0.0672)	(0.0085)
Essay Positive Terms (%)			-0.0689	-0.0031			0.0977*	0.0124*
			(0.0968)	(0.0043)			(0.0539)	(0.0068)
Essay Negative Terms (%)			-0.0863	-0.0039			-0.0441	-0.0056
			(0.1368)	(0.0061)			(0.0745)	(0.0094)
Missing Committee Score							-0.2179	-0.0276
							(0.1515)	(0.0191)
Missing Application Pdf			0.2614	0.0117			0.2498	0.0316
			(0.7005)	(0.0311)			(0.2756)	(0.0348)
Program Rank					-0.0084***	-0.0011***	-0.0086***	-0.0011***
					(0.0020)	(0.0003)	(0.0021)	(0.0003)
Missing Program Rank					-3.1385***	-0.4018***	-3.1438***	-0.3976***
					(0.4051)	(0.0517)	(0.4076)	(0.0514)
Obs | Param | LL | AIC	2329	133	-2964.5	6195.0				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 14: Restricted Static Estimates

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	-0.8781***	29.36%	-0.4389***	39.20%	-1.2521***	-0.9262**	
	(0.1628)	(23.20%, 36.38%)	(0.1432)	(32.75%, 46.05%)	(0.4466)	(0.4540)	
2016	-0.0091	29.17%	0.0554	40.53%	-0.0178	0.0421	
	(0.2466)	(22.21%, 37.27%)	(0.1767)	(33.55%, 47.92%)	(0.2477)	(0.1771)	
2017	0.3482*	37.06%	0.1035	41.69%	0.3447*	0.0698	
	(0.1963)	(32.10%, 42.29%)	(0.1727)	(34.83%, 48.90%)	(0.1964)	(0.1725)	
2018	0.3936**	38.12%	0.0040	39.30%	0.4390**	-0.0912	
	(0.1856)	(33.98%, 42.44%)	(0.1908)	(31.80%, 47.33%)	(0.1859)	(0.1872)	
2019	0.2163	34.04%	0.0175	39.62%	0.2258	-0.0617	
	(0.2017)	(28.67%, 39.85%)	(0.1791)	(32.54%, 47.16%)	(0.2018)	(0.1778)	
Winsorized Age					0.0018	0.0000	
					(0.0012)	(0.0016)	
Female					0.0118	0.0233	
					(0.0073)	(0.0146)	
U.S.					-0.0034	0.0022	
					(0.0058)	(0.0131)	
International Asian					-0.0006	-0.0302	
					(0.0041)	(0.0187)	
GRE Verbal Percentile					0.0010*	0.0001	
					(0.0005)	(0.0003)	
GRE Quantitative Percentile					0.0055*	0.0011	
					(0.0032)	(0.0011)	
GRE Analytic Percentile					0.0005*	0.0002	
					(0.0003)	(0.0002)	
TOEFL/IELTS					-0.0009	-0.0015	
					(0.0047)	(0.0144)	
Elite U.S. University					0.0252	-0.0131	
					(0.0155)	(0.0178)	
Elite U.S. Liberal Arts College					0.0087	0.0095	
					(0.0113)	(0.0216)	
Elite Foreign University					0.0013	-0.0229	
					(0.0086)	(0.0272)	
Other Foreign University					0.0038	0.0075	
					(0.0062)	(0.0180)	
Missing Undergraduate Institution					-0.0119	0.0030	
					(0.0084)	(0.0173)	
Graduate Degree					0.0306*	-0.0103	
					(0.0175)	(0.0117)	
Work Experience					0.0235*	0.0168	
					(0.0141)	(0.0130)	
#Professor Recommenders					-0.0004	-0.0057	
					(0.0025)	(0.0074)	
#Prolific Recommenders					0.0039	0.0052	
					(0.0026)	(0.0053)	
Missing GRE					-0.0566	0.0458	
					(0.0361)	(0.0532)	
Missing TOEFL/IELTS					0.0029	0.0123	
					(0.0041)	(0.0127)	
Committee Score						0.0961*	
						(0.0491)	
#Math Courses						-0.0064	
						(0.0069)	
Advanced Math						0.0287	
						(0.0244)	
Letters Length/100						0.0019	
						(0.0029)	
Letters Terms (%) PC1						0.0063	
						(0.0072)	
Letters Terms (%) PC2						-0.0103	
						(0.0104)	
CV Length/100						0.0010	
						(0.0027)	
CV Coding						-0.0054	
						(0.0043)	
CV Publication/Working Paper						0.0207	
						(0.0245)	
Essay Length/100						0.0050	
						(0.0089)	
Essay Specific Professor						0.0106	
						(0.0168)	
Essay Economics Terms (%)						0.0002	
						(0.0021)	
Essay Research Terms (%)						-0.0019	
						(0.0039)	
Essay Positive Terms (%)						-0.0014	
						(0.0025)	
Essay Negative Terms (%)						-0.0023	
						(0.0038)	
Missing Committee Score						-0.2328**	
						(0.1162)	
Missing Application Pdf						0.0079	
						(0.0185)	
Program Rank					-0.0085***	-0.0078***	
					(0.0021)	(0.0020)	
Missing Program Rank					-3.0030***	-3.1171***	
					(0.4080)	(0.4041)	
Sigma							0.0064*
							(0.0035)
Obs | Param | LL | AIC	2329	80	-2992.0	6144.1			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 15: Unrestricted Myopic Estimates

	Committee Score	Transition		Accept		Success	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	4.0405***	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.4553**	-0.5635**
	(0.9990)	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(2.1831)	(0.2752)
2016	0.1701	-0.2896	-0.0435	-0.5294	-0.0237	0.0762	0.0096
	(0.1966)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1883)	(0.0238)
2017	0.4355**	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1266	0.0160
	(0.2214)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1929)	(0.0244)
2018	0.8410***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0095	0.0012
	(0.1873)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.2011)	(0.0254)
2019	0.7560***	0.2905	0.0437	-3.0765***	-0.1377***	0.0114	0.0014
	(0.1987)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1974)	(0.0250)
Winsorized Age		0.0628***	0.0094***	-0.0133	-0.0006	0.0428*	0.0054*
		(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0235)	(0.0030)
Female		0.3919***	0.0589***	0.8909**	0.0399***	-0.0422	-0.0053
		(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1246)	(0.0158)
U.S.		-0.1124	-0.0169	0.0293	0.0013	0.1597	0.0202
		(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2219)	(0.0280)
International Asian		0.0087	0.0013	-1.1484***	-0.0514***	-0.0921	-0.0116
		(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1540)	(0.0195)
GRE Verbal Percentile		0.0326***	0.0049***	0.0039	0.0002	0.0006	0.0001
		(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0036)	(0.0005)
GRE Quantitative Percentile		0.1829***	0.0275***	0.0402	0.0018	0.0028	0.0004
		(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0080)	(0.0010)
GRE Analytic Percentile		0.0152***	0.0023***	0.0078	0.0004	-0.0022	-0.0003
		(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0029)	(0.0004)
TOEFL/IELTS		-0.0097	-0.0015	-0.1286	-0.0058	0.3375*	0.0427*
		(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1752)	(0.0221)
Elite U.S. University		0.9066***	0.1363***	-0.5317	-0.0238	-0.0941	-0.0119
		(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2319)	(0.0293)
Elite U.S. Liberal Arts College		0.3268	0.0491	0.3253	0.0146	-0.0866	-0.0110
		(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3922)	(0.0496)
Elite Foreign University		0.0518	0.0078	-0.9107	-0.0408	0.0958	0.0121
		(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3439)	(0.0435)
Other Foreign University		0.1161	0.0174	0.2293	0.0103	0.1627	0.0206
		(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2168)	(0.0274)
Missing Undergraduate Institution		-0.4337**	-0.0652**	0.0505	0.0023	0.1768	0.0224
		(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2321)	(0.0294)
Graduate Degree		1.0229***	0.1537***	-0.3419	-0.0153	-0.1684	-0.0213
		(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1367)	(0.0172)
Work Experience		0.7870***	0.1183***	0.5914*	0.0265*	0.2740*	0.0347*
		(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1424)	(0.0179)
#Professor Recommenders		-0.0211	-0.0032	-0.1896	-0.0085	-0.0622	-0.0079
		(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0859)	(0.0109)
#Prolific Recommenders		0.1325***	0.0199***	0.1903	0.0085	0.0667	0.0084
		(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0582)	(0.0073)
Missing GRE		-1.8142***	-0.2727***	1.6533	0.0740	0.0247	0.0031
		(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3772)	(0.0477)
Missing TOEFL/IELTS		0.0947	0.0142	0.4511	0.0202	-0.0036	-0.0005
		(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1435)	(0.0181)
Committee Score				3.7089***	0.1660***	-0.0006	-0.0001
				(0.3984)	(0.0065)	(0.0486)	(0.0062)
#Math Courses	-0.0804			-0.2494	-0.0112	0.0007	0.0001
	(0.0520)			(0.2386)	(0.0108)	(0.0879)	(0.0111)
Advanced Math	0.3560*			1.1003	0.0492	-0.2220	-0.0281
	(0.1909)			(0.7916)	(0.0347)	(0.2926)	(0.0370)
Letters Length/100	0.1225***			0.0770	0.0034	-0.0135	-0.0017
	(0.0242)			(0.1080)	(0.0048)	(0.0385)	(0.0049)
Letters Terms (%) PC1	0.0974*			0.2451	0.0110	0.1278	0.0162
	(0.0536)			(0.2489)	(0.0112)	(0.0911)	(0.0115)
Letters Terms (%) PC2	-0.0925			-0.3767	-0.0169	-0.1364	-0.0172
	(0.0856)			(0.3450)	(0.0153)	(0.1257)	(0.0159)
CV Length/100	0.0588*			0.0391	0.0017	0.0347	0.0044
	(0.0323)			(0.1060)	(0.0047)	(0.0592)	(0.0075)
CV Coding	0.0551			-0.2228*	-0.0100*	0.1024	0.0130
	(0.0417)			(0.1301)	(0.0060)	(0.0651)	(0.0082)
CV Publication/Working Paper	0.5324***			0.7611	0.0341	0.1440	0.0182
	(0.1975)			(0.8128)	(0.0364)	(0.3359)	(0.0425)
Essay Length/100	0.0763			0.2047	0.0092	-0.0718	-0.0091
	(0.0567)			(0.3483)	(0.0157)	(0.0818)	(0.0103)
Essay Specific Professor	0.1442			0.4125	0.0185	0.2894	0.0366
	(0.1767)			(0.6138)	(0.0275)	(0.2770)	(0.0350)
Essay Economics Terms (%)	0.0042			0.0154	0.0007	-0.0854	-0.0108
	(0.0314)			(0.0814)	(0.0036)	(0.0614)	(0.0077)
Essay Research Terms (%)	0.0304			-0.0751	-0.0034	0.0146	0.0018
	(0.0446)			(0.1492)	(0.0067)	(0.0672)	(0.0085)
Essay Positive Terms (%)	0.0175			-0.0689	-0.0031	0.0977*	0.0124*
	(0.0373)			(0.0968)	(0.0043)	(0.0539)	(0.0068)
Essay Negative Terms (%)	0.0429			-0.0863	-0.0039	-0.0441	-0.0056
	(0.0460)			(0.1368)	(0.0061)	(0.0745)	(0.0094)
Missing Committee Score						-0.2179	-0.0276
						(0.1515)	(0.0191)
Missing Application Pdf	-0.4006**			0.2614	0.0117	0.2498	0.0316
	(0.1850)			(0.7005)	(0.0311)	(0.2756)	(0.0348)
Program Rank						-0.0086***	-0.0011***
						(0.0021)	(0.0003)
Missing Program Rank						-3.1438***	-0.3976***
						(0.4076)	(0.0514)
$\sqrt{MSE}$	1.7525***						
	(0.0403)						
Obs | Param | LL | AIC	2329	297	-14470.7	29535.4			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 16: Restricted Myopic Estimates

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	3.4796***	-0.9771***	0.3764	-0.5522***	36.54%	-1.2859**	
	(1.2648)	(0.0960)	(0.3118, 0.4544)	(0.1553)	(29.80%, 43.84%)	(0.5090)	
2016	0.0978	-0.0005	0.3762	-0.0048	36.42%	-0.0164	
	(0.1990)	(0.1320)	(0.2992, 0.4731)	(0.2135)	(28.21%, 45.51%)	(0.2130)	
2017	0.7100***	0.1077	0.4192	0.1614	40.35%	0.1384	
	(0.2332)	(0.1059)	(0.3701, 0.4749)	(0.1760)	(35.00%, 45.95%)	(0.1772)	
2018	1.0737***	0.0146	0.3820	0.0286	37.20%	-0.0305	
	(0.1730)	(0.1219)	(0.3237, 0.4506)	(0.1968)	(31.07%, 43.76%)	(0.1993)	
2019	0.8095***	0.0005	0.3766	0.0004	36.54%	-0.0654	
	(0.1948)	(0.1254)	(0.3086, 0.4596)	(0.2030)	(29.43%, 44.30%)	(0.2045)	
Winsorized Age						0.0010	
						(0.0007)	
Female						0.0132**	
						(0.0063)	
U.S.						0.0010	
						(0.0048)	
International Asian						-0.0075	
						(0.0051)	
GRE Verbal Percentile						0.0008**	
						(0.0003)	
GRE Quantitative Percentile						0.0039**	
						(0.0017)	
GRE Analytic Percentile						0.0004**	
						(0.0002)	
TOEFL/IELTS						0.0009	
						(0.0042)	
Elite U.S. University						0.0105	
						(0.0071)	
Elite U.S. Liberal Arts College						0.0064	
						(0.0086)	
Elite Foreign University						-0.0012	
						(0.0077)	
Other Foreign University						0.0066	
						(0.0059)	
Missing Undergraduate Institution						-0.0034	
						(0.0053)	
Graduate Degree						0.0194**	
						(0.0090)	
Work Experience						0.0189**	
						(0.0089)	
#Professor Recommenders						-0.0014	
						(0.0022)	
#Prolific Recommenders						0.0016	
						(0.0016)	
Missing GRE						-0.0449	
						(0.0282)	
Missing TOEFL/IELTS						0.0012	
						(0.0035)	
Committee Score						0.0686**	
						(0.0302)	
#Math Courses	-0.0627					-0.0011	
	(0.0689)					(0.0071)	
Advanced Math	0.3298					0.0023	
	(0.2533)					(0.0161)	
Letters Length/100	0.1467***					0.0023	
	(0.0317)					(0.0029)	
Letters Terms (%) PC1	0.1155					0.0016	
	(0.0772)					(0.0087)	
Letters Terms (%) PC2	-0.1020					-0.0069	
	(0.1152)					(0.0089)	
CV Length/100	0.0628					0.0022	
	(0.0407)					(0.0026)	
CV Coding	0.0466					-0.0051	
	(0.0580)					(0.0036)	
CV Publication/Working Paper	0.7501***					-0.0009	
	(0.2653)					(0.0155)	
Essay Length/100	0.1355					0.0083	
	(0.0858)					(0.0131)	
Essay Specific Professor	0.1387					0.0023	
	(0.2579)					(0.0205)	
Essay Economics Terms (%)	-0.0024					-0.0018	
	(0.0396)					(0.0021)	
Essay Research Terms (%)	0.0663					0.0023	
	(0.0596)					(0.0044)	
Essay Positive Terms (%)	0.0403					0.0011	
	(0.0476)					(0.0033)	
Essay Negative Terms (%)	0.0459					-0.0009	
	(0.0575)					(0.0048)	
Missing Committee Score						-0.1177	
						(0.1482)	
Missing Application Pdf	-0.9529***					-0.0359**	
	(0.2274)					(0.0157)	
Program Rank						-0.0078***	
						(0.0020)	
Missing Program Rank						-3.0747***	
						(0.4058)	
$\sqrt{\text{MSE}}$	1.8070***						
	(0.0436)						
Sigma							0.0027**
							(0.0012)
Obs | Param | LL | AIC	2329	244	-14536.1	29560.2			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 17: Unrestricted Matriculation-Dynamic Estimates

	Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-2.7966	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.4553**	-0.5635**	21.0916***	3.3374***
	(2.1751)	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(2.1831)	(0.2752)	(6.9868)	(1.0163)
2016	0.0915	-0.2896	-0.0435	-0.5294	-0.0237	0.0762	0.0096	0.3206	0.0507
	(0.1856)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1883)	(0.0238)	(0.6530)	(0.1030)
2017	0.2354	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1266	0.0160	0.9971	0.1578
	(0.2130)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1929)	(0.0244)	(0.6410)	(0.1008)
2018	0.5999***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0095	0.0012	1.6496***	0.2610***
	(0.1777)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.2011)	(0.0254)	(0.5920)	(0.0886)
2019	0.6118***	0.2905	0.0437	-3.0765***	-0.1377***	0.0114	0.0014	1.0916*	0.1727*
	(0.1907)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1974)	(0.0250)	(0.6190)	(0.0971)
Winsorized Age	-0.0621***	0.0628***	0.0094***	-0.0133	-0.0006	0.0428*	0.0054*	-0.1396*	-0.0221*
	(0.0235)	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0235)	(0.0030)	(0.0738)	(0.0113)
Female	-0.0543	0.3919***	0.0589***	0.8909**	0.0399***	-0.0422	-0.0053	-0.4562	-0.0722
	(0.1125)	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1246)	(0.0158)	(0.3550)	(0.0556)
U.S.	-0.5071**	-0.1124	-0.0169	0.0293	0.0013	0.1597	0.0202	0.9160	0.1449
	(0.2043)	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2219)	(0.0280)	(0.5700)	(0.0896)
International Asian	-0.7530***	0.0087	0.0013	-1.1484***	-0.0514***	-0.0921	-0.0116	1.1667**	0.1846***
	(0.1452)	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1540)	(0.0195)	(0.4531)	(0.0689)
GRE Verbal Percentile	0.0066*	0.0326***	0.0049***	0.0039	0.0002	0.0006	0.0001	-0.0043	-0.0007
	(0.0036)	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0036)	(0.0005)	(0.0115)	(0.0018)
GRE Quantitative Percentile	0.0447***	0.1829***	0.0275***	0.0402	0.0018	0.0028	0.0004	-0.1358***	-0.0215***
	(0.0111)	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0080)	(0.0010)	(0.0444)	(0.0064)
GRE Analytic Percentile	0.0029	0.0152***	0.0023***	0.0078	0.0004	-0.0022	-0.0003	-0.0185**	-0.0029**
	(0.0027)	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0029)	(0.0004)	(0.0087)	(0.0013)
TOEFL/IELTS	0.4407***	-0.0097	-0.0015	-0.1286	-0.0058	0.3375*	0.0427*	-0.5140	-0.0813
	(0.1696)	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1752)	(0.0221)	(0.5175)	(0.0807)
Elite U.S. University	0.5507***	0.9066***	0.1363***	-0.5317	-0.0238	-0.0941	-0.0119	-0.4342	-0.0687
	(0.1994)	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2319)	(0.0293)	(0.5475)	(0.0858)
Elite U.S. Liberal Arts College	-0.0711	0.3268	0.0491	0.3253	0.0146	-0.0866	-0.0110	1.0020	0.1586
	(0.3683)	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3922)	(0.0496)	(0.7518)	(0.1185)
Elite Foreign University	0.2902	0.0518	0.0078	-0.9107	-0.0408	0.0958	0.0121	-0.2759	-0.0437
	(0.2991)	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3439)	(0.0435)	(0.9777)	(0.1548)
Other Foreign University	0.4797**	0.1161	0.0174	0.2293	0.0103	0.1627	0.0206	0.4536	0.0718
	(0.2022)	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2168)	(0.0274)	(0.6285)	(0.0992)
Missing Undergraduate Institution	0.5097**	-0.4337**	-0.0652**	0.0505	0.0023	0.1768	0.0224	-0.8208	-0.1299
	(0.2157)	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2321)	(0.0294)	(0.6805)	(0.1058)
Graduate Degree	-0.0757	1.0229***	0.1537***	-0.3419	-0.0153	-0.1684	-0.0213	-0.0653	-0.0103
	(0.1279)	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1367)	(0.0172)	(0.4719)	(0.0746)
Work Experience	0.3458**	0.7870***	0.1183***	0.5914*	0.0265*	0.2740*	0.0347*	0.9166**	0.1450**
	(0.1345)	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1424)	(0.0179)	(0.4411)	(0.0667)
#Professor Recommenders	0.0618	-0.0211	-0.0032	-0.1896	-0.0085	-0.0622	-0.0079	0.3588	0.0568
	(0.0769)	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0859)	(0.0109)	(0.2789)	(0.0433)
#Prolific Recommenders	0.1667***	0.1325***	0.0199***	0.1903	0.0085	0.0667	0.0084	-0.2845	-0.0450
	(0.0548)	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0582)	(0.0073)	(0.1875)	(0.0288)
Missing GRE	-1.6087*	-1.8142***	-0.2727***	1.6533	0.0740	0.0247	0.0031		
	(0.9422)	(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3772)	(0.0477)		
Missing TOEFL/IELTS	0.3199**	0.0947	0.0142	0.4511	0.0202	-0.0036	-0.0005	-0.0091	-0.0014
	(0.1319)	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1435)	(0.0181)	(0.5591)	(0.0885)
Committee Score				3.7089***	0.1660***	-0.0006	-0.0001	-0.1637	-0.0259
				(0.3984)	(0.0065)	(0.0486)	(0.0062)	(0.2099)	(0.0336)
#Math Courses	-0.0790			-0.2494	-0.0112	0.0007	0.0001	-0.0182	-0.0029
	(0.0519)			(0.2386)	(0.0108)	(0.0879)	(0.0111)	(0.1671)	(0.0265)
Advanced Math	0.2979			1.1003	0.0492	-0.2220	-0.0281	-1.1143**	-0.1763**
	(0.1889)			(0.7916)	(0.0347)	(0.2926)	(0.0370)	(0.5578)	(0.0877)
Letters Length/100	0.0858***			0.0770	0.0034	-0.0135	-0.0017	-0.0744	-0.0118
	(0.0245)			(0.1080)	(0.0048)	(0.0385)	(0.0049)	(0.0711)	(0.0113)
Letters Terms (%) PC1	0.1027*			0.2451	0.0110	0.1278	0.0162	0.3997**	0.0632**
	(0.0551)			(0.2489)	(0.0112)	(0.0911)	(0.0115)	(0.1903)	(0.0287)
Letters Terms (%) PC2	-0.0208			-0.3767	-0.0169	-0.1364	-0.0172	0.2897	0.0458
	(0.0823)			(0.3450)	(0.0153)	(0.1257)	(0.0159)	(0.2460)	(0.0388)
CV Length/100	0.0513			0.0391	0.0017	0.0347	0.0044	-0.1109	-0.0175
	(0.0329)			(0.1060)	(0.0047)	(0.0592)	(0.0075)	(0.1105)	(0.0175)
CV Coding	0.0381			-0.2228*	-0.0100*	0.1024	0.0130	-0.2901**	-0.0459**
	(0.0424)			(0.1301)	(0.0060)	(0.0651)	(0.0082)	(0.1176)	(0.0181)
CV Publication/Working Paper	0.5482***			0.7611	0.0341	0.1440	0.0182	-0.3451	-0.0546
	(0.1912)			(0.8128)	(0.0364)	(0.3359)	(0.0425)	(0.5102)	(0.0812)
Essay Length/100	0.0927			0.2047	0.0092	-0.0718	-0.0091	0.0882	0.0140
	(0.0597)			(0.3483)	(0.0157)	(0.0818)	(0.0103)	(0.1386)	(0.0220)
Essay Specific Professor	0.1411			0.4125	0.0185	0.2894	0.0366	0.6812	0.1078
	(0.1744)			(0.6138)	(0.0275)	(0.2770)	(0.0350)	(0.4976)	(0.0773)
Essay Economics Terms (%)	0.0287			0.0154	0.0007	-0.0854	-0.0108	0.2244**	0.0355**
	(0.0312)			(0.0814)	(0.0036)	(0.0614)	(0.0077)	(0.1075)	(0.0169)
Essay Research Terms (%)	0.0396			-0.0751	-0.0034	0.0146	0.0018	-0.1904	-0.0301
	(0.0426)			(0.1492)	(0.0067)	(0.0672)	(0.0085)	(0.1440)	(0.0226)
Essay Positive Terms (%)	0.0171			-0.0689	-0.0031	0.0977*	0.0124*	-0.0170	-0.0027
	(0.0345)			(0.0968)	(0.0043)	(0.0539)	(0.0068)	(0.1151)	(0.0182)
Essay Negative Terms (%)	0.0441			-0.0863	-0.0039	-0.0441	-0.0056	-0.0176	-0.0028
	(0.0443)			(0.1368)	(0.0061)	(0.0745)	(0.0094)	(0.1043)	(0.0165)
Missing Committee Score						-0.2179	-0.0276		
						(0.1515)	(0.0191)		
Missing Application Pdf	-0.5376***			0.2614	0.0117	0.2498	0.0316	-0.7440	-0.1177
	(0.1841)			(0.7005)	(0.0311)	(0.2756)	(0.0348)	(0.5803)	(0.0919)
Program Rank						-0.0086***	-0.0011***		
						(0.0021)	(0.0003)		
Missing Program Rank						-3.1438***	-0.3976***		
						(0.4076)	(0.0514)		
$\sqrt{\text{MSE}}$	1.6467***								
	(0.0370)								
Obs | Param | LL | AIC	2329	621	-14259.1	29760.1					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 18: Restricted Matriculation-Dynamic Estimates

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-6.5631***	-4.9411***	0.0071	-0.5602***	36.35%	-0.8176**	1.5446	
	(2.5333)	(0.5082)	(0.0026, 0.0194)	(0.1594)	(29.47%, 43.84%)	(0.3643)	(5.3407)	
2016	0.2912	-0.0124	0.0071	0.0811	38.25%	0.0776	-1.2678*	
	(0.2079)	(0.1800)	(0.0028, 0.0180)	(0.1961)	(31.05%, 46.00%)	(0.1966)	(0.7295)	
2017	0.6424***	0.3850**	0.0105	0.1816	40.65%	0.1644	-0.3048	
	(0.2488)	(0.1788)	(0.0041, 0.0267)	(0.1820)	(35.15%, 46.39%)	(0.1821)	(0.8727)	
2018	0.8514***	0.2894	0.0095	0.0102	36.59%	-0.0518	1.3060*	
	(0.1924)	(0.1890)	(0.0038, 0.0240)	(0.2053)	(30.01%, 43.71%)	(0.2096)	(0.7318)	
2019	0.5883***	0.2802	0.0095	0.0456	37.41%	-0.0066	0.7687	
	(0.2154)	(0.2153)	(0.0035, 0.0255)	(0.1960)	(31.04%, 44.26%)	(0.1975)	(0.8962)	
Winsorized Age	-0.0569**					-0.0001	-0.0312	
	(0.0252)					(0.0011)	(0.0547)	
Female	-0.0776					0.0116	0.0927	
	(0.1152)					(0.0083)	(0.2112)	
U.S.	-0.3707*					-0.0033	0.4696	
	(0.2114)					(0.0084)	(0.4504)	
International Asian	-0.5051***					-0.0103	0.5968*	
	(0.1631)					(0.0083)	(0.3430)	
GRE Verbal Percentile	0.0082**					0.0000	0.0098	
	(0.0037)					(0.0002)	(0.0075)	
GRE Quantitative Percentile	0.0704***					0.0010	-0.0610*	
	(0.0105)					(0.0009)	(0.0338)	
GRE Analytic Percentile	0.0016					0.0001	0.0021	
	(0.0029)					(0.0002)	(0.0066)	
TOEFL/IELTS	0.2848					-0.0023	-0.0629	
	(0.1746)					(0.0101)	(0.2753)	
Elite U.S. University	0.6391***					-0.0098	-0.0637	
	(0.1892)					(0.0142)	(0.3925)	
Elite U.S. Liberal Arts College	-0.3528					0.0127	-0.2757	
	(0.3489)					(0.0174)	(0.6783)	
Elite Foreign University	0.1906					-0.0187	0.4536	
	(0.2875)					(0.0195)	(0.5403)	
Other Foreign University	0.5511***					0.0038	0.2515	
	(0.2049)					(0.0148)	(0.3764)	
Missing Undergraduate Institution	0.5458***					0.0033	-0.3893	
	(0.2109)					(0.0140)	(0.4067)	
Graduate Degree	-0.1826					-0.0069	0.4784	
	(0.1589)					(0.0082)	(0.3581)	
Work Experience	0.0765					0.0080	0.5662	
	(0.1486)					(0.0075)	(0.3731)	
#Professor Recommenders	0.0824					-0.0021	0.2152	
	(0.0822)					(0.0054)	(0.1592)	
#Prolific Recommenders	0.0919					-0.0002	-0.1769*	
	(0.0590)					(0.0032)	(0.1041)	
Missing GRE	-1.3931					0.0140		
	(0.8753)					(0.0192)		
Missing TOEFL/IELTS	0.2632**					0.0023	0.0851	
	(0.1311)					(0.0066)	(0.2494)	
Committee Score						0.0649**	0.5117***	
						(0.0299)	(0.1497)	
#Math Courses	-0.1660					-0.0048	-0.1664	
	(0.1024)					(0.0057)	(0.1619)	
Advanced Math	0.7224*					0.0257	0.0920	
	(0.4161)					(0.0208)	(0.5297)	
Letters Length/100	0.1095**					0.0013	-0.0122	
	(0.0471)					(0.0022)	(0.0646)	
Letters Terms (%) PC1	0.2635***					0.0089	0.4342***	
	(0.0988)					(0.0071)	(0.1642)	
Letters Terms (%) PC2	-0.3026*					-0.0108	-0.2680	
	(0.1764)					(0.0091)	(0.2167)	
CV Length/100	0.0263					0.0004	-0.0601	
	(0.0752)					(0.0023)	(0.0819)	
CV Coding	0.0417					-0.0049	-0.0663	
	(0.1086)					(0.0044)	(0.1017)	
CV Publication/Working Paper	1.3411***					0.0190	-0.2251	
	(0.3435)					(0.0192)	(0.5320)	
Essay Length/100	0.2771*					0.0065	0.1436	
	(0.1475)					(0.0103)	(0.1533)	
Essay Specific Professor	0.4040					0.0133	0.6561	
	(0.3655)					(0.0168)	(0.4106)	
Essay Economics Terms (%)	0.0881					0.0010	0.0869	
	(0.0633)					(0.0022)	(0.0809)	
Essay Research Terms (%)	0.1320					0.0005	-0.1728	
	(0.0843)					(0.0030)	(0.1284)	
Essay Positive Terms (%)	0.0538					-0.0006	0.0066	
	(0.0761)					(0.0025)	(0.0966)	
Essay Negative Terms (%)	0.0781					-0.0021	-0.0494	
	(0.0830)					(0.0035)	(0.1032)	
Missing Committee Score						-0.1920		
						(0.1290)		
Missing Application Pdf	-0.4098					0.0124	-1.5977***	
	(0.2846)					(0.0125)	(0.2218)	
Program Rank						-0.0079***		
						(0.0020)		
Missing Program Rank						-3.0808***		
						(0.4053)		
$\sqrt{\text{MSE}}$	1.7632***							
	(0.0469)							
Sigma								0.0020**
								(0.0010)
Obs | Param | LL | AIC	2329	568	-14260.9	29657.8				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 19: Deviation Gains Attribution Table

(a) Acceptance Set

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.8788	25.9910	26.2840	29.9572	29.5331	1.1674	1.0326
Work Experience	2.3065	0.5247	0.7276	0.9844	0.9650	0.5141	0.4752
Essay Positive Terms (%)	1.9410	11.7499	11.6521	12.5376	12.1183	0.7858	0.4137
Missing TOEFL/IELTS	1.7635	0.6076	0.6654	0.0739	0.4086	-1.2110	-0.5258
CV Coding	1.3478	2.8001	2.7820	3.3399	2.9415	0.6244	0.1785
Elite U.S. University	1.3179	0.1267	0.2335	0.0078	0.0389	-0.6784	-0.5848
TOEFL/IELTS	1.2400	7.4376	7.5675	7.7759	7.6827	0.5957	0.3294
Essay Economics Terms (%)	1.2043	9.7228	9.7423	9.0284	9.5388	-0.6411	-0.1828
#Professor Recommenders	1.1455	2.6230	2.6148	1.3852	2.1362	-1.6680	-0.6493
Letters Terms (%) PC1	1.0074	-0.0106	0.1374	0.4969	0.1387	0.5202	0.0018

(b) Matriculation-Set

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.7099	25.9910	26.3284	30.9159	29.7737	1.4580	1.0950
Essay Positive Terms (%)	2.2339	11.7499	11.6107	12.9136	12.2763	1.1534	0.5905
Work Experience	2.0220	0.5247	0.7938	0.9949	0.9799	0.4077	0.3725
TOEFL/IELTS	1.7179	7.4376	7.4939	7.8501	7.6128	1.0238	0.3397
Essay Economics Terms (%)	1.3503	9.7228	9.9994	9.0856	9.5738	-0.8271	-0.3821
CV Coding	1.3100	2.8001	2.6661	3.3395	2.8782	0.7618	0.2374
#Professor Recommenders	1.3005	2.6230	2.6171	1.4173	2.2255	-1.6321	-0.5312
Missing TOEFL/IELTS	1.1999	0.6076	0.6858	0.0442	0.4368	-1.3075	-0.5099

D Simulated Structural Results

In this appendix, I present structural estimation results from two simulated datasets, one in which the committee is nearly optimal and one in which it is not. The data-generating parameters for the means, variances, and correlations across the independent variables, as well as the parameters governing the effects of the independent variables on success, are meant to resemble those of an actual admissions dataset. In the near-optimal dataset, the parameters governing the effect of the independent variables on acceptance are chosen to be similar to the success parameters. In contrast, the suboptimal committee discriminates against *Demographic 1*, undervalues *Advanced Math*, and overvalues *Committee Score*. The goal of this exercise is to verify that the optimality test rejects optimality more strongly for the suboptimal dataset than the near-optimal dataset.

In Tables 20 and 21, I present the unrestricted and restricted dynamic estimates, respectively. Because I have two z variables, namely *Advanced Math* and *Committee Score*, I have two PDFs to estimate. Committee score is a continuous z, and since I parameterize continuous z's with the normal distribution, I have an additional σ_F parameter that equals the square root of the distribution's mean squared error. I also let *Committee Score* depend on *Advanced Math*, so I can estimate the joint distribution of the two z's without assuming the z's are independent. As usual, I do not assume independent z's. The simulated data are such that *GRE Quantitative Score*, *Advanced Math*, and *Program Rank* significantly predict success at the 1% levels.[17]

The unrestricted and restricted static estimates are in Tables 22 and 23, respectively. Although the data are simulated, Table 23 contains an interesting result. There are two years in each of the simulated datasets: one in which the committee transitions 45–50% of the applicants to Round 2, and one in which it only transitions 25–30%. Essentially, in *Year Transition More*, the committee is more concerned with admitting a strong class than saving time and effort.[18] And in that year, as expected, the Round 1 cutoff probability is lower than the Round 2 cutoff probability. That is, it is easier to move on to Round 2 than it ultimately is to get into the program. However, in the other year, which is represented by the intercept term, the Round 1 cutoff is greater than the Round 2 cutoff, meaning the $P(\text{Success}|x,t,\bar{k})$ of the marginal applicant rejected in Round 1 is greater than the $P(\text{Success}|x,z,t,\bar{k})$ of the marginal applicant accepted into the program.

How can this be? Sometimes when a committee transitions a seemingly strong applicant to Round 2, the applicant's z reveals that they are weaker than previously thought. For example, the applicant may have a negative recommendation letter. If the committee transitions too few applicants to Round 2, it can end up in an unfortunate situation where it threw out so many files that it now has to accept those weaker applicants. As a simple example, suppose there are three applicants: A, B, and C. The committee has time to read two applicants' files and space to accept one applicant. Suppose that overall, A is the best applicant and C is the worst, but A

[17] As in the actual data, *Program Rank*'s sign is negative because the best rank is 1, and the worst rank is 150.

[18] In *Year Transition More*, the applicants also have higher matriculation probabilities on average, so this variable is called *Year Transition/Matriculate More* in the matriculation models in this appendix.

is the worst in objective variables. Then the committee will reject A in Round 1 and accept B, who is actually worse than A. In the simulated data, the minimum percentage of applications the committee must read in Round 2 to avoid this situation is between 25–30%, where the situation occurs, and 45–50%, where it does not. However, this situation does not occur in the actual data.

In Tables 24 and 25, I present estimates from the myopic model. Recall that the myopic model is a special case of the dynamic model in which the parameters for x in every block of $f_{z|x,t}$ are restricted to equal zero, while the intercepts, parameters for t, dependence parameters among the components of z, and variance parameters remain unrestricted. In this model, if an objective variable is associated with z in addition to being associated with success in its own right, the committee puts no extra weight on that variable in Round 1. Finally, the matriculation-dynamic estimates are in Tables 26 and 27. As with the dynamic estimates without matriculation, the simulated data are such that *GRE Quantitative Score*, *Advanced Math*, and *Program Rank* significantly predict success at the 1% level. But *Committee Score* significantly negatively predicts matriculation at the 1% level. The rationale is that applicants with higher committee scores likely have more acceptances from other universities and so are less likely to matriculate if accepted.

I conclude this appendix by presenting the deviation gains test results on the simulated datasets. Per Figure 7, in the dataset where the committee is nearly optimal, the models' CDFs for the acceptance set are close to those of the committee's. But in the suboptimal dataset, all the models' CDFs stochastically dominate the committee's CDF, and the test always rejects with $p < 0.0002$. The reason is that the committee is discriminating against *Demographic 1*, not putting enough weight on *Advanced Math*, and putting too much weight on *Committee Score*. The results for the matriculation-dynamic and static models on the matriculation set are in Figure 8, and those results are essentially the same as the results in Figure 7. For the optimal dataset, the optimality test never rejects, and for the suboptimal dataset, it always rejects with $p < 0.0002$. In this figure, I also plot the CDF for the matriculation set of the dynamic model without matriculation. In the near-optimal dataset, *GRE Quantitative Percentile* negatively predicts matriculation, so the matriculation-dynamic committee selects relatively lower GREs in Round 1. This decision ends up not paying off for the committee because for simplicity, the simulation does not allow undershooting the target cohort size. However, in the actual data, even though the simulation still does not allow undershooting, the matriculation-dynamic CDF is to the right of the dynamic CDF.

Table 20: Simulated Unrestricted Dynamic Estimates

(a) Near-Optimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577**	-0.4303**	0.3695	-12.0690***	-2.2483***	-8.3663***	-1.6540***	-7.3172***	-1.0451***
	(0.7805)	(0.1891)	(1.4912)	(0.9758)	(0.1337)	(1.7459)	(0.3133)	(1.2885)	(0.1661)
GRE Quantitative Percentile	0.0251***	0.0061***	0.0510***	0.1260***	0.0235***	0.0911***	0.0180***	0.1028***	0.0147***
	(0.0094)	(0.0023)	(0.0170)	(0.0114)	(0.0017)	(0.0200)	(0.0036)	(0.0142)	(0.0017)
Gender	-0.2435*	-0.0596*	0.1807	0.0400	0.0074	0.0150	0.0030	0.2259	0.0323
	(0.1312)	(0.0319)	(0.2034)	(0.1525)	(0.0284)	(0.2415)	(0.0477)	(0.1762)	(0.0251)
Demographic 1	-0.0504	-0.0123	-0.6989**	0.0163	0.0030	-0.0503	-0.0100	-0.4384*	-0.0626*
	(0.1730)	(0.0424)	(0.2889)	(0.2031)	(0.0378)	(0.3722)	(0.0736)	(0.2358)	(0.0334)
Demographic 2	-0.3121	-0.0764	-0.4226	0.1678	0.0313	-0.2878	-0.0569	-0.4682*	-0.0669*
	(0.1930)	(0.0470)	(0.3254)	(0.2315)	(0.0431)	(0.4145)	(0.0818)	(0.2734)	(0.0387)
Advanced Math			1.1347***			0.2066	0.0408	0.7127***	0.1018***
			(0.1992)			(0.2496)	(0.0491)	(0.1791)	(0.0247)
Committee Score						0.0600	0.0119	-0.0790	-0.0113
						(0.0639)	(0.0126)	(0.0688)	(0.0098)
Missing Committee Score								0.0486	0.0069
								(0.1969)	(0.0281)
Year Transition More	-0.2523**	-0.0618**	0.1483	1.1441***	0.2131***	-1.1114***	-0.2197***	-0.3543**	-0.0506**
	(0.1282)	(0.0311)	(0.2025)	(0.1501)	(0.0250)	(0.2446)	(0.0435)	(0.1794)	(0.0254)
Program Rank								-0.0302***	-0.0043***
								(0.0018)	(0.0002)
$\sqrt{\text{MSE}}$			1.8393***						
			(0.0614)						
Obs | Param | LL | AIC	1000	38	-2602.7	5281.5					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577**	-0.4303**	1.2187	-10.3454***	-2.0151***	-4.8695***	-1.0694***	-7.3017***	-1.0448***
	(0.7805)	(0.1891)	(1.4016)	(0.9474)	(0.1440)	(1.5490)	(0.3252)	(1.2721)	(0.1635)
GRE Quantitative Percentile	0.0251***	0.0061***	0.0431***	0.1103***	0.0215***	0.0488***	0.0107***	0.1021***	0.0146***
	(0.0094)	(0.0023)	(0.0164)	(0.0111)	(0.0018)	(0.0180)	(0.0038)	(0.0140)	(0.0017)
Gender	-0.2435*	-0.0596*	0.1707	-0.0472	-0.0092	0.1746	0.0383	0.2087	0.0299
	(0.1312)	(0.0319)	(0.2037)	(0.1485)	(0.0289)	(0.2312)	(0.0506)	(0.1755)	(0.0251)
Demographic 1	-0.0504	-0.0123	-0.7422***	-0.4655**	-0.0907**	-0.5625*	-0.1235*	-0.4120*	-0.0590*
	(0.1730)	(0.0424)	(0.2732)	(0.1978)	(0.0382)	(0.3192)	(0.0689)	(0.2351)	(0.0334)
Demographic 2	-0.3121	-0.0764	-0.4868*	0.1277	0.0249	-0.1757	-0.0386	-0.3798	-0.0543
	(0.1930)	(0.0470)	(0.2925)	(0.2186)	(0.0426)	(0.3381)	(0.0742)	(0.2696)	(0.0384)
Advanced Math			1.1482***			-0.0098	-0.0021	0.7178***	0.1027***
			(0.1999)			(0.2366)	(0.0520)	(0.1795)	(0.0248)
Committee Score						0.1714***	0.0376***	-0.0769	-0.0110
						(0.0604)	(0.0128)	(0.0684)	(0.0098)
Missing Committee Score								0.1129	0.0162
								(0.1909)	(0.0273)
Year Transition More	-0.2523**	-0.0618**	0.0871	1.0879***	0.2119***	-0.7286***	-0.1600***	-0.3741**	-0.0535**
	(0.1282)	(0.0311)	(0.2014)	(0.1466)	(0.0255)	(0.2324)	(0.0485)	(0.1781)	(0.0253)
Program Rank								-0.0309***	-0.0044***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.8362***						
			(0.0603)						
Obs | Param | LL | AIC	1000	38	-2648.4	5372.9					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 21: Simulated Restricted Dynamic Estimates

(a) Near-Optimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-1.9394**	0.3771	-0.1457**	0.8644	0.2458	56.11%	-7.8112***	
	(0.7791)	(1.4942)	(0.0663)	(0.7591, 0.9843)	(0.2300)	(44.89%, 66.74%)	(1.1221)	
GRE Quantitative Percentile	0.0273***	0.0509***					0.1044***	
	(0.0094)	(0.0171)					(0.0128)	
Gender	-0.2478*	0.1808					0.0937	
	(0.1310)	(0.2034)					(0.1163)	
Demographic 1	-0.0432	-0.6993**					-0.2023	
	(0.1727)	(0.2904)					(0.1559)	
Demographic 2	-0.3026	-0.4231					-0.2735	
	(0.1930)	(0.3260)					(0.1817)	
Advanced Math		1.1347***					0.4597***	
		(0.1992)					(0.1452)	
Committee Score							-0.0021	
							(0.0436)	
Missing Committee Score							0.0240	
							(0.1891)	
Year Transition More	-0.2448*	0.1481	-0.1684***	0.7304	0.4096	65.82%	-0.3690**	
	(0.1281)	(0.2023)	(0.0637)	(0.6462, 0.8256)	(0.2744)	(57.07%, 73.62%)	(0.1763)	
Program Rank							-0.0300***	
							(0.0018)	
$\sqrt{\text{MSE}}$		1.8393***						
		(0.0614)						
Sigma								0.1647***
								(0.0242)
Obs | Param | LL | AIC	1000	29	-2642.8	5343.7				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-1.9242**	1.0267	-0.0633	0.9387	0.1263	53.15%	-7.7219***	
	(0.7773)	(1.4057)	(0.0892)	(0.7882, 1.1179)	(0.2508)	(40.97%, 64.97%)	(1.1783)	
GRE Quantitative Percentile	0.0271***	0.0453***					0.1006***	
	(0.0093)	(0.0165)					(0.0134)	
Gender	-0.2513*	0.1636					0.1366	
	(0.1310)	(0.2039)					(0.1297)	
Demographic 1	-0.0530	-0.7462***					-0.5186***	
	(0.1728)	(0.2730)					(0.1682)	
Demographic 2	-0.3004	-0.4758					-0.2123	
	(0.1928)	(0.2928)					(0.1992)	
Advanced Math		1.1476***					0.4443***	
		(0.1999)					(0.1595)	
Committee Score							0.0635	
							(0.0480)	
Missing Committee Score							0.1011	
							(0.1862)	
Year Transition More	-0.2466*	0.0910	-0.2559***	0.7268	0.3083	60.70%	-0.3596**	
	(0.1281)	(0.2010)	(0.0710)	(0.6177, 0.8551)	(0.3058)	(50.39%, 70.13%)	(0.1749)	
Program Rank							-0.0303***	
							(0.0018)	
$\sqrt{\text{MSE}}$		1.8362***						
		(0.0603)						
Sigma								0.2258***
								(0.0411)
Obs | Param | LL | AIC	1000	29	-2686.1	5430.2				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 22: Simulated Unrestricted Static Estimates

(a) Near-Optimal Committee

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-12.0690***	-2.2483***	-8.3663***	-1.6540***	-7.3875***	-1.0821***	-7.3172***	-1.0451***
	(0.9758)	(0.1337)	(1.7459)	(0.3133)	(1.1197)	(0.1406)	(1.2885)	(0.1661)
GRE Quantitative Percentile	0.1260***	0.0235***	0.0911***	0.0180***	0.1043***	0.0153***	0.1028***	0.0147***
	(0.0114)	(0.0017)	(0.0200)	(0.0036)	(0.0134)	(0.0016)	(0.0142)	(0.0017)
Gender	0.0400	0.0074	0.0150	0.0030	0.1580	0.0231	0.2259	0.0323
	(0.1525)	(0.0284)	(0.2415)	(0.0477)	(0.1732)	(0.0254)	(0.1762)	(0.0251)
Demographic 1	0.0163	0.0030	-0.0503	-0.0100	-0.4556**	-0.0667**	-0.4384*	-0.0626*
	(0.2031)	(0.0378)	(0.3722)	(0.0736)	(0.2269)	(0.0328)	(0.2358)	(0.0334)
Demographic 2	0.1678	0.0313	-0.2878	-0.0569	-0.5153*	-0.0755**	-0.4682*	-0.0669*
	(0.2315)	(0.0431)	(0.4145)	(0.0818)	(0.2641)	(0.0383)	(0.2734)	(0.0387)
Advanced Math			0.2066	0.0408			0.7127***	0.1018***
			(0.2496)	(0.0491)			(0.1791)	(0.0247)
Committee Score			0.0600	0.0119			-0.0790	-0.0113
			(0.0639)	(0.0126)			(0.0688)	(0.0098)
Missing Committee Score							0.0486	0.0069
							(0.1969)	(0.0281)
Year Transition More	1.1441***	0.2131***	-1.1114***	-0.2197***	-0.3942**	-0.0577**	-0.3543**	-0.0506**
	(0.1501)	(0.0250)	(0.2446)	(0.0435)	(0.1689)	(0.0244)	(0.1794)	(0.0254)
Program Rank					-0.0299***	-0.0044***	-0.0302***	-0.0043***
					(0.0018)	(0.0002)	(0.0018)	(0.0002)
Obs | Param | LL | AIC	1000	31	-1632.5	3327.1				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-10.3454***	-2.0151***	-4.8695***	-1.0694***	-7.2362***	-1.0632***	-7.3017***	-1.0448***
	(0.9474)	(0.1440)	(1.5490)	(0.3252)	(1.1209)	(0.1418)	(1.2721)	(0.1635)
GRE Quantitative Percentile	0.1103***	0.0215***	0.0488***	0.0107***	0.1026***	0.0151***	0.1021***	0.0146***
	(0.0111)	(0.0018)	(0.0180)	(0.0038)	(0.0134)	(0.0016)	(0.0140)	(0.0017)
Gender	-0.0472	-0.0092	0.1746	0.0383	0.1408	0.0207	0.2087	0.0299
	(0.1485)	(0.0289)	(0.2312)	(0.0506)	(0.1727)	(0.0254)	(0.1755)	(0.0251)
Demographic 1	-0.4655**	-0.0907**	-0.5625*	-0.1235*	-0.4199*	-0.0617*	-0.4120*	-0.0590*
	(0.1978)	(0.0382)	(0.3192)	(0.0689)	(0.2285)	(0.0332)	(0.2351)	(0.0334)
Demographic 2	0.1277	0.0249	-0.1757	-0.0386	-0.4343*	-0.0638*	-0.3798	-0.0543
	(0.2186)	(0.0426)	(0.3381)	(0.0742)	(0.2605)	(0.0380)	(0.2696)	(0.0384)
Advanced Math			-0.0098	-0.0021			0.7178***	0.1027***
			(0.2366)	(0.0520)			(0.1795)	(0.0248)
Committee Score			0.1714***	0.0376***			-0.0769	-0.0110
			(0.0604)	(0.0128)			(0.0684)	(0.0098)
Missing Committee Score							0.1129	0.0162
							(0.1909)	(0.0273)
Year Transition More	1.0879***	0.2119***	-0.7286***	-0.1600***	-0.4350**	-0.0639***	-0.3741**	-0.0535**
	(0.1466)	(0.0255)	(0.2324)	(0.0485)	(0.1689)	(0.0244)	(0.1781)	(0.0253)
Program Rank					-0.0305***	-0.0045***	-0.0309***	-0.0044***
					(0.0019)	(0.0002)	(0.0019)	(0.0002)
Obs | Param | LL | AIC	1000	31	-1673.0	3408.1				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 23: Simulated Restricted Static Estimates

(a) Near-Optimal Committee

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	1.2114***	77.05%	0.3765	59.30%	-7.7989***	-6.9427***	
	(0.2502)	(67.28%, 84.58%)	(0.2666)	(46.35%, 71.08%)	(1.0685)	(1.2241)	
GRE Quantitative Percentile					0.1065***	0.0957***	
					(0.0128)	(0.0139)	
Gender					0.0694	0.1374	
					(0.1077)	(0.1373)	
Demographic 1					-0.1638	-0.3053	
					(0.1410)	(0.1944)	
Demographic 2					-0.0994	-0.3969*	
					(0.1623)	(0.2189)	
Advanced Math						0.5127***	
						(0.1456)	
Committee Score						-0.0104	
						(0.0458)	
Missing Committee Score						-0.0145	
						(0.1924)	
Year Transition More	-1.3691***	46.07%	0.6127*	72.89%	-0.3712**	-0.3819**	
	(0.2738)	(40.01%, 52.24%)	(0.3291)	(61.87%, 81.67%)	(0.1649)	(0.1814)	
Program Rank					-0.0298***	-0.0300***	
					(0.0018)	(0.0018)	
Sigma							0.2021***
							(0.0252)
Obs | Param | LL | AIC	1000	22	-1639.7	3323.5			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	1.5071***	81.86%	0.2881	57.15%	-7.9222***	-6.4145***	
	(0.3814)	(68.13%, 90.50%)	(0.2858)	(43.24%, 70.02%)	(1.1110)	(1.2821)	
GRE Quantitative Percentile					0.1102***	0.0864***	
					(0.0132)	(0.0144)	
Gender					0.0340	0.2000	
					(0.1208)	(0.1433)	
Demographic 1					-0.4496***	-0.5033***	
					(0.1586)	(0.1917)	
Demographic 2					-0.1170	-0.3225	
					(0.1795)	(0.2194)	
Advanced Math						0.4844***	
						(0.1548)	
Committee Score						0.0524	
						(0.0491)	
Missing Committee Score						0.0419	
						(0.1918)	
Year Transition More	-1.6799***	45.69%	0.3660	65.79%	-0.4035**	-0.3931**	
	(0.3946)	(39.03%, 52.51%)	(0.3462)	(53.86%, 76.01%)	(0.1655)	(0.1786)	
Program Rank					-0.0303***	-0.0304***	
					(0.0019)	(0.0018)	
Sigma							0.2573***
							(0.0392)
Obs | Param | LL | AIC	1000	22	-1687.8	3419.7			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 24: Simulated Unrestricted Myopic Estimates

(a) Near-Optimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.1765**	0.0439**	4.4154***	-12.0690***	-2.2483***	-8.3663***	-1.6540***	-7.3172***	-1.0451***
	(0.0898)	(0.0222)	(0.1909)	(0.9758)	(0.1337)	(1.7459)	(0.3133)	(1.2885)	(0.1661)
GRE Quantitative Percentile				0.1260***	0.0235***	0.0911***	0.0180***	0.1028***	0.0147***
				(0.0114)	(0.0017)	(0.0200)	(0.0036)	(0.0142)	(0.0017)
Gender				0.0400	0.0074	0.0150	0.0030	0.2259	0.0323
				(0.1525)	(0.0284)	(0.2415)	(0.0477)	(0.1762)	(0.0251)
Demographic 1				0.0163	0.0030	-0.0503	-0.0100	-0.4384*	-0.0626*
				(0.2031)	(0.0378)	(0.3722)	(0.0736)	(0.2358)	(0.0334)
Demographic 2				0.1678	0.0313	-0.2878	-0.0569	-0.4682*	-0.0669*
				(0.2315)	(0.0431)	(0.4145)	(0.0818)	(0.2734)	(0.0387)
Advanced Math			1.1642***			0.2066	0.0408	0.7127***	0.1018***
			(0.1981)			(0.2496)	(0.0491)	(0.1791)	(0.0247)
Committee Score						0.0600	0.0119	-0.0790	-0.0113
						(0.0639)	(0.0126)	(0.0688)	(0.0098)
Missing Committee Score								0.0486	0.0069
								(0.1969)	(0.0281)
Year Transition More	-0.2485**	-0.0618**	0.1348	1.1441***	0.2131***	-1.1114***	-0.2197***	-0.3543**	-0.0506**
	(0.1268)	(0.0313)	(0.2040)	(0.1501)	(0.0250)	(0.2446)	(0.0435)	(0.1794)	(0.0254)
Program Rank								-0.0302***	-0.0043***
								(0.0018)	(0.0002)
$\sqrt{\text{MSE}}$			1.8730***						
			(0.0618)						
Obs | Param | LL | AIC	1000	30	-2617.5	5295.0					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.1765**	0.0439**	4.5276***	-10.3454***	-2.0151***	-4.8695***	-1.0694***	-7.3017***	-1.0448***
	(0.0898)	(0.0222)	(0.1915)	(0.9474)	(0.1440)	(1.5490)	(0.3252)	(1.2721)	(0.1635)
GRE Quantitative Percentile				0.1103***	0.0215***	0.0488***	0.0107***	0.1021***	0.0146***
				(0.0111)	(0.0018)	(0.0180)	(0.0038)	(0.0140)	(0.0017)
Gender				-0.0472	-0.0092	0.1746	0.0383	0.2087	0.0299
				(0.1485)	(0.0289)	(0.2312)	(0.0506)	(0.1755)	(0.0251)
Demographic 1				-0.4655**	-0.0907**	-0.5625*	-0.1235*	-0.4120*	-0.0590*
				(0.1978)	(0.0382)	(0.3192)	(0.0689)	(0.2351)	(0.0334)
Demographic 2				0.1277	0.0249	-0.1757	-0.0386	-0.3798	-0.0543
				(0.2186)	(0.0426)	(0.3381)	(0.0742)	(0.2696)	(0.0384)
Advanced Math			1.1789***			-0.0098	-0.0021	0.7178***	0.1027***
			(0.1973)			(0.2366)	(0.0520)	(0.1795)	(0.0248)
Committee Score						0.1714***	0.0376***	-0.0769	-0.0110
						(0.0604)	(0.0128)	(0.0684)	(0.0098)
Missing Committee Score								0.1129	0.0162
								(0.1909)	(0.0273)
Year Transition More	-0.2485**	-0.0618**	0.0712	1.0879***	0.2119***	-0.7286***	-0.1600***	-0.3741**	-0.0535**
	(0.1268)	(0.0313)	(0.2044)	(0.1466)	(0.0255)	(0.2324)	(0.0485)	(0.1781)	(0.0253)
Program Rank								-0.0309***	-0.0044***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.8655***						
			(0.0600)						
Obs | Param | LL | AIC	1000	30	-2662.4	5384.8					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 25: Simulated Restricted Myopic Estimates

(a) Near-Optimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.1822**	4.4131***	-0.1459**	0.8642	0.2450	56.09%	-7.8232***	
	(0.0898)	(0.1915)	(0.0669)	(0.7581, 0.9852)	(0.2303)	(44.86%, 66.74%)	(1.1263)	
GRE Quantitative Percentile							0.1055***	
							(0.0128)	
Gender							0.0872	
							(0.1160)	
Demographic 1							-0.2089	
							(0.1545)	
Demographic 2							-0.2850	
							(0.1804)	
Advanced Math		1.1642***					0.4385***	
		(0.1982)					(0.1452)	
Committee Score							-0.0125	
							(0.0436)	
Missing Committee Score							0.0243	
							(0.1889)	
Year Transition More	-0.2418*	0.1344	-0.1671***	0.7312	0.4173	65.98%	-0.3689**	
	(0.1267)	(0.2039)	(0.0640)	(0.6463, 0.8273)	(0.2757)	(57.16%, 73.81%)	(0.1764)	
Program Rank							-0.0300***	
							(0.0018)	
$\sqrt{\text{MSE}}$		1.8730***						
		(0.0618)						
Sigma								0.1652***
								(0.0244)
Obs | Param | LL | AIC	1000	21	-2658.4	5358.7				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.1801**	4.5328***	-0.0597	0.9420	0.1255	53.13%	-7.7935***	
	(0.0898)	(0.1919)	(0.0912)	(0.7878, 1.1264)	(0.2525)	(40.87%, 65.03%)	(1.1728)	
GRE Quantitative Percentile							0.1025***	
							(0.0132)	
Gender							0.1327	
							(0.1299)	
Demographic 1							-0.5335***	
							(0.1674)	
Demographic 2							-0.2291	
							(0.1982)	
Advanced Math		1.1784***					0.4297***	
		(0.1973)					(0.1610)	
Committee Score							0.0501	
							(0.0488)	
Missing Committee Score							0.1041	
							(0.1858)	
Year Transition More	-0.2436*	0.0726	-0.2565***	0.7289	0.3183	60.92%	-0.3616**	
	(0.1267)	(0.2043)	(0.0714)	(0.6175, 0.8605)	(0.3098)	(50.44%, 70.47%)	(0.1752)	
Program Rank							-0.0303***	
							(0.0018)	
$\sqrt{\text{MSE}}$		1.8654***						
		(0.0599)						
Sigma								0.2283***
								(0.0424)
Obs | Param | LL | AIC	1000	21	-2701.0	5444.0				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 26: Simulated Unrestricted Matriculation-Dynamic Estimates

(a) Near-Optimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577**	-0.4303**	0.3695	-12.0690***	-2.2483***	-8.3663***	-1.6540***	-7.3172***	-1.0451***	10.8487*	1.1240**
	(0.7805)	(0.1891)	(1.4912)	(0.9758)	(0.1337)	(1.7459)	(0.3133)	(1.2885)	(0.1661)	(5.7339)	(0.5653)
GRE Quantitative Percentile	0.0251***	0.0061***	0.0510***	0.1260***	0.0235***	0.0911***	0.0180***	0.1028***	0.0147***	-0.1049*	-0.0109*
	(0.0094)	(0.0023)	(0.0170)	(0.0114)	(0.0017)	(0.0200)	(0.0036)	(0.0142)	(0.0017)	(0.0630)	(0.0063)
Gender	-0.2435*	-0.0596*	0.1807	0.0400	0.0074	0.0150	0.0030	0.2259	0.0323	-0.9225	-0.0956*
	(0.1312)	(0.0319)	(0.2034)	(0.1525)	(0.0284)	(0.2415)	(0.0477)	(0.1762)	(0.0251)	(0.5834)	(0.0575)
Demographic 1	-0.0504	-0.0123	-0.6989**	0.0163	0.0030	-0.0503	-0.0100	-0.4384*	-0.0626*	1.2693	0.1315
	(0.1730)	(0.0424)	(0.2889)	(0.2031)	(0.0378)	(0.3722)	(0.0736)	(0.2358)	(0.0334)	(0.8109)	(0.0828)
Demographic 2	-0.3121	-0.0764	-0.4226	0.1678	0.0313	-0.2878	-0.0569	-0.4682*	-0.0669*	0.2958	0.0306
	(0.1930)	(0.0470)	(0.3254)	(0.2315)	(0.0431)	(0.4145)	(0.0818)	(0.2734)	(0.0387)	(0.9734)	(0.1014)
Advanced Math			1.1347***			0.2066	0.0408	0.7127***	0.1018***	0.3819	0.0396
			(0.1992)			(0.2496)	(0.0491)	(0.1791)	(0.0247)	(0.5414)	(0.0557)
Committee Score						0.0600	0.0119	-0.0790	-0.0113	-1.0758***	-0.1115***
						(0.0639)	(0.0126)	(0.0688)	(0.0098)	(0.2376)	(0.0153)
Missing Committee Score								0.0486	0.0069		
								(0.1969)	(0.0281)		
Year Transition/Matriculate More	-0.2523**	-0.0618**	0.1483	1.1441***	0.2131***	-1.1114***	-0.2197***	-0.3543**	-0.0506**	5.2621***	0.5452***
	(0.1282)	(0.0311)	(0.2025)	(0.1501)	(0.0250)	(0.2446)	(0.0435)	(0.1794)	(0.0254)	(0.8695)	(0.0260)
Program Rank								-0.0302***	-0.0043***		
								(0.0018)	(0.0002)		
$\sqrt{\text{MSE}}$			1.8393***								
			(0.0614)								
Obs | Param | LL | AIC	1000	46	-2644.2	5380.5							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577**	-0.4303**	1.2187	-10.3454***	-2.0151***	-4.8695***	-1.0694***	-7.3017***	-1.0448***	5.7452	0.5714
	(0.7805)	(0.1891)	(1.4016)	(0.9474)	(0.1440)	(1.5490)	(0.3252)	(1.2721)	(0.1635)	(4.2604)	(0.4164)
GRE Quantitative Percentile	0.0251***	0.0061***	0.0431***	0.1103***	0.0215***	0.0488***	0.0107***	0.1021***	0.0146***	-0.0443	-0.0044
	(0.0094)	(0.0023)	(0.0164)	(0.0111)	(0.0018)	(0.0180)	(0.0038)	(0.0140)	(0.0017)	(0.0468)	(0.0046)
Gender	-0.2435*	-0.0596*	0.1707	-0.0472	-0.0092	0.1746	0.0383	0.2087	0.0299	-1.0062*	-0.1001*
	(0.1312)	(0.0319)	(0.2037)	(0.1485)	(0.0289)	(0.2312)	(0.0506)	(0.1755)	(0.0251)	(0.5693)	(0.0535)
Demographic 1	-0.0504	-0.0123	-0.7422***	-0.4655**	-0.0907**	-0.5625*	-0.1235*	-0.4120*	-0.0590*	0.8111	0.0807
	(0.1730)	(0.0424)	(0.2732)	(0.1978)	(0.0382)	(0.3192)	(0.0689)	(0.2351)	(0.0334)	(0.7062)	(0.0712)
Demographic 2	-0.3121	-0.0764	-0.4868*	0.1277	0.0249	-0.1757	-0.0386	-0.3798	-0.0543	0.7188	0.0715
	(0.1930)	(0.0470)	(0.2925)	(0.2186)	(0.0426)	(0.3381)	(0.0742)	(0.2696)	(0.0384)	(0.7190)	(0.0724)
Advanced Math			1.1482***			-0.0098	-0.0021	0.7178***	0.1027***	-0.7520	-0.0748
			(0.1999)			(0.2366)	(0.0520)	(0.1795)	(0.0248)	(0.5452)	(0.0534)
Committee Score						0.1714***	0.0376***	-0.0769	-0.0110	-1.0032***	-0.0998***
						(0.0604)	(0.0128)	(0.0684)	(0.0098)	(0.2212)	(0.0150)
Missing Committee Score								0.1129	0.0162		
								(0.1909)	(0.0273)		
Year Transition/Matriculate More	-0.2523**	-0.0618**	0.0871	1.0879***	0.2119***	-0.7286***	-0.1600***	-0.3741**	-0.0535**	5.6048***	0.5574***
	(0.1282)	(0.0311)	(0.2014)	(0.1466)	(0.0255)	(0.2324)	(0.0485)	(0.1781)	(0.0253)	(0.9335)	(0.0301)
Program Rank								-0.0309***	-0.0044***		
								(0.0019)	(0.0002)		
$\sqrt{\text{MSE}}$			1.8362***								
			(0.0603)								
Obs | Param | LL | AIC	1000	46	-2693.0	5478.1							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 27: Simulated Restricted Matriculation-Dynamic Estimates

(a) Near-Optimal Committee

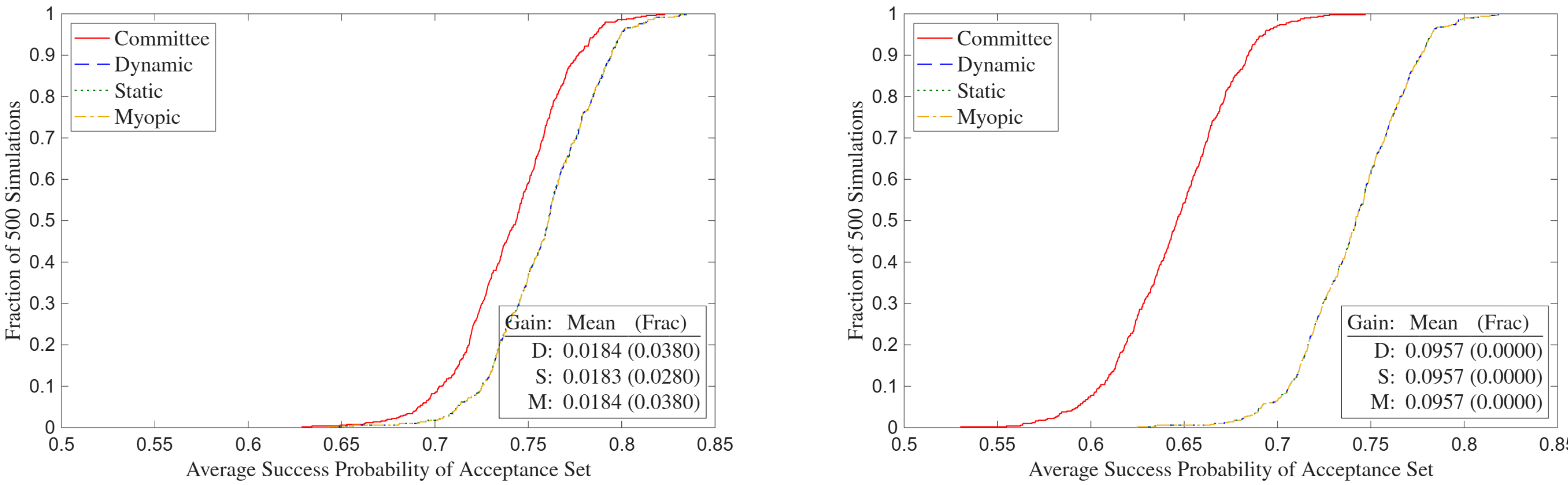
	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-1.8113**	1.2405	-1.4922***	0.2249	0.3506	58.68%	-7.7566***	2.9160	
	(0.7828)	(1.5031)	(0.1913)	(0.1546, 0.3272)	(0.2619)	(45.94%, 70.35%)	(1.1350)	(5.1107)	
GRE Quantitative Percentile	0.0257***	0.0408**					0.1057***	-0.0133	
	(0.0094)	(0.0173)					(0.0129)	(0.0562)	
Gender	-0.2464*	0.1731					0.1159	-0.9424*	
	(0.1312)	(0.2034)					(0.1353)	(0.5543)	
Demographic 1	-0.0487	-0.7009**					-0.2877	1.2129	
	(0.1729)	(0.2890)					(0.1882)	(0.8220)	
Demographic 2	-0.3098	-0.4291					-0.3732*	0.5064	
	(0.1929)	(0.3258)					(0.2125)	(0.9746)	
Advanced Math		1.1261***					0.4974***	0.3171	
		(0.2002)					(0.1521)	(0.5299)	
Committee Score							-0.0238	-1.0550***	
							(0.0447)	(0.2347)	
Missing Committee Score							0.0211		
							(0.1952)		
Year Transition/Matriculate More	-0.2441*	0.1538	-1.2324***	0.0656	0.4094	68.14%	-0.3789**	4.9595***	
	(0.1282)	(0.2051)	(0.2881)	(0.0355, 0.1212)	(0.3147)	(57.83%, 76.93%)	(0.1823)	(0.9084)	
Program Rank							-0.0301***		
							(0.0018)		
$\sqrt{\text{MSE}}$		1.8440***							
		(0.0615)							
Sigma									0.1872***
									(0.0329)
Obs | Param | LL | AIC	1000	37	-2707.6	5489.2					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-1.7155**	1.8306	-1.0818***	0.3390	0.2612	56.49%	-7.7219***	0.2437	
	(0.7813)	(1.4228)	(0.2600)	(0.2036, 0.5643)	(0.3040)	(41.71%, 70.20%)	(1.1701)	(4.4569)	
GRE Quantitative Percentile	0.0246***	0.0357**					0.1020***	0.0200	
	(0.0094)	(0.0167)					(0.0133)	(0.0495)	
Gender	-0.2456*	0.1683					0.1877	-1.0474*	
	(0.1312)	(0.2035)					(0.1499)	(0.5402)	
Demographic 1	-0.0469	-0.7099***					-0.5368***	0.5144	
	(0.1731)	(0.2735)					(0.1983)	(0.7157)	
Demographic 2	-0.3097	-0.4873*					-0.3087	0.7381	
	(0.1929)	(0.2925)					(0.2284)	(0.7585)	
Advanced Math		1.1443***					0.5177***	-0.8396	
		(0.2001)					(0.1729)	(0.5352)	
Committee Score							0.0412	-0.9473***	
							(0.0545)	(0.2114)	
Missing Committee Score							0.0852		
							(0.1912)		
Year Transition/Matriculate More	-0.2478*	0.0862	-1.0168***	0.1226	0.3707	65.29%	-0.3780**	5.3618***	
	(0.1281)	(0.2022)	(0.2514)	(0.0646, 0.2327)	(0.4109)	(50.97%, 77.29%)	(0.1794)	(0.9378)	
Program Rank							-0.0305***		
							(0.0018)		
$\sqrt{\text{MSE}}$		1.8366***							
		(0.0602)							
Sigma									0.2943***
									(0.0717)
Obs | Param | LL | AIC	1000	37	-2747.6	5569.2					

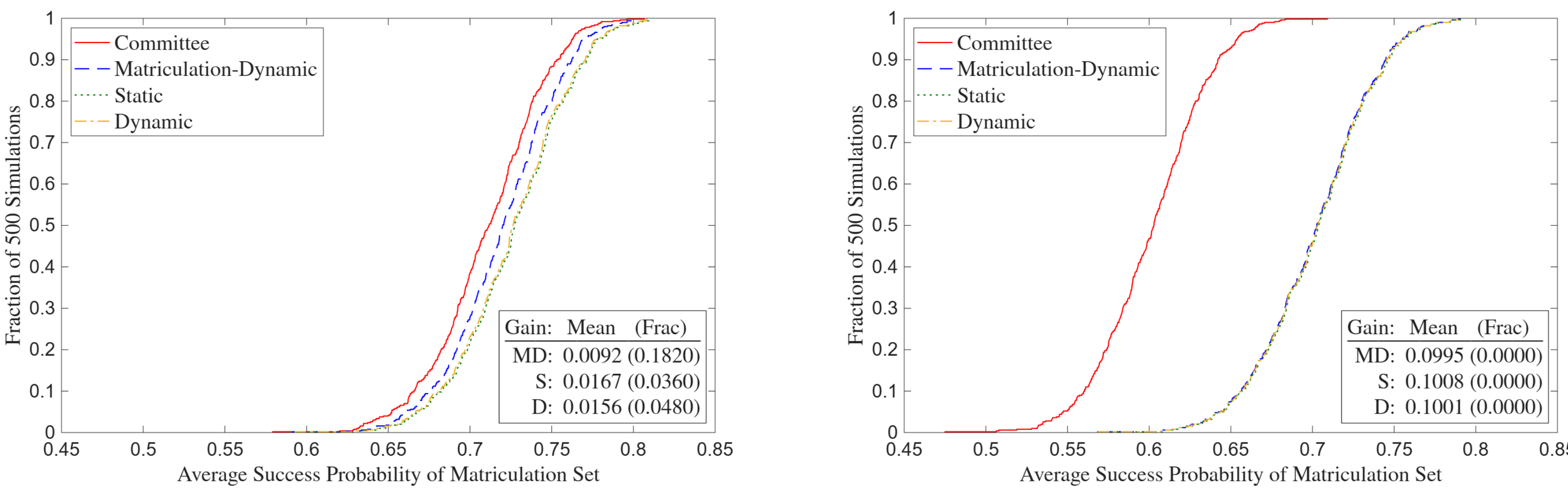
Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.



(a) Near-Optimal Committee

(b) Suboptimal Committee

Figure 7: Simulated Deviation Gains



(a) Near-Optimal Committee

(b) Suboptimal Committee

Figure 8: Simulated Matriculation Deviation Gains

E Two-Round Portfolio Choice

E.1 Without Matriculation

In Section 4.1, I show that the committee's portfolio choice problem reduces to a cutoff rule problem, keeping it to one round for simplicity. Then in Sections 4.1.1 and 4.1.2, I write out dynamic and static two-round cutoff rule problems with Round 1 filtering and Round 2 final selections. In this appendix, I solve a two-round portfolio choice problem by backward induction, showing that each round's solution takes the cutoff rule form in the paper. I start with Round 2:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A. \tag{33}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{2,n} = A$, which means that the constraint binds. The Lagrangian with multiplier λ_2 is:

$$\begin{aligned}\mathcal{L}(d_{2,1},\ldots,d_{2,N_T};\lambda_2) &= \sum_{n=1}^{N_T} P(y_n = S|x_n,z_n)\mathbb{1}(d_{2,n} = A) + \lambda_2\left\{N_A - \left[N_T - \sum_{n=1}^{N_T}\mathbb{1}(d_{2,n} = R)\right]\right\} \\ &= \sum_{n=1}^{N_T}\left[P(y_n = S|x_n,z_n)\mathbb{1}(d_{2,n} = A) + \lambda_2\mathbb{1}(d_{2,n} = R)\right] + \lambda_2(N_A - N_T)\end{aligned} \tag{34}$$

By the same logic in Section 4.1, the Round 2 solution is as follows with set-identified cutoff probability $\lambda_2^* = P(y_2^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$:

$$\forall n \in \{1,\ldots,N_T\},\, d_{2,n}(x_n,z_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|x_n,z_n) > \lambda_2^* \\ \{A,R\} & \text{if } P(y_n = S|x_n,z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|x_n,z_n) < \lambda_2^*. \end{cases} \tag{35}$$

When the committee solves Round 1, it must anticipate λ_2^*, which is a function of the $(\mathbf{x},\mathbf{z})$ of the entire transitioned pool. I make two assumptions so that this anticipation is tractable. First, conditional on $\mathbf{x}$, I assume that $\mathbf{z}$ is independent across applicants. Second, I assume that N is large. Under these assumptions, the same Glivenko-Cantelli argument as in the proof of Theorem 1, with $\mathbf{z}$ in place of the preference shocks, makes λ_2^* deterministic for large N.

Denote $\hat{G}_{N_T}(c) = \frac{1}{N_T}\sum_{n=1}^{N}\mathbb{1}\{P(y_n = S|x_n,z_n) \leq c\}\mathbb{1}\{d_{1,n} = T\}$ as the empirical distribution of the success probabilities of the N_T transitioned applicants. The cutoff λ_2^* is the $(1 - N_A/N_T)$ quantile of this distribution, since the committee accepts the N_A such applicants with the highest success probabilities. Because only $\mathbf{x}$ is observed in Round 1, transition depends on x_n alone; the Round 1 cutoff rule derived below transitions applicant n when x_n lies in transition region $\mathcal{X}_T$. For large N, the cutoff λ_1^* converges to a constant by the same argument, so $\mathcal{X}_T$ is deterministic.

Since the pairs (x_n, z_n) are iid and membership in $\mathcal{X}_T$ depends only on x_n, the transitioned success probabilities are themselves iid, with conditional distribution $G(c) = P(P(y = S|x,z) \leq$

$c|x \in \mathcal{X}_T)$. So by the Glivenko-Cantelli theorem, $\hat{G}_{N_T}$ converges almost surely to G for large N_T. Therefore, as $N_A/N_T \to q_2$, $\hat{G}_{N_T}$'s $(1-N_A/N_T)$ quantile converges to G's deterministic $(1-q_2)$ quantile. In turn, the set-identified interval for λ_2^* collapses to a point, denoted $\overline{\lambda_2^*}$. As in the proof of Theorem 1, independence is essential; a common component across applicants would slide $\hat{G}_{N_T}$ in unison and leave λ_2^* random in the limit. This convergence is a fixed point because the transition region $\mathcal{X}_T$ is pinned down by the Round 1 cutoff λ_1^*, which ranks applicants by their anticipated Round 2 payoff, which itself embeds $\overline{\lambda_2^*}$. The pair $(\lambda_1^*, \overline{\lambda_2^*})$ is jointly determined. Because $\overline{\lambda_2^*}$ is deterministic, no individual applicant's signal moves the cutoff, the committee's anticipated Round 2 payoff separates across applicants, and Round 1 reduces to a cutoff rule.

Let $EV_{2,n}(x_n) = \mathbb{E}[\max\{P(y_n = S|x_n, z_n), \overline{\lambda_2^*}\}|x_n] = \int_{\widetilde{z}_n} \max\{P(y_n = S|x_n, \widetilde{z}_n), \overline{\lambda_2^*}\} dF_{z|x}(\widetilde{z}_n|x_n)$. Then the Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n}=T) \leq N_T. \tag{36}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian with multiplier λ_1 is:

$$\begin{aligned}
\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) &= \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T) + \lambda_1 \left\{ N_T - \left[N - \sum_{n=1}^{N} \mathbb{1}(d_{1,n}=R) \right] \right\} \\
&= \sum_{n=1}^{N} \{EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T) + \lambda_1 \mathbb{1}(d_{1,n}=R)\} + \lambda_1(N_T - N).
\end{aligned} \tag{37}$$

By the same logic in Section 4.1, the Round 1 solution is as follows with set-identified cutoff level $\lambda_1^* = L(y_1^* = S) \in \left[EV_{2,N_T+1}(x_{N_T+1}), EV_{2,N_T}(x_{N_T})\right]$:

$$\forall n \in \{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{38}$$

There is one discrepancy between $\lambda_1^* = L(y_1^* = S)$ in this appendix vs. Section 4.1.1. Here, λ_1^* is bounded above by 1 because for all n, it is the case that $\max\{P(y_n = S|x_n, \widetilde{z}_n), \overline{\lambda_2^*}\} \leq 1$ pointwise in $\widetilde{z}_n$. Therefore, $EV_{2,n}(x_n) \leq \int_{\widetilde{z}_n} dF_{z|x}(\widetilde{z}_n|x_n) = 1$. But in Section 4.1.1, the type 1 extreme value preference shocks increase this upper bound to $1 + \sigma \log 2$, where I now denote $\overline{\lambda_2^*}$ by $P(y_2^* = S)$:

$$\begin{aligned}
\mathbb{E}[v_2(x_n, z_n, \varepsilon_{2,n})|x_n] &= \int_{\widetilde{z}_n} \sigma \log\left(e^{P(y_n=S|x_n,\widetilde{z}_n)/\sigma} + e^{P(y_2^*=S)/\sigma}\right) dF(\widetilde{z}_n|x_n) \\
&\leq \int_{\widetilde{z}_n} \sigma \log\left(2e^{\max\{P(y_n=S|x_n,\widetilde{z}_n),P(y_2^*=S)\}/\sigma}\right) dF(\widetilde{z}_n|x_n) \\
&= \int_{\widetilde{z}_n} \max\{P(y_n=S|x_n,\widetilde{z}_n), P(y_2^*=S)\} dF(\widetilde{z}_n|x_n) + \sigma \log 2 \\
&\leq 1 + \sigma \log 2.
\end{aligned} \tag{39}$$

In the static version of the two-round portfolio choice model, the Round 1 problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} P(y_n=S|x_n)\mathbb{1}(d_{1,n}=T), \text{ such that: } \sum_{n=1}^{N}\mathbb{1}(d_{1,n}=T)\leq N_T. \tag{40}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{1,n}=T$, which means that the constraint binds. The Lagrangian with multiplier λ_1 is:

$$\begin{aligned}\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) &= \sum_{n=1}^{N} P(y_n=S|x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\left\{N_T-\left[N-\sum_{n=1}^{N}\mathbb{1}(d_{1,n}=R)\right]\right\}\\ &= \sum_{n=1}^{N}\left[P(y_n=S|x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\mathbb{1}(d_{1,n}=R)\right]+\lambda_1(N_T-N). \end{aligned}\tag{41}$$

By the same logic in Section 4.1, the solution is as follows with set-identified cutoff probability $\lambda_1^*=P(y_1^*=S)\in\left[P(y_{N_T+1}=S|x_{N_T+1}),P(y_{N_T}=S|x_{N_T})\right]$:

$$\forall n\in\{1,\ldots,N\},\, d_{1,n}(x_n|\lambda_1^*)=\begin{cases}T & \text{if } P(y_n=S|x_n)>\lambda_1^*\\ \{T,R\} & \text{if } P(y_n=S|x_n)=\lambda_1^*\\ R & \text{if } P(y_n=S|x_n)<\lambda_1^*.\end{cases}\tag{42}$$

E.2 With Matriculation

In Section 4.2, I show that because the committee's capacity constraint is on the number of matriculations rather than acceptances, a one-round portfolio choice problem with matriculation reduces to a cutoff rule problem on success probabilities. However, a two-round portfolio choice problem with matriculation is less straightforward because matriculation ends up mattering in Round 1. In this appendix, I present such a model with Round 1 filtering and Round 2 final selections. For readability, I keep finite-sum notation rather than the integrals in the proof of Theorem 3 in Appendix F. But I invoke that theorem's large-N framework throughout because it removes the finite-N edge cases and, as in Appendix E.1, collapses λ_2^* to the deterministic $\overline{\lambda_2^*}$. Working backward, I start with Round 2, where for brevity, I denote $w=(x,z)$:

$$\begin{aligned}&\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n=S|w_n)P(m_n=M|w_n)\mathbb{1}(d_{2,n}=A),\\ &\text{such that: } \sum_{n=1}^{N_T} P(m_n=M|w_n)\mathbb{1}(d_{2,n}=A)\leq N_M.\end{aligned}\tag{43}$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{2,n}=A$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned}\mathcal{L}(d_{2,1},\ldots,d_{2,N_T};\lambda_2) &= \sum_{n=1}^{N_T} P(y_n=S|w_n)P(m_n=M|w_n)\mathbb{1}(d_{2,n}=A)+\lambda_2\left\{N_M-\left[\overline{N_M}-\sum_{n=1}^{N_T}P(m_n=M|w_n)\mathbb{1}(d_{2,n}=R)\right]\right\}\\ &= \sum_{n=1}^{N_T}\left[P(y_n=S|w_n)\mathbb{1}(d_{2,n}=A)+\lambda_2\mathbb{1}(d_{2,n}=R)\right]P(m_n=M|w_n)+\lambda_2(N_M-\overline{N_M}), \end{aligned}\tag{44}$$

where $\overline{N_M} = \sum_{n=1}^{N_T} P(m_n = M|w_n)$ is the expected number of matriculants if everyone is accepted.

Edge cases aside, the Round 2 solution, with cutoff probability $\lambda_2^* = P(y_2^* = S)$, is:

$$\forall n \in \{1,\ldots,N_T\},\, d_{2,n}(w_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A,R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^*. \end{cases} \tag{45}$$

Let $EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S|x_n, \tilde{z}_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|x_n, \tilde{z}_n) dF_{z|x}(\tilde{z}_n|x_n)$. This netted form, and not $\int_{\tilde{z}_n} \max\{P(y_n = S|x_n, \tilde{z}_n), \overline{\lambda_2^*}\} P(m_n = M|x_n, \tilde{z}_n) dF_{z|x}(\tilde{z}_n|x_n)$, is the correct transition payoff. To see why, evaluate the Round 2 Lagrangian at the cutoff rule solution with deterministic cutoff $\overline{\lambda_2^*}$, where $P(y_n = S|w_n)\mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)$ becomes $\max\{P(y_n = S|w_n), \overline{\lambda_2^*}\}$:

$$\mathcal{L}(d_2^*; \overline{\lambda_2^*}) = \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n), \overline{\lambda_2^*}\} P(m_n = M|w_n) + \overline{\lambda_2^*}(N_M - \overline{N_M}). \tag{46}$$

By $\max\{P(y_n = S|w_n), \overline{\lambda_2^*}\} = \max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\} + \overline{\lambda_2^*}$, and $\overline{N_M} = \sum_{n=1}^{N_T} P(m_n = M|w_n)$:

$$\begin{aligned} \mathcal{L}(d_2^*; \overline{\lambda_2^*}) &= \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|w_n) + \overline{\lambda_2^*}\,\overline{N_M} + \overline{\lambda_2^*}(N_M - \overline{N_M}) \\ &= \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|w_n) + \overline{\lambda_2^*} N_M. \end{aligned} \tag{47}$$

Although $\overline{N_M}$ varies with the Round 1 transition decisions, the $\overline{\lambda_2^*}\,\overline{N_M}$ terms cancel, leaving the constant $\overline{\lambda_2^*} N_M$. The Round 1 transition payoff is therefore the per-applicant netted term $\max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|w_n)$, whose expectation over $\tilde{z}_n$ defines $EV_{2,n}$.

The netting distinguishes the matriculation case from the case without it. Without matriculation, the netted and non-netted forms differ by $\int_{\tilde{z}_n} \overline{\lambda_2^*} dF_{z|x}(\tilde{z}_n|x_n) = \overline{\lambda_2^*}$, a constant across applicants that leaves the Round 1 ordering unchanged. With matriculation, they differ by $\int_{\tilde{z}_n} \overline{\lambda_2^*} P(m_n = M|x_n, \tilde{z}_n) dF_{z|x}(\tilde{z}_n|x_n)$, which varies with x_n. The non-netted form would therefore order applicants differently, and incorrectly. The Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N}) \in \{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \le N_T. \tag{48}$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned} \mathcal{L}(d_{1,1}, \ldots, d_{1,N}; \lambda_1) &= \sum_{n=1}^{N} EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \left\{ N_T - \left[N - \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = R) \right] \right\} \\ &= \sum_{n=1}^{N} [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1 (N_T - N). \end{aligned} \tag{49}$$

Edge cases aside, the Round 1 solution with cutoff level $\lambda_1^* = L(y_1^* = S, m_1^* = M)$ is:

$$\forall n \in \{1, \ldots, N\},\, d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{50}$$

The Round 1 transition payoff with Type 1 extreme value shocks is therefore:

$$\mathbb{E}[v_2(x_n, z_n, \varepsilon_{2,n})|x_n] = \int_{\widetilde{z}_n} \sigma \log\left(e^{[P(y_n=S|x_n,\widetilde{z}_n)-P(y_2^*=S)]/\sigma} + 1\right) P(m_n = M|x_n, \widetilde{z}_n)\, dF(\widetilde{z}_n|x_n). \tag{51}$$

The committee prefers to transition likely matriculants so that it can be more selective in Round 2. As a simple example, suppose that $N = 4$, $N_T = 2$, and $N_M = 1$, and suppose that recommendation letters matter more for success than GRE scores. Suppose also that Applicant 1 has a high GRE and strong recommendation letter, Applicant 2 has a high GRE and weak letter, Applicant 3 has a low GRE and strong letter, and Applicant 4 has a low GRE and weak letter. If the committee ignores matriculation probabilities in Round 1, it will transition Applicants 1 and 2 and then accept Applicant 1. But suppose that because Applicant 1 is so strong, they choose a higher-ranked university. Then the committee has to settle for Applicant 2. Whereas if the committee had transitioned Applicants 3 and 4, it could have accepted and landed Applicant 3.

In the static version of the model with matriculation, the Round 1 problem, with $\alpha \in [0, 1]$, is:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \le N_T. \tag{52}$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned} \mathcal{L}(d_{1,1}, \ldots, d_{1,N}; \lambda_1) &= \sum_{n=1}^{N} P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} \mathbb{1}(d_{1,n} = T) + \lambda_1 \left\{ N_T - \left[N - \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = R) \right] \right\} \\ &= \sum_{n=1}^{N} \left[P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R) \right] + \lambda_1 (N_T - N). \end{aligned} \tag{53}$$

Edge cases aside, the solution is as follows with cutoff probability $\lambda_1^* = P(y_1^* = S, m_1^* = M)$:

$$\forall n \in \{1, \ldots, N\},\, d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} > \lambda_1^* \\ \{T, R\} & \text{if } P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} = \lambda_1^* \\ R & \text{if } P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha} < \lambda_1^*. \end{cases} \tag{54}$$

But per the dynamic $EV_{2,n}$ integrand without shocks, $\max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|w_n)$, matriculation probabilities matter less than success probabilities. For applicants with $P(y_n = S|w_n) \le \overline{\lambda_2^*}$, matriculation has a weight of zero. Even for the other applicants, the weight is only large when $P(y_n = S|w_n) \gg \overline{\lambda_2^*}$. Thus, the static model without matriculation approximates the matriculation-dynamic model well enough empirically that α can be set to 1. Doing so also aligns with committee members indicating that they do not consider matriculation in Round 1.

F Proofs of Theorems

Theorem 1. *For large N, the committee's portfolio choice problem with preference shocks has a cutoff rule solution with logistic conditional choice probabilities.*

Proof. The portfolio choice problem with type 1 extreme value preference shocks is:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \left[(P(y_n=S|x_n,z_n)+\varepsilon_n(A))\mathbb{1}(d_n=A)+\varepsilon_n(R)\mathbb{1}(d_n=R)\right], \text{ such that } \sum_{n=1}^{N}\mathbb{1}(d_n=A)\leq N_A. \quad (55)$$

Because success probabilities are never negative, the unconstrained objective increases in expectation every time $d_n=A$, and so the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned}\mathcal{L}(d_1,\ldots,d_N;\lambda) &= \sum_{n=1}^{N}\left[(P(y_n=S|x_n,z_n)+\varepsilon_n(A))\mathbb{1}(d_n=A)+\varepsilon_n(R)\mathbb{1}(d_n=R)\right]+\lambda\left\{N_A-\left[N-\sum_{n=1}^{N}\mathbb{1}(d_n=R)\right]\right\}\\ &= \sum_{n=1}^{N}\left[(P(y_n=S|x_n,z_n)+\varepsilon_n(A))\mathbb{1}(d_n=A)+(\lambda+\varepsilon_n(R))\mathbb{1}(d_n=R)\right]+\lambda(N_A-N). \quad (56)\end{aligned}$$

Suppose without loss of generality that the applicants are sorted in decreasing order of their shock-perturbed success probabilities such that $P(y_1=S|x_1,z_1)+\varepsilon_1(A)-\varepsilon_1(R)\geq\cdots\geq P(y_N=S|x_N,z_N)+\varepsilon_N(A)-\varepsilon_N(R)$. To ensure that the committee accepts exactly N_A applicants, it must be the case that $\lambda^*=\lambda^*(\boldsymbol{\varepsilon}(A),\boldsymbol{\varepsilon}(R))\in\left[P(y_{N_A+1}=S|x_{N_A+1},z_{N_A+1})+\varepsilon_{N_A+1}(A)-\varepsilon_{N_A+1}(R),P(y_{N_A}=S|x_{N_A},z_{N_A})+\varepsilon_{N_A}(A)-\varepsilon_{N_A}(R)\right]$. That is, in this setup, λ^* is set identified. Then the solution is to accept all applicants whose shock-perturbed success probability exceeds λ^* and reject the rest:

$$\forall n\in\{1,\ldots,N\},\, d_n(x_n,z_n,\varepsilon_n|\lambda^*)=\begin{cases}A & \text{if } P(y_n=S|x_n,z_n)+\varepsilon_n(A)>\lambda^*+\varepsilon_n(R)\\ \{A,R\} & \text{if } P(y_n=S|x_n,z_n)+\varepsilon_n(A)=\lambda^*+\varepsilon_n(R)\\ R & \text{if } P(y_n=S|x_n,z_n)+\varepsilon_n(A)<\lambda^*+\varepsilon_n(R).\end{cases} \quad (57)$$

This cutoff rule holds conditional on the realized shocks. Given the shocks, the committee computes λ^* and applies the cutoff, so there is no difficulty for the committee. But there is a complication regarding estimation, namely that unlike the committee, the econometrician does not observe the shocks. Therefore, the conditional choice probabilities used in estimation require taking expectations across each applicant's shocks, which would be fine except λ^* is a function of each applicant's shocks simultaneously. Suppose for now that λ^* is instead deterministic. Then:

$$\begin{aligned}P(d_n=A|x_n,z_n) &= P(P(y_n=S|x_n,z_n)+\varepsilon_n(A)\geq\lambda^*+\varepsilon_n(R))\\ &= P(\varepsilon_n(R)-\varepsilon_n(A)\leq P(y_n=S|x_n,z_n)-\lambda^*)\\ &= \frac{\exp(P(y_n=S|x_n,z_n)/\sigma)}{\exp(P(y_n=S|x_n,z_n)/\sigma)+\exp(\lambda^*/\sigma)},\end{aligned} \quad (58)$$

and the likelihood becomes the product of individual binary logistics. The third equality holds because the difference of two independent T1EV random variables has a logistic distribution.

That said, λ^* is not deterministic for finite N, so the logistic conditional choice probabilities do not hold exactly. But for large N, λ^* converges to a constant. Denote $\hat{F}_N(c) = \frac{1}{N}\sum_{n=1}^{N} \mathbb{1}\{P(y_n = S|x_n, z_n) + \varepsilon_n(A) - \varepsilon_n(R) \leq c\}$ as the empirical distribution of the shock-perturbed success probabilities. The cutoff λ^* is the $(1 - N_A/N)$ quantile of this distribution. Because the shocks are iid, the perturbed indices in $\hat{F}_N$ are iid. So by the Glivenko-Cantelli theorem, $\hat{F}_N$ converges almost surely to the population distribution F for large N. Therefore, as $N_A/N \to q$, $\hat{F}_N$'s $(1 - N_A/N)$ quantile converges to F's deterministic $(1-q)$ quantile. The iid assumption is essential, and not only because each applicant's shock individually moves the cutoff by $O(1/N)$. A common perturbation component across applicants would have the same vanishing per-applicant influence, but it would slide the entire empirical distribution in unison, leaving λ^* random in the limit. Independence makes the simultaneous fluctuations of the N shocks cancel, so $\hat{F}_N$ and λ^* converge. With λ^* deterministic, no individual applicant's shock affects λ^*, the expectation defining each conditional choice probability factors across applicants, and the above logistic form holds.

Theorem 2. *For simplicity, suppose there is one round. The maximum likelihood estimate $\hat{P}_{y^*} = \hat{P}(y^* = S|t)$ is such that the number of acceptances N_A equals the expected N_A. Also, for any $\hat{P}_{y^*}$ and N_A, $\mathcal{L}^R$ is highest when the committee accepts the N_A highest $P_{y_n} = P(y_n = S|x_n, z_n, t_n, \bar{k}_n)$.*

Proof. Supposing one round, the restricted likelihood is:

$$\mathcal{L}^R = \prod_{n=1}^{N} \left[\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right]^{\mathbb{1}(d_n=A)} \left[\frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right]^{\mathbb{1}(d_n=R)} \left[\frac{e^{P_{y_n}}}{1+e^{P_{y_n}}}\right]^{\mathbb{1}(y_n=S)} \left[\frac{1}{1+e^{P_{y_n}}}\right]^{\mathbb{1}(y_n=F)} \tag{59}$$

When an applicant is accepted, the resulting term in $\mathcal{L}^R$ decreases in P_{y^*}, providing downward pressure on $\hat{P}_{y^*}$. Also, when one is rejected, the term increases in P_{y^*}, providing upward pressure.

Since $\mathbb{1}(d_n = A) + \mathbb{1}(d_n = R) = \mathbb{1}(y_n = S) + \mathbb{1}(y_n = F) = 1$, the restricted log-likelihood is:

$$\begin{aligned}
\ell^R = \sum_{n=1}^{N} \Bigg[&\mathbb{1}(d_n = A)\log\left(\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right) + \mathbb{1}(d_n = R)\log\left(\frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right) \\
&+ \mathbb{1}(y_n = S)\log\left(\frac{e^{P_{y_n}}}{1+e^{P_{y_n}}}\right) + \mathbb{1}(y_n = F)\log\left(\frac{1}{1+e^{P_{y_n}}}\right)\Bigg] \\
= \sum_{n=1}^{N} \Bigg[&\frac{1}{\sigma}\left(\mathbb{1}(d_n = A)P_{y_n} + \mathbb{1}(d_n = R)P_{y^*}\right) - \log\left(e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}\right) + \mathbb{1}(y_n = S)P_{y_n} - \log\left(1 + e^{P_{y_n}}\right)\Bigg].
\end{aligned} \tag{60}$$

Since ℓ^R is strictly concave in P_{y^*}, I find the unique maximizer $\hat{P}_{y^*}$ by the first-order condition:

$$\frac{1}{\sigma}\left[N - N_A - \sum_{n=1}^{N} \frac{e^{\hat{P}_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}}\right] = 0 \quad \Rightarrow \quad \sum_{n=1}^{N} \frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}} = N_A. \tag{61}$$

The left side of the latter equation is each applicant's acceptance probability summed over the number of applicants; that is, expected N_A. Therefore, $\hat{P}_{y^*}$ is such that the expected N_A equals N_A.

Given that $\hat{P}_{y^*}$ is identified, I will show that ℓ^R, and therefore $\mathcal{L}^R$, are highest when the committee accepts the N_A applicants with the highest P_{y_n}. The only part of ℓ^R that depends on d_n is

$\sum_{n=1}^{N}[\mathbb{1}(d_n=A)P_{y_n}+\mathbb{1}(d_n=R)P_{y^*}]=NP_{y^*}+\sum_{n=1}^{N}\mathbb{1}(d_n=A)(P_{y_n}-P_{y^*})$; this equality holds since $\mathbb{1}(d_n=A)+\mathbb{1}(d_n=R)=1$. As such, the only part depending on d_n is $\sum_{n=1}^{N}\mathbb{1}(d_n=A)(P_{y_n}-P_{y^*})$.

This expression is highest when $d_n=A$ for the N_A highest values of P_{y_n}, so the likelihood is highest when the committee accepts the N_A highest P_{y_n}. Intuitively, the committee's selectivity identifies $\hat{P}_{y^*}$, and it is not necessary for identification for the committee to accept applicants with higher P_{y_n}. However, if the committee engages in positive sorting, the model's fit will improve.

Theorem 3. *For large N, the committee's portfolio choice problem with matriculation probabilities has a cutoff rule solution on success probabilities.*

Proof. For brevity, let $w=(x,z)$. Then for large N, the portfolio choice problem is:

$$\max_{\delta\in\{\delta|\delta:\mathbb{R}^K\to[0,1]\}}\int_{w\in\mathbb{R}^K}P(y=S|w)P(m=M|w)\delta(w)f(w)dw,$$

$$\text{such that: }\int_{w\in\mathbb{R}^K}P(m=M|w)\delta(w)f(w)dw\leq N_M,\tag{62}$$

where δ is the committee's decision rule, and f is the density of w. Because success and matriculation probabilities, as well as densities, are never negative, the unconstrained objective increases in δ, which means that the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned}\mathcal{L}(\delta;\lambda)&=\int_{w\in\mathbb{R}^K}P(y=S|w)P(m=M|w)\delta(w)f(w)dw\\&\quad+\lambda\left\{N_M-\left[\int_{w\in\mathbb{R}^K}P(m=M|w)f(w)dw-\int_{w\in\mathbb{R}^K}P(m=M|w)(1-\delta(w))f(w)dw\right]\right\}\\&=\int_{w\in\mathbb{R}^K}\left[P(y=S|w)\delta(w)+\lambda(1-\delta(w))\right]P(m=M|w)f(w)dw+\lambda\left[N_M-\int_{w\in\mathbb{R}^K}P(m=M|w)f(w)dw\right].\end{aligned}\tag{63}$$

In the discrete case, there is not necessarily a cutoff rule solution. But in the continuous case, the committee can accept the necessary fraction of applicants such that $\int_{w\in\mathbb{R}^K}P(m=M|w)\delta(w|\lambda^*)f(w)dw=N_M$, which results in a cutoff rule solution on success probabilities alone:

$$\forall w\in\mathbb{R}^K,\ \delta(w|\lambda^*)=\begin{cases}1 & \text{if } P(y=S|w)>\lambda^*\\ [0,1] & \text{if } P(y=S|w)=\lambda^*\\ 0 & \text{if } P(y=S|w)<\lambda^*.\end{cases}\tag{64}$$

The committee accepts all applicants with success probabilities $P(y=S|w)$ strictly greater than the cutoff probability λ^*, rejects all applicants with $P(y=S|w)$ strictly less than λ^*, and may accept or reject any applicants with $P(y=S|w)$ equal to λ^*, if such a mass of applicants exists. All that remains is to identify λ^*. Like Bai et al. (2022), I define the following two functions:

$$\begin{aligned}M^H(\lambda)&=\int_{\{w\in\mathbb{R}^K|P(y=S|w)\geq\lambda\}}P(m=M|w)f(w)dw\\M^L(\lambda)&=\int_{\{w\in\mathbb{R}^K|P(y=S|w)>\lambda\}}P(m=M|w)f(w)dw.\end{aligned}\tag{65}$$

The upper bound function M^H is the fraction of applicants, scaled by their matriculation probabilities, whose success probabilities are greater than or equal to some λ, and the lower bound function M^L is the fraction whose success probabilities are strictly greater than λ. Both functions are between 0 and 1, both are decreasing in λ, and for any λ, it must be that $M^H(\lambda) \geq M^L(\lambda)$.

Then it is the case that $\lambda^* = \inf_{\lambda}\{\lambda | M^L(\lambda) \leq N_M\}$. Intuitively, λ^* is the largest cutoff probability where the fraction of matriculation-probability-scaled applicants with $P(y = S|w) > \lambda$ does not exceed the maximum number of matriculants N_M. If there is a positive mass of matriculation-probability-scaled applicants with $P(y = S|w) = \lambda^*$ (that is, if $M^H(\lambda^*) - M^L(\lambda^*) > 0$), then the committee will randomly accept the exact fraction $\alpha^* \in [0,1]$ of that mass of applicants such that $M^L(\lambda^*) + \alpha^*[M^H(\lambda^*) - M^L(\lambda^*)] = N_M$. That is, the committee will accept every applicant with $P(y = S|w) > \lambda^*$ and accept just enough applicants with $P(y = S|w) = \lambda^*$ to fill the cohort.

Theorem 4. *Let d^R be the model's decision rule with $\sigma = 0$ in the CCPs, and let d^U be the committee's decision rule. By optimality:*

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$
$$\geq \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\} \quad (66)$$

Proof. This theorem states that conditional on a fixed Round 1 decision rule $d_1[P_y(\hat{\theta}^U)]$ and corresponding transition set of exogenously-determined size N_T, the sample average success probability of the also-exogenous N_A applicants that the model accepts in Round 2 weakly exceeds the sample average success probability of the N_A applicants that any actual committee accepts in Round 2. The inequality holds since the model's Round 2 decision rule is to accept the N_A applicants with the highest success probabilities. Therefore, the sample average success probability of the model's acceptances must weakly exceed that of any other set of N_A acceptances.

However, the inequality would not always hold if it were written as follows (changes in bold):

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{\mathbf{d_{1,n}^R}\left[\mathbf{P_{y_n}}\left(\hat{\boldsymbol{\theta}}_{\mathbf{S}}^{\mathbf{U}}\right)\right] = \mathbf{T}, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$
$$\textbf{vs.} \ \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{\mathbf{d_{1,n}^U}\left[\mathbf{P_{y_n}}\left(\hat{\boldsymbol{\theta}}_{\mathbf{T}}^{\mathbf{U}}\right)\right] = \mathbf{T}, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\} \quad (67)$$

The difference is that the Round 1 transition set is no longer fixed. Now, the model's transition set is based on choosing the N_T applicants in each year with the highest $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log(e^{P(y=S|x,z,t,\bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma}) dF(\tilde{z}|x,t)$. In contrast, the committee may have a different Round 1 decision rule. Crucially, although the model is choosing applicants akin to those with the highest expected success probabilities, it is ignoring any value added from upside potential.

For example, suppose $N_A = 1$, and there is an applicant in Round 1 with a low expected success probability but a known high degree of variance in the distribution of z conditional on their (x_n, t_n). That is, $F(z|x,t)$ is heteroskedastic. Also suppose that their low expected success probability yields an $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]$ too low to survive Round 1, even though their actual success probability ends up being the highest of any applicant once z is realized in Round 2. If the committee's decision rule considers upside potential, then the committee may accept this applicant, while the model will not. In most cases, the model's Round 1 decision rule should lead to a better acceptance set than the committee's rule. But such an outcome is not guaranteed.

Theorem 5. *For large B, the fraction of $\tilde{\theta}^U_{S,b}$ with gain $g(\tilde{\theta}^U_{S,b}) \leq 0$ is a p-value for $H_0\colon\, g_0 \leq 0$.*

Proof. Define deviation gain $g(\theta_S) = \frac{1}{N_A}\sum_{n=1}^N P\Big(y_n = S\Big|x_n, z_n, t_n, \bar{k}_n; \theta_S\Big)\mathbb{1}(d^R_{1,n} = T, d^R_{2,n} = A) - \frac{1}{N_A}\sum_{n=1}^N P\Big(y_n = S\Big|x_n, z_n, t_n, \bar{k}_n; \theta_S\Big)\mathbb{1}(d^U_{1,n} = T, d^U_{2,n} = A)$. That is, the gain is the average success probability of the model's acceptance set minus that of the committee's acceptance set. Since the model and committee's sets are determined before performing the test, the proof works the same way for matriculation sets as acceptance sets. Suppose the maximum likelihood estimate $\hat{\theta}^U_S \overset{a}{\sim} \mathcal{N}(\theta^0_S, \Sigma_0)$, and each perturbation draw $\tilde{\theta}^U_{S,b} \sim \mathcal{N}(\hat{\theta}^U_S, \hat{\Sigma})$, where $\hat{\Sigma}$ is consistent for Σ_0.

Because g is a smooth function of θ_S, it is true by the delta method that $\hat{g} = g(\hat{\theta}^U_S) \overset{a}{\sim} \mathcal{N}(g_0, \sigma^2_g)$ and $\tilde{g} = g(\tilde{\theta}^U_{S,b}) \mid \hat{g} \overset{a}{\sim} \mathcal{N}(\hat{g}, \sigma^2_g)$, where $g_0 = g(\theta^0_S)$ and $\sigma^2_g = \nabla g(\theta^0_S)'\Sigma_0 \nabla g(\theta^0_S)$. As the number of perturbations B becomes large, it is true by the law of large numbers that the fraction of perturbations with $\tilde{g} \leq 0$ converges to the probability that a single perturbation is weakly negative:

$$\hat{p} = P(\tilde{g} \leq 0 \mid \hat{g}) = \Phi\left(\frac{0 - \hat{g}}{\sigma_g}\right) = \Phi(-\hat{g}/\sigma_g). \tag{68}$$

For $\hat{p}$ to be a p-value, it must equal the probability, if the null is true with $g_0 = 0$, of an estimated gain at least as large as the observed gain. Denote $\hat{g}' = \sigma_g Z$ as a hypothetical estimated gain generated under $g_0 = 0$, where $Z \sim \mathcal{N}(0,1)$. Then, using $P(Z \geq c) = \Phi(-c)$:

$$P(\hat{g}' \geq \hat{g} \mid g_0 = 0) = P(Z \geq \hat{g}/\sigma_g) = \Phi(-\hat{g}/\sigma_g) = \hat{p}. \tag{69}$$

That is, the probability, if the null is true with $g_0 = 0$, of an estimated gain at least as large as the observed gain equals the probability to which the fraction of perturbations with $\tilde{g} \leq 0$ converges. Finally, I show that when $g_0 \leq 0$, the probability of a Type 1 error is bounded above by the statistical significance threshold α. That is, $P(\hat{p} \leq \alpha) \leq \alpha$ when $g_0 \leq 0$. Because inverses of strictly increasing functions preserve inequalities, Φ is strictly increasing, and $\hat{p} = \Phi(-\hat{g}/\sigma_g)$; it is the case that $\hat{p} \leq \alpha$ holds exactly when $\hat{g} \geq -\sigma_g\Phi^{-1}(\alpha)$. Substituting $\hat{g} = g_0 + \sigma_g Z$:

$$P(\hat{p} \leq \alpha) = P(Z \geq -\Phi^{-1}(\alpha) - g_0/\sigma_g) = \Phi(\Phi^{-1}(\alpha) + g_0/\sigma_g). \tag{70}$$

For $g_0 = 0$, it is the case that $P(\hat{p} \leq \alpha) = \Phi(\Phi^{-1}(\alpha)) = \alpha$. For $g_0 < 0$, because Φ is strictly increasing, it is the case that $P(\hat{p} \leq \alpha) = \Phi(\Phi^{-1}(\alpha) + g_0/\sigma_g) < \alpha$. Therefore, rejecting H_0 when $\hat{p} \leq \alpha$ produces Type 1 errors with probability at most α everywhere that H_0 is true.